Three Essays on Resource Economics

by

Elyse Jayne Adamic

B.A., University of Alberta, 2015

M.A., University of Toronto, 2016

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submitted by **Elyse Jayne Adamic** in partial fulfillment of the requirements for the degree of **Doctor of Philosophy** in **Land and Food Systems**.

Examining Committee:

Frederik Noack, Associate Professor, Food and Resource Economics, UBC
Supervisor

Patrick Baylis, Associate Professor, Economics, UBC
Supervisory Committee Member

Sumeet Gulati, Professor, Food and Resource Economics, UBC
Supervisory Committee Member

TBA, UBC
University Examiner

TBA, UBC
University Examiner

TBA
External Examiner

Abstract

This dissertation consists of three essays concerning topics in resource economics, with a focus on the oil and gas and fishing sectors. Each essay quantifies the impacts of extraction on both resource users and broader local populations with relevance for designing policies that balance effective management with damage mitigation.

In the first essay, I investigate the effects of oil and gas wells on housing prices across the life cycle of the well. I combine detailed geospatial well production data with property characteristics and transactions in Pennsylvania to identify all wells within specified distances of each home. The effects of active wells on home prices reflect short-term production activity, while the effect of abandoned wells capture the long-term perceived environmental risk that can persist well after extraction ends. I also consider whether well-plugging, a growing focus of environmental policy, can reclaim those losses. I find inactive wells have a negative impact that does not diminish with distance, and plugging does not fully reverse these losses.

In the second essay, we use the universe of individual level catch and revenue data for Maritime Quebec to quantify the distributional impacts of an ITQ system. We look at fishers who differ in skill and outside option to speak to the efficiency-equity tradeoff that is a central concern in property-rights based regulation. We implement a difference-in-difference approach to look at heterogeneous changes in income, effort, and exit along these distributions. We find the introduction of a particular ITQ system raised incomes without harming resource dependent communities.

In the third essay, I develop a predictive framework to identify Pennsylvania wells at highest risk of becoming orphaned. I first construct a naive, production-based classification and then estimate the probability that a well becomes orphaned within a fixed time horizon using a logit model. Age, declining production profiles, and small-operator ownership are consistent predictors of future orphaning.

Together, these chapters show that while resource-based industries generate substantial economic value, they also create environmental risks, distributional consequences, and legacy liabilities that require policies to anticipate and internalize these costs.

Lay Summary

This dissertation consists of three essays relating to the impacts of resource extraction on the local economy. In the first essay, I quantify the long-term impacts of oil and gas wells on the housing market in Pennsylvania. I consider the effect of active, abandoned and plugged wells within a given radius on sale prices to show that negative externalities persist after production ceases. In the second essay, I study the distributional impacts of individual transferable quotas in Québec on revenues, effort, and exit across fishers who differ in skill and outside option. In the third essay, I develop a framework to predict the future orphan well burden in Pennsylvania counties, identifying the set of unplugged wells most at risk based on well and operator characteristics. Together, these essays provide insights into the economic and environmental consequences of extractive industries and the policies intended to address them.

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Dedication

Chapter 1

Introduction

Effective natural resource management requires balancing efficient use with the mitigation of economic and environmental costs. Achieving these objectives depends on designing policies with a clear understanding of how extraction and regulations affect economic outcomes across time horizons and broader populations. Resource policies shape both the distribution of benefits and the incidence of costs, raising questions about who gains, who bears the losses, and how short-run gains trade off against long-run consequences. This dissertation is a collection of research that applies empirical techniques to quantify how natural resource extraction and management institutions shape economic outcomes.

A central challenge in natural resource management is internalizing long-run damages, particularly when extraction generates short-term benefits that accrue to a narrow population while future costs are borne widely. Resource industries often provide immediate local economic gains through increased investment and employment, while associated costs are diffuse, delayed, or not directly priced in markets. These challenges are compounded by the fact that for both renewable and non-renewable resources, policy must address not only efficiency in extraction but also conservation, environmental protection, and site restoration. Such objectives are especially difficult to balance in regions with resource-dependent labor markets, where economic and environmental systems are tightly linked and mutually reinforcing. Despite substantial attention to maximizing economic value from resource use, less is often known about how the associated benefits and costs are distributed across populations and over time. Identifying these impacts and trade-offs empirically is complicated by spatial heterogeneity, time trends, and endogenous policy responses. Each chapter combines micro-level data and quasi-experimental methods to quantify costs that are typically overlooked, focusing on latent risks and distributional shifts.

I study these questions in two resource industries: oil and gas in Chapter 2 and Chapter 4, and commercial fisheries in Chapter 3. Despite differing in physical characteristics and regulatory environments, both illustrate how extraction generates economic benefits alongside delayed environmental costs and distributional consequences. The oil and gas sector is characterized by boom and bust cycles, with substantial drilling and production during periods of high prices and persistent environmental liabilities after production ends. In Pennsylvania, this legacy includes a large stock of abandoned and orphaned wells that pose ongoing environmental and fiscal risks, particularly because wells are widely dispersed and frequently located near residential communities. In fisheries, the central regulatory challenge lies in preventing overexploitation of a renewable resource while sustaining the livelihoods of resource-dependent communities. Policy responses often involve the assignment of harvesting privileges to promote efficiency and sustainability, yet these can reshape access, income distribution, and labor outcomes among heterogeneous fishers. Together, these settings provide complementary perspectives on how resource policies allocate benefits and costs across space, agents, and time. Chapter 2 examines the long-term impacts of oil and gas wells on the housing market. The drilling of oil and gas wells creates large upfront economic benefits for drillers, but the economic and environmental costs only appear years or decades later, often long after the original owners of the well are legally required or financially able to repay them. This arrangement creates the potential for economic inefficiency along several dimensions: "judgement-proof" drillers who do not account for such long-run costs in their drilling decision, myopic homeowners who fail to anticipate the capital costs to their land, and neighbors onto whom the environmental costs of drilled, uncapped wells often spill over. Although the possibility of these types of market failures is well-documented, relatively little evidence is available about the magnitude of such costs and the degree to which they persist over time. Wells present a hazard to homes by being open-ended in duration, uncertain in remediation, and regulatorily ambiguous. I consider both whether homeowners with wells on their property suffer long-term property value losses, specifically due to abandoned, non-producing wells, and whether well-plugging, a growing focus of environmental policy, can reclaim those losses. I combine detailed geospatial well location and production data with individual home characteristics and sales in Pennsylvania. I find that nearby oil and gas wells generate moderate yet economically meaningful reductions in housing values, which persist after active production. Abandoned wells are associated with similar to slightly larger price penalties than active wells, reflecting the willingness to pay to avoid long-run environmental risk and uncertainty. I find that plugging does not fully reverse this loss, suggesting persistent perceptions of risk or environmental degradation. These results support environmental policies that increase setback requirements and strengthen incentives for timely well plugging and land reclamation.

Chapter 3 uses the universe of individual level catch and revenue data for Maritime Québec to quantify the distributional impacts of introducing individual transferable quotas. Global overfishing causes substantial economic losses. ITQ systems are seen by economists as a promising solution, creating a market mechanism which limits catch to the socially optimal level and ensures fish are caught at the least cost. However, despite increasing overall rents, fishers benefit differently from the ITQ system depending on their skill and outside income option, and efficiency can come at the cost of equity. This is particularly important for small resource dependent communities. We use matching methods to find appropriate control fleets, and then implement a difference-in-difference event-study approach to first look at distributional changes in incomes and effort on the intensive margin and then exit rates on the extensive margin. Overall, we find that fishing incomes increase following the introduction of an ITQ regulation while fishing effort remains constant, supporting the efficiency increasing role of ITQ systems. Revenue gains are concentrated among the lower skilled and those with lower outside options, suggesting a decrease in the revenue dispersion within the fishery for those most dependent. On the extensive margin we find substantial exit across all labour market strengths but primarily among low-skill fishers, indicating that consolidation is driven primarily by productivity differences rather than differential reallocation opportunities. Taken together, ITQs compress earnings along the skill distribution and accrue to those without strong outside options among active fishers, but substantial consolidation implies that the overall equity implications will depend on the post-exit outcomes of displaced participants.

Chapter 4 takes on the task of predicting the future orphan well burden. Abandoned wells impose increasing environmental and fiscal risks across the United States, yet policymakers lack reliable tools to anticipate where future liabilities will arise. I develop a predictive framework to identify Pennsylvania wells at highest risk of becoming orphaned. Using detailed well-level production and ownership records, I first construct a naive, production-based classification of the current well stock and aggregate these categories to the operator level, highlighting the concentration of high-risk wells among small, low-production firms with limited operational capacity. I then estimate the probability that a well becomes orphaned within a fixed time horizon using a logit model incorporating well age, cumulative and recent production, operator scale, and drilling vintage. Age, declining production profiles, and small-operator ownership emerge as strong and consistent predictors of future orphaning. Mapping these probabilities reveals that the highest-risk wells are concentrated in older conventional oil fields in northwestern and southwestern Pennsylvania, rather than in the state's shale-gas regions.

While united under a common theme and sharing empirical methodologies, the three chapters in this dissertation can be read independently. It is the author's hope that each provides policy insights. In addition to ex-post evaluation, the dissertation also informs ex-ante policy design for mitigating damages, including plugging incentives and setback limits, bonding requirements, and ITQ designs that consider labor mobility and inequality. More broadly, the dissertation speaks to the central objective of resource economics of promoting efficient and sustainable resource use while equitably allocating the benefits and burdens of extraction across agents and over time.

Chapter 2

The Long Term Impacts of Oil and Gas Extraction

2.1 Introduction

Oil and gas activity is known to generate immediate economic benefits to the surrounding communities through increased employment, higher wages, and improvements to local infrastructure (Allcott & Keniston, 2018; Maniloff & Mastromonaco, 2017). This pattern is evident in both early conventional development and modern shale booms, however these short run gains may be offset by long run liabilities if firms delay the costs of plugging. Mounting evidence documents the environmental risks associated with abandoned wells, including methane emissions, explosions, groundwater contamination, and land subsidence. In Pennsylvania alone, more than 90 percent of abandoned wells emit methane, contributing an estimated 5 to 8 percent of the state's total methane emissions (Kang et al., 2021). The sheer scale of the orphan well problem has also become increasingly apparent, with tens of thousands of wells across the United States lacking any financially responsible party on record. These environmental and financial burdens often emerge decades later during the post-production phase and are not internalized during the drilling decision. Consequently, costs that should be borne by operators are shifted onto communities and state governments imposing substantial public liabilities.

The externalities associated with oil and gas wells also change significantly over the life cycle. While active wells generate noise, traffic, and pollution, landowners may be partially compensated through lease payments. Over time, as production declines and operator incentives to maintain the well diminish, reduced monitoring can cause environmental and safety risks to persist or intensify. In contrast to active drilling where the effects are more immediate and visible, abandoned wells represent a more pervasive and less perceptible environmental threat while also far outnumbering producing wells. Recent state and federal initiatives, including the 2021 Infrastructure Investment and Jobs Act (IIJA), have sought to expand plugging efforts. However, even with expanded funding, remediation efforts lag far behind the rate at which wells become inactive and fall far short of addressing the vast legacy inventory of unplugged wells. Over time the stock of abandoned and orphan wells grows, transforming industrial activity into a persistent environmental cost.

Despite growing policy attention, there is limited empirical evidence on the long-term economic impacts of oil and gas wells. While the short-term gains can be widespread across the local economy, most enduring risks associated with abandoned or improperly maintained wells are concentrated among those residents in direct proximity. Using property market data from Pennsylvania, I quantify the extent to which environmental hazards are capitalized into housing prices, paying particular attention to the effect of distance and unplugged wells. I also examine whether plugging reverses any of the associated price penalties. I first outline how the hedonic framework capitalizes a dynamic environmental good to demonstrate how housing prices reflect households' willingness to pay for environmental quality and local amenities. This allows me to recover the implicit cost that nearby residents bear from proximity to wells while controlling for the vicinity. This approach follows closely the methodology of Muehlenbachs, Spiller, and Timmins (2015), and I extend the analysis by focusing explicitly on the post-production phase and on the persistent risks associated with long-term inactivity and abandonment.

To study these dynamics, I construct a novel matched dataset of well production histories and property transactions in Pennsylvania over 1980 - 2023. I identify homes that were exposed to oil and gas activity at the time of sale and use variation in both distance and intensity to wells of differing statuses within a specified radius. For each well status, I construct exposure in a set of non-overlapping concentric distance bins around each home. I use the production histories to track transitions in status, namely differentiating between inactivity and abandonment in the post-production period. I supplement the production histories with the complete documented inventory of legacy wells, which have remained inactive over the entire period. I employ a two-way fixed effects model to address potential confounding from well placement, neighborhood characteristics, and temporal shocks. This empirical strategy captures the cumulative risks associated with prolonged environmental exposure.

I find meaningful and persistent effects of drilling on property values. Active wells are associated with a 0.2 to 0.6 percent reduction in home value, but only for the furthest distance bin suggesting royalty benefits or lease payments. When looking at repeat sales, this could be as high as 1.1 percent. These losses do not dissipate when wells become abandoned: abandoned wells are associated with similarly sized or larger price penalties from 0.7 to 1.2 percent, consistent with the interpretation that abandonment combines the legacy effects of prior production with new, long-run environmental risk and uncertainty. Interestingly, I do not find well plugging recovers home values, where plugged wells within 500 m can be associated with up to 2.1% reduction in nearby home prices. These effects diminish with distance, but this suggests that remediation may not fully eliminate perceived or actual risks. While small in magnitude, each home is surrounded by a varying landscape of wells. However, these effects are largely driven by older wells plugged prior to modern regulations. However home prices may not fully capture all disamenities if risks are uncertain, not salient to buyers at the time of purchase, or diffuse such as broader climate impacts. In this sense, these estimates represent local perceived costs.

By quantifying the long-run damages imposed on housing markets, this research informs regulatory efforts and highlights the financial burdens associated with legacy environmental risks. The results speak to the inadequacy of bonding requirements intended to ensure operators pay for plugging, setback rules that determine residential exposure, and financing mechanisms such as Pennsylvania's impact fee. Strengthening these policies is critical not only for addressing existing liabilities but also for preventing the emergence of a new generation of abandoned wells.

In Section 2.2 I provide a detailed background of the oil and gas industry in Pennsylvania and current regulations. In Section 2.3 I provide an overview of the hedonic method and in Section 2.4 I describe the data. I present the empirical strategy in Section 2.5, with results in Section 2.6. I present a sub-group analysis in 2.7. Section 2.8 provides a series of robustness checks. I present a policy discussion in 2.9 and conclude with 2.10.

2.2 Background

2.2.1 Oil and Gas Drilling in Pennsylvania

Oil and gas extraction has played a significant role in Pennsylvania's economy since the drilling of the first commercial oil well in the United States in 1859. Over the following century and a half, the industry expanded geographically and technologically, leaving behind a dispersed network of wells that span multiple regulatory eras. Many early wells were drilled prior to modern permitting or reporting requirements, and as operators exhausted production or exited the industry, large numbers were left unplugged. The legacy of this early drilling is still visible today, as modern communities live among a dispersed and aging network of wells with varying production statuses. Pennsylvania contains over 300,000 documented wells, and estimates suggest that an additional 200,000 - 500,000 undocumented wells may exist statewide (Kang et al., 2016).

Historically, Pennsylvania's industry relied on conventional vertical wells drilled by small operators, producing relatively low volumes over long periods. However, beginning in the mid-2000s, advances in hydraulic fracturing and horizontal drilling enabled the development of the Marcellus Shale, transforming the state's industry. Unconventional wells peak early in production, while conventional wells tend to decline slowly, often continuing to produce marginal quantities for decades.

The lifecycle of a well includes site preparation, drilling, production and closure. During the active production phase, wells extract oil or gas and generate revenue. As a well nears the end of its productive life, operators face decisions about whether to extend production through additional interventions, temporarily idle the well, or plug it. Many older conventional wells transition into "stripper wells," producing only marginal quantities of hydrocarbons but remaining active to postpone costly closure requirements. Other wells may be temporarily idled as operators wait for favorable market conditions. Once a well is no longer viable, proper plugging and abandonment are required to prevent environmental hazards and regain use of the land. If these are not followed, the well is abandoned.

Pennsylvania's oil and gas regulations have evolved considerably, reflecting increasing awareness of environmental risks. Early drilling was mostly unregulated, where operators were not required to report the drilling location or plug wells and many wells drilled during this time are still unaccounted for. The Oil and Gas Act of 1984 introduced more stringent permitting requirements and standards for well construction, operation, and plugging, however enforcement was limited. Regulatory updates such as Act 13 of 2012, sought to address the unique challenges of shale gas extraction. Act 13 introduced impact fees, stricter well construction standards, and measures to protect water resources. Act 13 also introduced higher bond requirements for producers that use fracking, although these requirements often fall short of covering full plugging costs. Further amendments in 2016 mandated disclosure of chemicals used in hydraulic fracturing and imposed stricter zoning restrictions on oil and gas operations.

Despite these regulatory improvements, the stock of unplugged wells is increasing, particularly during periods of low oil prices when operators face financial distress or bankruptcy. Firms often have limited financial incentives to plug wells, especially when bankruptcy allows them to offload environmental liabilities (Mitchell & Casman, 2011). Boomhower (2019) shows that weak bonding and limited liability protections can lead operators to under-invest in cleanup and exit before remediation is complete. Black, McCoy, and Weber (2018) show that impact fees often fall short of addressing long-term environmental liabilities such as unplugged wells, although intended to internalize some external costs.

The persistence of orphan wells reflects historical regulatory inadequacies and underscores their potential economic impacts, including effects on housing markets. Understanding this context is essential for analyzing the economic and policy dimensions of orphan wells.

2.2.2 Housing Market and Oil and Gas Activity

The economic literature of the effects of on oil and gas development on housing markets has largely focused on active wells.[1] Most notably, Muehlenbachs et al. (2015) estimate the impact of shale gas development in Pennsylvania differentiating between ground-water dependent properties and piped-water properties. They find that properties within 1.5 km of a shale gas well experience a 13% reduction in sales price if they rely on groundwater, reflecting concerns about contamination. In contrast, similar properties with access to piped water show a small positive effect, likely due to lease payments or expected economic gains from development. Boxall, Chan, and McMillan (2005) find a negative impact of hazardous wells on rural residential properties in Alberta, Canada. Property values are reduced between 4 and 8 percent within 4 km of a facility emitting lethal gases. Gopalakrishnan and Klaiber (2014) find that proximity to shale gas wells consistently diminishes property values in Washington County, Pennsylvania. The losses increase with each additional well, with particularly large losses for agricultural land. Delgado, Guilfoos, and Boslett (2016) only find weak evidence for a few select Pennsylvania counties that are isolated from other resource extraction or large urban areas. In contrast, Balthrop and Hawley (2017) study a densely populated area in Texas and find robust evidence that unconventional wells drilled within 3500 ft of a property reduce property values by 1.5 - 3 percent with an additional loss from construction. Weber and Hitaj (2015) study the link between shale development and farm real estate values, which reflects the value of the undeveloped land. They find small positive effects, suggesting the leasing potential of the land - any long-term disamenities associated with drilling were not large enough to outweigh the positive effects of gas development. Bennett and Loomis (2015) use a hedonic pricing model to examine the impact of fracked oil and gas wells on housing values in

[1] All of these papers use some form of the hedonic method developed by Rosen (1979). I build on the extensive literature using housing market to measure environmental disamenities (Greenstone and Gallagher (2008), Gamper-Rabindran and Timmins (2013), Cassidy, Meeks, and Moore (2023).)

Weld County, Colorado, finding that proximity to wells can have both positive and negative effects depending on the distance and density of development.

While this literature has established effects of active wells, little attention has been given to the impacts of abandoned and orphan wells. My research extends this body of work by quantifying the economic costs of abandoned wells on housing markets, addressing a critical gap in understanding long-term externalities.

2.2.3 Local Economy and Oil and Gas Activity

The benefits that accrue to local economies during periods of high oil and gas extraction are well documented. These effects are broad in scope, short and medium run by nature, and are distributed equally among the proximal population.

Evidence across multiple settings demonstrates persistent increases in wages, employment, and royalty income, with the largest gains occurring in areas experiencing the most intensive drilling (Feyrer, Mansur, & Sacerdote, 2017). Maniloff and Mastromonaco (2017) attribute over 500,000 new local jobs to fracking. Komarek (2016) finds that shale gas development in the Marcellus region increased county-level employment by 1.5% to 2.5% and average wages by 1% to 2%. While these booms broadly improve local economic indicators, including government revenue and local public finances, they may simultaneously reduce certain local amenities and reveal considerable heterogeneity in resident's willingness to pay for fracking (Bartik, Currie, Greenstone, & Knittel, 2019). Patterns of residential mobility likewise shift in response to energy booms, reflecting both new economic opportunities and changing local conditions (Wilson, 2022).

At the same time, there is evidence of distributional and social costs. Increased labor demand can raise the opportunity cost of schooling for some workers, leading to declines in educational attainment among certain demographic groups, specifically low-skilled males (Cascio & Narayan, 2022). In terms of environmental justice, marginalized and lower-income communities are often disproportionately located near fracking sites, reinforcing patterns of environmental inequality (Proville et al., 2022; Zwickl, 2019). Smith and Wills (2018) further document that oil booms can benefit urban populations while leaving behind the rural poor. Additional documented risks include adverse health impacts, increased traffic and trucking activity, and elevated noise levels (Hill, 2018).

This paper contributes to this literature by introducing a long-run margin of economic cost which contrasts these temporary economic effects associated with active drilling.

2.2.4 Environmental Risks of Abandoned Wells

While the effects of active wells may be more perceptible to property owners, abandoned wells significantly outnumber producing wells in both number and risk. It is estimated there are over 80,000 documented orphan wells across the country with no solvent owner of record (Boutot, Peltz, McVay, & Kang, 2022). Recent work also uses data-driven methods including machine learning on satellite images to identify undocumented orphan wells, suggesting that the true inventory of abandoned wells is far larger than official records imply and that associated environmental risks may be substantially underestimated (Ciulla et al., 2024; Jahan et al., 2025; Kim, Kadeethum, Downs, Viswanathan, & O'Malley, 2024). Direct measurements show that many abandoned wells emit methane at levels that meaningfully contribute to greenhouse gas inventories (Kang et al., 2016; Townsend-Small, Ferrara, Lyon, Fries, & Lamb, 2016). Emissions from abandoned wells may even be underestimated by 150% in Canada and by 20% in the U.S (Williams, Regehr, & Kang, 2020).

In Pennsylvania, abandoned wells have been shown to depress land investment and hinder ecological recovery. Harleman, Weber, and Berkowitz (2022) find that land surrounding decades-old abandoned wells exhibits lower investment activity relative to areas with properly plugged wells. Restoration studies further demonstrate that abandoned wells can impede the recovery of ecosystem services following reclamation efforts (Nallur, McClung, & Moran, 2020).

Overall, the evidence shows that abandoned wells impose persistent environmental and human risks across multiple dimensions. My work complements this literature by translating these substantial risks into measurable economic losses.

2.3 Hedonic Pricing Model with an Environmental Good

Here I adopt the standard hedonic pricing model to include an environmental good. The hedonic pricing model assumes that the market transaction price P_i depends on a vector of its characteristics X_i, including structural features and neighborhood attributes. In the case where there is an evolving environmental component E_{it} with changing levels of disamenities, we can assume the price P_{it} equals the present value of an infinite stream of per-period housing utility flows:

$$P_{it} = f(X_i) + f(\sum_{\tau=0}^{\infty} \delta^\tau E_{it+\tau}) + \epsilon_{it}$$

where $0 < \delta < 1$ is the household's discount factor. Let home i be within the specified radius of J wells $W_{it} = (w_{1t}...w_{Jt})$, such that the total environmental component is equal to the sum of the disamenity of each well weighted by a distance function:

$$E_{it} = f(W_{it}) = \sum_{j=1}^{J} d(w_{ij})v(w_{jt})$$

At a given time, the well is defined by one of the mutually exclusive states $s \in S = \{a, u, p\}$ or active, unplugged, and plugged. Let $m(s)$ denote the per-period marginal disamenity from having the well in state s, such that $v(w_{jt} = s) = m(s)$. We now have:

$$P_{it} = f(X_i) + f(\sum_{j=1}^{J} \sum_{\tau=0}^{\infty} d(w_{ij})\delta^\tau v(w_{jt})) + \epsilon_{it}$$

Perfect foresight Suppose households have perfect information about the future path of the well's state $\{w_{jt+\tau}\}_{\tau \geq 0}$ and treat it as deterministic. For example, suppose it is known that an active well becomes abandoned after T_U periods, and plugged after T_P periods, after which it is plugged forever.

Then the present value of disamenities for well j is:

$$v(w_{jt} = a) = \sum_{k=0}^{T^U-1} \delta^k m(a) + \sum_{k=T^U}^{T^P-1} \delta^k m(u) + \sum_{k=T^P}^{\infty} \delta^k m(p).$$

These sums evaluate to:

$$v(w_{jt} = a) = \frac{1-\delta^{T^U}}{1-\delta} m(a) + \frac{\delta^{T^U} - \delta^{T^U+T^P}}{1-\delta} m(u) + \frac{\delta^{T^U+T^P}}{1-\delta} m(p).$$

Similarly, for an unplugged well with a deterministic plugging date:

$$v(w_{jt} = u) = \frac{1-\delta^{T^P}}{1-\delta} m(u) + \frac{\delta^{T^U+T^P}}{1-\delta} m(p).$$

Finally, for a plugged well:

$$v(w_{jt} = p) = \frac{1}{1-\delta} m(p).$$

Therefore, the valuation of a specific well just depends on the duration in each status, which in turn can be a function of age, operator, and county. A well need not complete its full life cycle if it is known, for example, that it will remain unplugged indefinitely. Therefore, given perfect foresight, the coefficients capture the discounted value of all future disamenities given the current state and the observed transaction price is then:

$$P_{it} = f(X_i) + f(W_{it}) + \epsilon_{it}$$

Zero foresight Since the future path of the well's life-cycle is highly uncertain in reality, homeowners instead must form expectations, given the current state at time t which can be included in the hedonic function:

$$P_{it} = f(X_i) + f(W_{it}) + f(E_{t+1}^{\infty}(W_{it})) + \epsilon_{it}$$

While location disclosure is mandatory, it can also be assumed homeowners lack well-specific information or knowledge that would help to form expectations over the life-cycle. Consequently, all wells of an observed status within the radius of interest are treated identically. Two types of homeowners can be considered: naive, who would assume the observed status persists forever, and optimistic, who would assume all wells will be plugged within a chosen timeframe. In either case, the expectations become deterministic, and the hedonic framework still fully captures the discounted stream of future amenities or disamenities:

$$P_{it} = f(X_i) + f(W_{it}) + \epsilon_{it}$$

In this case, the estimated coefficients capture the perceived long-run disamenity associated with the current state, which may or may not equal the true underlying dynamic hazard. Thus, the way expectations are formed matters for interpretation where the empirical estimates reflect the perceived discounted harm of an abandoned or unplugged well. Importantly however, it is these perceptions that are realized in market prices and therefore determine observed welfare impacts. The following analysis therefore reflects the degree to which buyers internalize environmental hazards. Systematic undercapitalization would suggest low salience, information frictions, or optimistic beliefs among buyers, whereas over-capitalization could indicate heightened perception of risk, stigma, or precautionary pricing.

The hedonic model assumes equal willingness to pay across individuals and stability in the hedonic gradient over time. Homeowners across time periods capitalize wells identically, there is no dynamic learning or uncertainty resolution. Under these assumptions, transaction prices represent the discounted value of all perceived present and future amenities and disamenities as outlined.

Taken together, this framework clarifies how the estimated hedonic coefficients should be interpreted in the empirical analysis when dealing with a dynamic environmental good. This is to contrast a temporary disamenity, such as noise from construction, which may have housing market impacts but only short-lived. Regardless of whether households possess perfect foresight or form expectations using simplified beliefs, observed transaction prices capitalize the discounted value of perceived future environmental disamenities associated with nearby wells, conditional on their current observable status. The coefficients recovered in the hedonic regressions should be understood as reduced form measures of the long-run disutility associated with active, abandoned, or plugged wells, reflecting how buyers internalize information, uncertainty, and stigma related to the well life cycle.

2.4 Data

2.4.1 Well Data

I compile data on oil and gas wells from several reports provided by the Pennsylvania Department of Environmental Protection (PADEP or DEP). The current well inventory identifies all wells known to the DEP. I link this inventory to historical production reports dating back to 1980 to identify the well status at each point in time. Oil and gas operators are required to report production details for each individual well they own. I also use the Plugged Wells Report to get the precise date of plugging when relevant.

Although there exists production evidence for wells not in the current DEP inventory, I choose to omit these in the interest of only addressing wells most likely known to a nearby homeowner. Undocumented wells would not be capitalized into home prices and I assume only wells known to the DEP would be subject to disclosure laws. However I do include legacy wells that are listed in the inventory but lack production histories for the same reason. These wells are either orphaned or plugged and I treat their status as fixed over my time period of interest. These wells contribute to the initial stock of abandoned and plugged wells. While it is estimated that there are hundreds of thousands of undocumented wells unknown to the DEP, I assume these are randomly scattered across the state or also unknown to homeowners and thus do not bias my estimates. Furthermore, in specifications using repeat sales, home fixed effects absorb any time-invariant characteristics.

Prior to 2010, these data are compiled annually, after which they are reported biannually until 2015, and then monthly until the present. These reports include complete information about the location, spud year, production volume, type (oil or gas), purpose (extraction, injection, storage), and well configuration (conventional vs. unconventional). Each well is identified using a unique permit number. I remove wells with inconsistent or missing latitude and longitude coordinates, as well as wells that were permitted but never spud. If the spud year is missing, I assign the minimum observed production year. Wells are found in 33 Pennsylvania counties as shown in 2.1. For consistency, I aggregate the post-2010 reports to the annual level by summing the production quantities and preserving time invariant characteristics. I maintain the DEP status which is the status declared by the operator for all baseline specifications. Since the majority of "active" wells are not producing large positive quantities in any given year, I also impute a well status based off of production quantity for alternative specifications. However, while there are certain disamenities associated with active production, temporarily inactive wells retain the capacity to produce with the associated surface equipment in place, so homeowners are unlikely to distinguish between short inactive periods and active production.

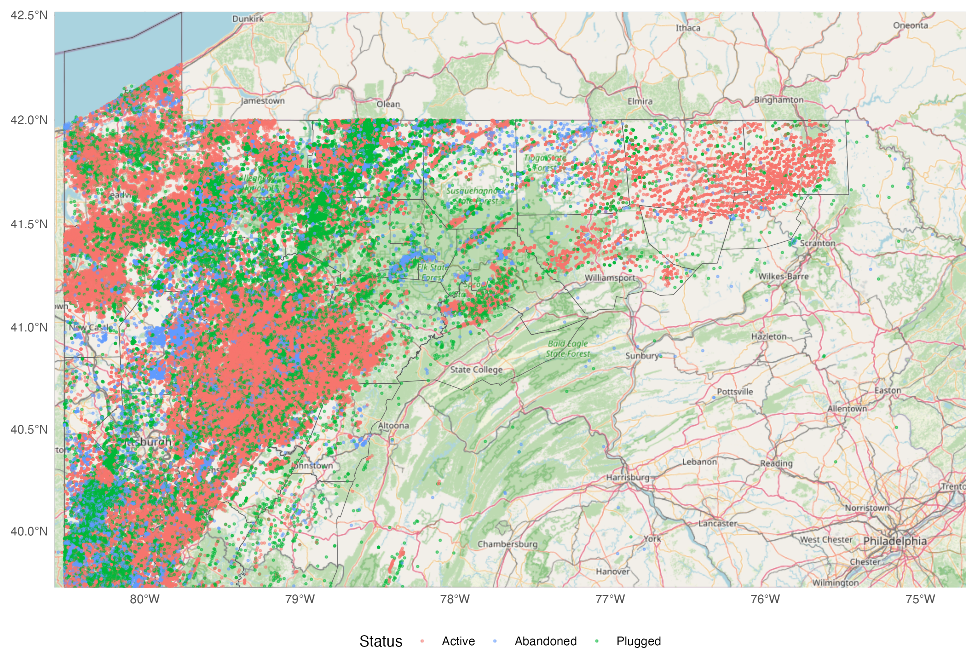

Figure 2.1: Oil and Gas Activity in Pennsylvania

This results in a well-level panel that tracks status transitions over four decades. Due to sparse observations for many wells in non-producing years, I impute the status in the following manner in order to balance the panel. Active wells become abandoned in the years following the last active observation until the plug year, or indefinitely if there is no plug year. If there is a non-missing reported plug year, the well is assigned a plugged status in all remaining years. Wells that temporarily stop reporting but later resume production are treated as continuously active, reflecting short-run operational interruptions rather than abandonment.

I do not impute the regulatory inactive status, as this designation must be formally declared by the operator and is used consistently by the DEP. Wells deemed regulatory inactive are not intended to sit idle for extended periods of time, however this is common in the data and for certain specifications I treat these wells as abandoned in accordance with terminology used by the DEP. An abandoned well is defined as "a well: (i) that has not been used to produce, extract or inject any gas, petroleum or other liquid within the preceding 12 months; (ii) for which equipment necessary for production, extraction or injection has been removed; or (iii) considered dry and not equipped for production within 60 days after drilling, redrilling or deepening".[2]

An orphan well is defined as a well abandoned prior to April 18, 1985. While all orphan wells are abandoned, not all abandoned wells are legally classified as orphaned. In practice, however, I do not distinguish between the two in most specifications, as both types of wells represent inactive sites with comparable implications for homeowners.[3] Finally, for wells with substantial gaps between the spud year and their first reported production, I classify them as temporarily inactive unless they never report positive production, in which case they are considered abandoned. I remove wells that were granted permits but not spud.

This leaves data on around 175, 000 wells. Figure 2.2 shows spudding over time by type. By resource, only 22% are oil compared to 66.7% gas and 7.1% combined. By type, only 6.2 % of wells are unconventional. All unconventional wells are gas, yet these account for the vast majority of production.

[2] 58 Pa. Cons. Stat. § 3203

[3] Throughout this chapter, the terms "abandoned", "inactive", and "unplugged" are used interchangeably to refer to wells that are no longer producing and have not been plugged. I only distinguish these from temporarily inactive wells when it is relevant for status classification.

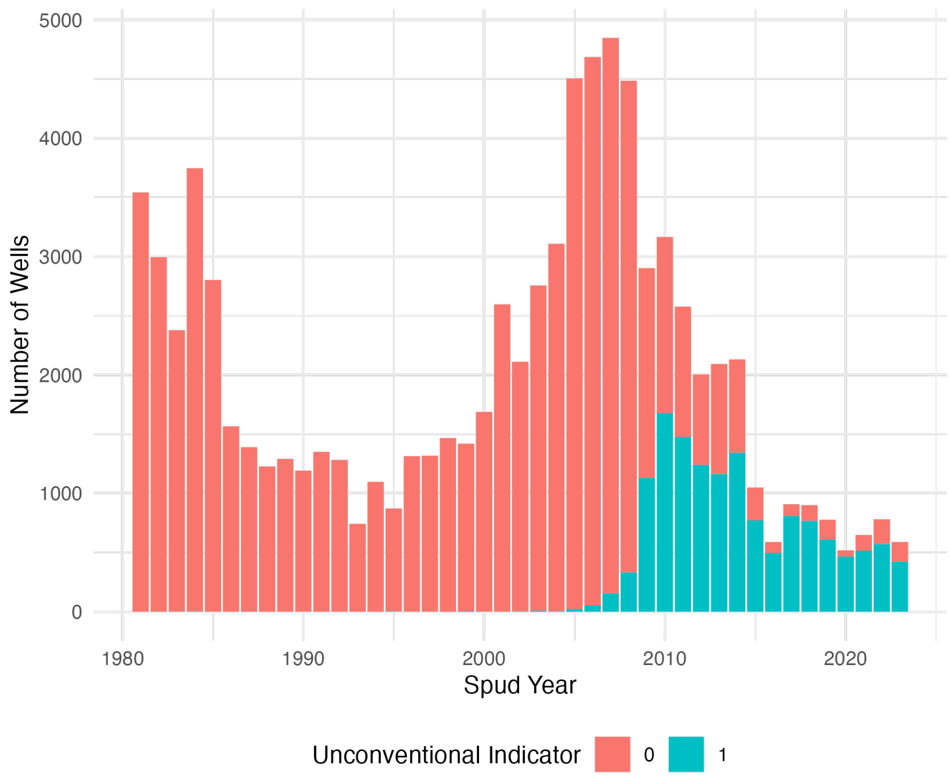

Figure 2.2: Wells Spud Over Time by Type

Figure 2.3 plots the number of wells at a given status each year using the balanced panel. This is a cumulative count where once a well becomes abandoned, it continues to contribute to the stock of abandoned wells in all subsequent years. The count of plugged wells increases over time showing that some abandoned and active wells are being plugged. Inactive refers to the DEP status ”Regulatory Inactive”, although at any given point a high percentage of active wells are also not producing positive quantities.

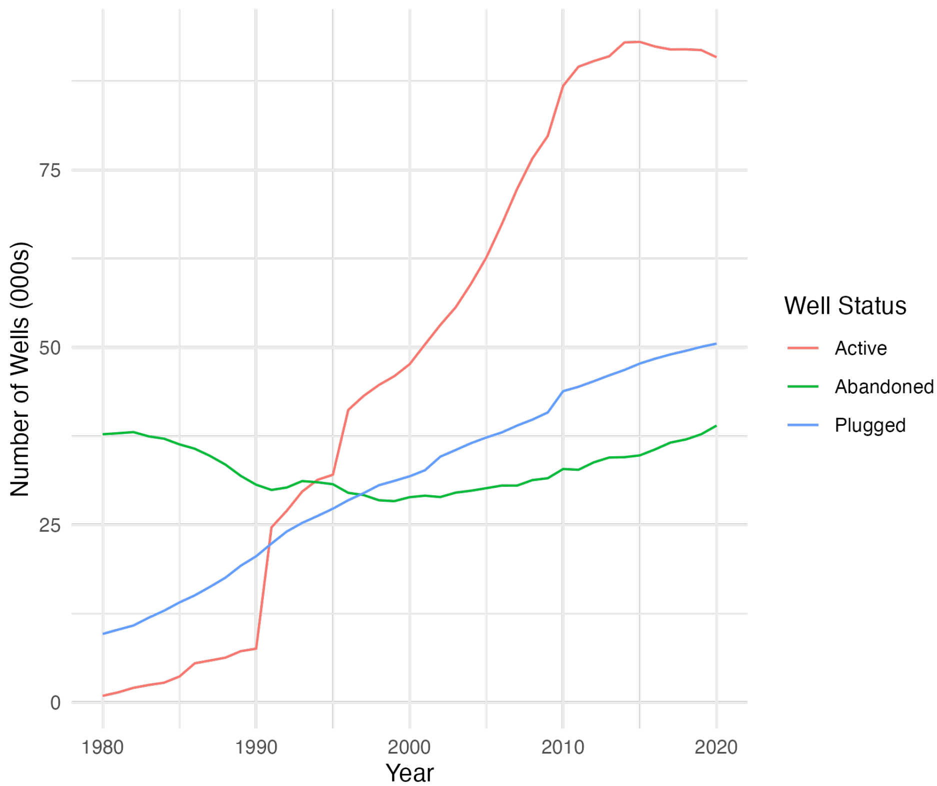

Figure 2.3: Well Status Over Time

Figure 2.4 displays the age-status distribution of wells. Although wells of every status appear at nearly every age, the composition shifts markedly as wells mature. Older wells are substantially more likely to be plugged or abandoned, while the share of active wells declines steadily with age. The distributions are cumulative in that a 35-year old well is also represented in all younger age bins, hence the figure illustrates the increasing probability that a well transitions out of active production as it ages.

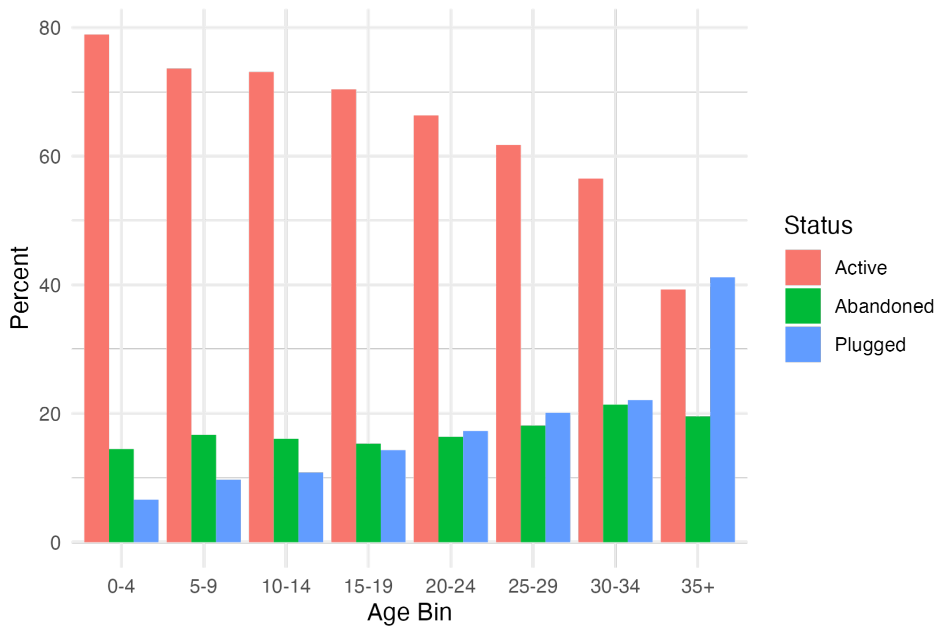

Figure 2.4: Age Distribution of Wells

2.4.2 Housing Data

Housing market data is obtained from Attom - a nationwide provider of proprietary property data. The data are the universe of housing transactions since 1980, although I limit the sample to sales from 1997 onwards when reporting becomes more reliable. These data include property characteristics, along with date of sale and location. I remove arms-length transactions with low transaction values, as well as homes built to order where the sale year is the same as the built year. I identify single family homes by limiting the number of transactions over the time period and deflate sale prices. I remove observations in which all structural characteristics are jointly zero and winsorize extreme values of building area, lot size, and bedroom and bathroom counts. I end up with 929,219 housing market transactions that occurred between 1995 and 2023 with 608,886 homes. The identifying variation comes from homes sold at different times and new wells or existing wells changing status. The same home sold in two different time periods is essentially treated as two separate observations subject to different oil and gas activity, unless the specification includes property fixed effects. Table 2.1 presents summary statistics.

Table 2.1: Home Sale Summary Stats

	Mean	SD	Min	Median	Max
Variables					
Price (K USD)	139	117	10	112	1220
Year Built	1948	34	1700	1950	2022
Lot Size (K sqft)	16	24	0.5	8.3	183
Building Size (K sqft)	1.6	0.67	0.5	1.5	4.9
Bedrooms	3	0.87	1	3	8
Bathrooms	1.9	0.91	0.25	2	6
Counts					
Transactions	738453				
Properties	472309				
Census Tracts	821				

Notes: Sample of homes sold between 1995 and 2023 in Pennsylvania counties that have oil and gas activity. Homes without complete characteristics have been removed and variables have been winsorized to be within reasonable bounds. Homes with more than 100 wells have also been removed.

2.5 Empirical Strategy

2.5.1 Matching

For each home, I create well counts of each status within a given distance bin. I begin with three status categories and four distance bins in 500 meter intervals up to 2000 meters. A home is considered treated if it has at least one well of any status within any bin, and homes with zero wells in all bins form the control group. Treatment status is determined by the proximity to wells, while treatment intensity is determined by the total number of wells within the radius. Although homes in the control group have no wells within the distance bins, they should still be located within the same local economy as treated homes so both groups are subject to the same changes in wages or employment, and local amenities, or "vicinity effects" (Muehlenbachs et al., 2015). For this reason, I limit the control group to homes in oil and gas counties, and no home in the sample is more than 20 km from the nearest well. In certain specifications, I adjust the treatment distance cutoff r and sum the counts in all distance bins to find the total wells within r meters.

Homes are matched to oil and gas wells using a fixed radius nearest-neighbour matching algorithm, identifying all wells within 2000 meters and assigning them to appropriate distance bins. As wells and homes are geocoded using latitude and longitude points, I use the Haversine formula to calculate distance after accounting for the Earth's curvature.

Figure 2.5 shows the concentric circles used to measure well exposure and the variation in well counts across two time for the same home. Between the two sale years new wells were drilled, particularly in the closest two distance bin, and wells were abandoned. This variation in well counts within fixed distance bands forms the core source of identifying variation in the empirical analysis. A property's exposure to nearby wells changes through new drilling activity or transitions in well status. The aim is to control for all confounders so any remaining price differences across transactions are attributed to changes in nearby well exposure. Note this figure depicts repeat sales of the same property for intuition while the empirical specifications primarily use cross-sectional variation across different homes sold in the same year.

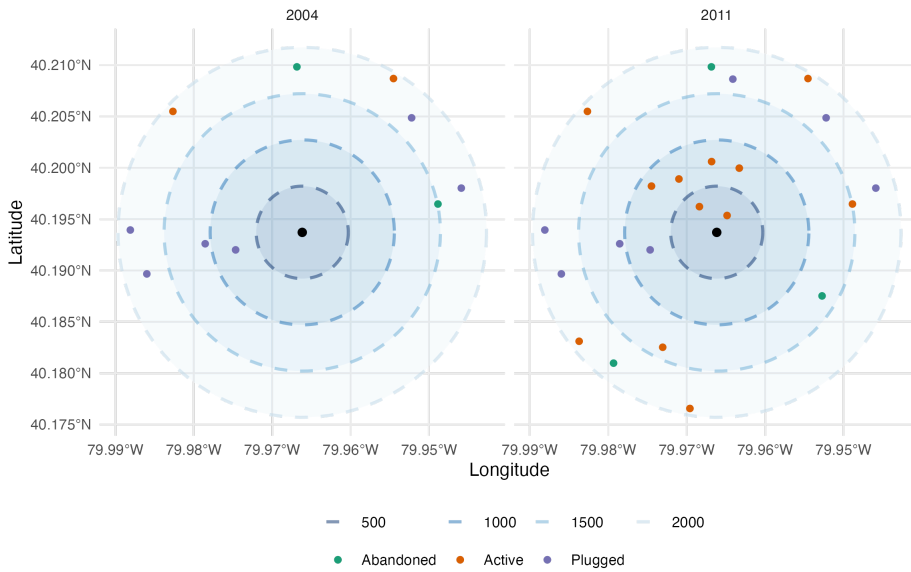

Figure 2.5: Example of Wells Matched to Home i

I begin by matching the full set of known wells to all residential property transactions occurring between 1997 and 2023. I then restrict the sample to only those wells that were already spudded by the year of each home sale. For each matched well, I require its status in the year of sale which comes from merging with the imputed balanced panel of production histories. Accordingly, wells are active as designated by the DEP, plugged when the recorded plug date falls before the sale year, or as abandoned or idle when they do not report production in the sale year and remain unplugged. I further reduce the sample to homes with less than 100 wells nearby to limit the influence of extreme outliers in well density.

At this stage, the status variable includes several forms of inactivity, including idle wells that later resume production, idle wells that never produce again, long-term abandoned wells, regulatory inactive wells, and official orphaned wells. Although these categories differ administratively and reflect distinct regulatory processes, they have broadly similar implications for nearby residents. I therefore aggregate all unplugged non-active statuses into a unified "abandoned/unplugged" category. However I can separate legacy unplugged wells from those that retain some operational capacity as required to assess whether these distinctions matter for the estimated effects.

Summary statistics of matched wells to homes are presented in Table 2.2. Figure 2.6 provides a visual depiction of these patterns over time by plotting the average number of wells of each status within each distance bin for homes sold in a given year.

Table 2.2: Matching Wells to Homes Summary Stats

		Share of Homes	Number of Wells if Treated			
Status	Distance Bin	Perc. Treated	Mean	SD	Median	Max
Active	1	0.12	1.95	1.42	1	19
	2	0.23	3.54	3.47	2	31
	3	0.31	4.98	5.35	3	49
	4	0.36	6.34	7.35	4	62
Abandoned	1	0.18	2.00	1.58	1	56
	2	0.35	3.07	3.13	2	48
	3	0.45	3.92	4.34	2	50
	4	0.53	4.74	5.51	3	68
Plugged	1	0.16	1.68	1.41	1	30
	2	0.33	2.42	2.22	2	42
	3	0.43	2.99	3.24	2	45
	4	0.52	3.55	4.56	2	84

Notes: Inactive wells include legacy wells, temporarily inactive, and abandoned wells.

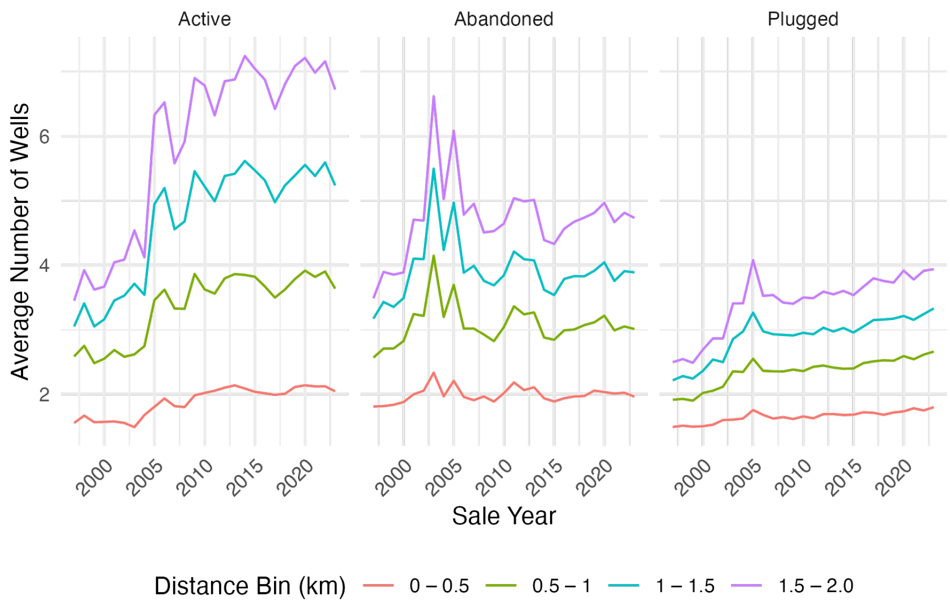

Figure 2.6: Number of Wells by Status

2.5.2 Identification

Within our matched sample of homes and wells, I follow Muehlenbachs et al. (2015) in considering several pairwise comparisons. Empirically, essentially all treated homes are located near multiple wells, yet the approach can be illustrated with simply comparing a treated home with a single nearby well versus a control home with no wells within proximity. Each such comparison reflects a composite of potential effects. Active wells may provide financial benefits through royalty payments to landowners (+), but also introduce local disamenities and environmental risks (–). When production ceases, these financial benefits stop, and some of the active phase externalities may diminish. However, the well can continue to pose environmental hazards, especially if left unplugged. Unplugged wells can degrade local aesthetics and, when visible, impose further disamenity costs. The comparison between a home near an unplugged well and a control home with no wells thus captures the long-run externalities of oil and gas extraction in the absence of proper plugging practices. Without intervention, unplugged wells remain features of the landscape indefinitely.

Wells in the post-production phase can either be left unaddressed in which case they become officially abandoned, or can be plugged. The longer a well remains unplugged, the lower the probability it will be remediated, especially if ownership becomes unclear or the responsible operator becomes insolvent. Whether homeowners incorporate well age or regulatory status into their assessments remains an open empirical question. A properly plugged well should eliminate environmental risks if all technical and regulatory requirements are satisfied. In this case, any residual impact on housing prices would reflect public perception rather than actual hazard. Therefore the comparison between a home near a plugged well and a control home captures whether homes homes with and without a history of extraction are valued similarly. Comparing these groups provides insight into whether market participants treat remediated well sites as fully restored.

Treated Home	Control Home	Comparison Captures
Near active producing well	No wells nearby	Net effect of production: royalties (+), disamenities (–)
Near inactive, unplugged well	No wells nearby	Long-run externalities without remediation
Near plugged, inactive well	No wells nearby	Residual effect of fully remediated site (perceived or real)
Near unplugged, inactive well	Near plugged, inactive well	Incremental effect of plugging (policy impact)
Near older legacy well	Near newer unplugged well	Effect of duration of abandonment

2.5.3 Baseline Specification

The following specification identifies relationship between residential property values and exposure to oil and gas wells, differentiated by status (Active, Abandoned, Plugged) and d distance bin: [0–500), [500–1000), [1000–1500), [1500–2000).

$$ln(price_{itg}) = \sum_{d=1}^{4} \alpha_d A_{ditg} + \sum_{d=1}^{4} \alpha_d U_{ditg} + \sum_{d=1}^{4} \alpha_d P_{ditg} + \gamma_t + \lambda_g + \mathbf{X}_i + \epsilon_{itg} \tag{2.1}$$

I consider a home i sold in year t of geography g. A range of fixed effects specifications to flexibly control for both spatial and temporal confounders. These can be separate or interacted. Year of sale fixed effects absorb aggregate shocks to the housing market, reflecting macroeconomic trends. I can interact year with county to allow for counties to have different trends over time, due to local demand or labour market polcies for example. Month of sale fixed effects capture seasonality in housing prices of demand over the year. Then I introduce some level of geographic fixed effect, either census tract or block. These effects account for time-invariant spatial characteristics, and I am dealing with within-tract variation in well exposure over time. More saturated specifications, such as tract-by-year plus month fixed effects, allow for highly granular control over localized trends in housing prices. I control for characteristics $\mathbf{X}_i$ in all specifications besides those with property fixed effects. By controlling for observable characteristics, any remaining variation in the housing price can be attributed to unobservables and the surrounding oil and gas activity. The unobservables are accounted for by comparing houses that are otherwise identical across everything except the proximity to wells, achieved through the inclusion of these fixed effects. Characteristics include year built, land square footage, building square footage, number of bedrooms and number of bathrooms.

2.5.4 Restricting Plugged Well Exposure Window

The number of plugged wells near a home can only increase over time by construction, as additional active or abandoned wells are sealed and add to the total. Plugged wells therefore cover an increasing fraction of the land over time. Consequently, once a well is plugged, its potential effect on property values persists indefinitely. However, it is plausible that older plugged wells become less relevant as time passes. Plugged wells may be most salient in the years immediately following remediation, while older wells are less likely to influence buyer perceptions. Since the key question is whether lost property value is possible to recover through land reclamation, this is may be most appropriately captured by focusing on recently plugged wells.

First, for each home i sold in year t, I take all plugged wells within 2 km and classify them according to the number of years since plugging. Instead of assuming each plugged well has the same marginal effect, I allow wells to differ by duration of years since plugging. Specifically, I construct counts for duration bins for well s: $d_{st} = (0-4], (5-14], (15-24], (25-50], (50, \infty)$. In the case that exact year of plugging is missing, I take the first year the well is reported plugged in the production reports. If the well came from the inventory with no production reports and is plugged, I use the spud year. In the case of a spud year of 1800, I automatically assume the highest duration bin. I show the distribution of these implied plugged years in Figure 2.7 and the distribution of plugging duration in Figure 2.8. Figure 2.8 excludes those wells with a spud year of 1800 to avoid plugging durations of over 200 years. I replace the total plugged count with these variables in my baseline specification. Finally, I rerun my main specification while restricting the count of plugged wells to only those plugged within the last ten years. I removes almost 7 million matches including over 40,000 wells.

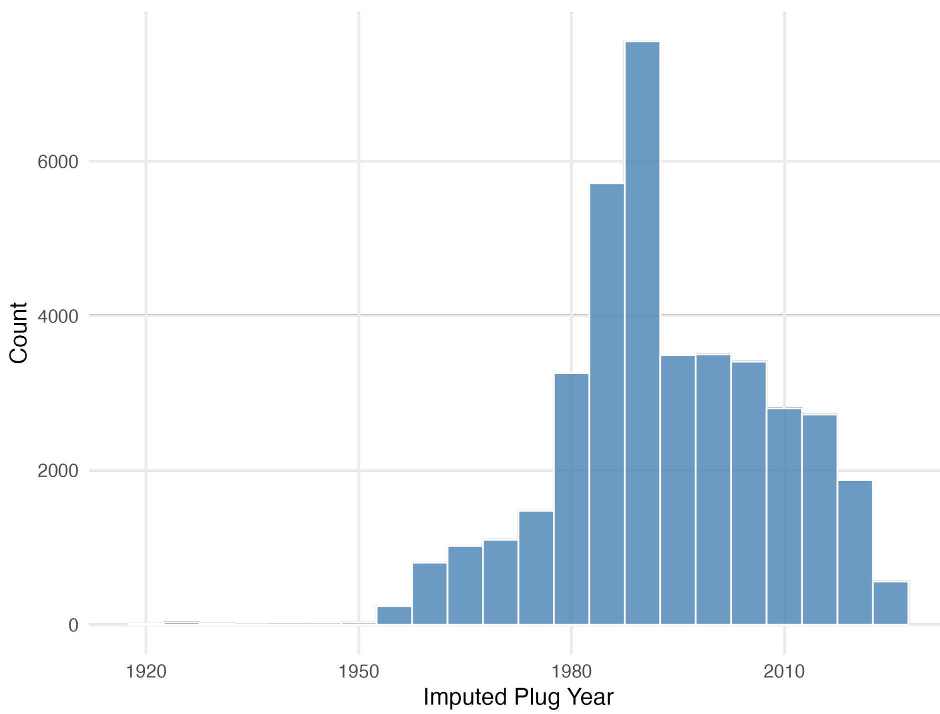

Figure 2.7: Distribution of Imputed Well Plugging Years

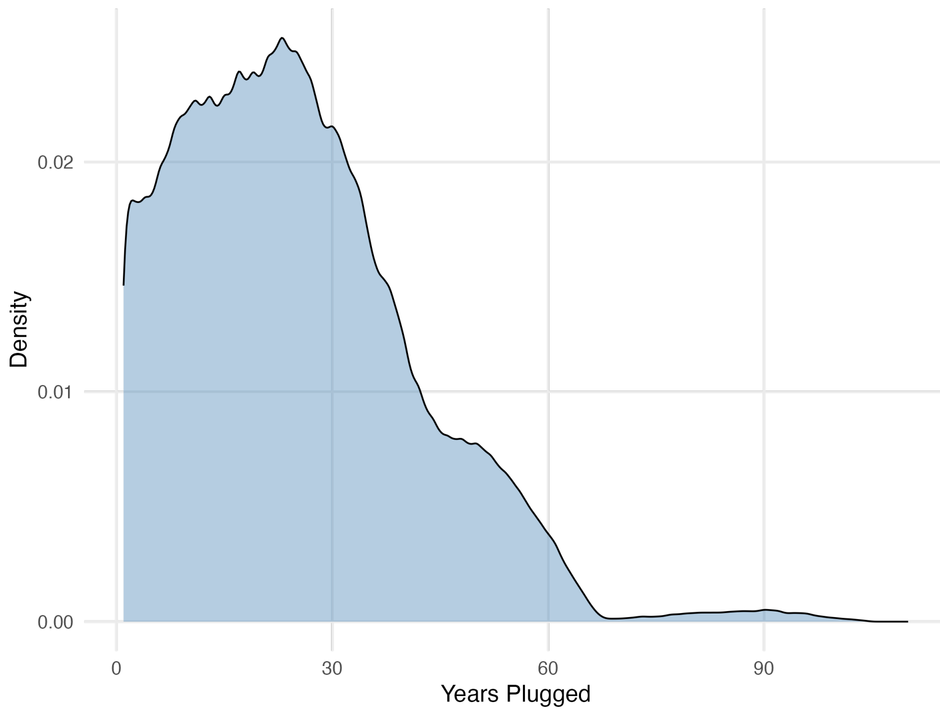

Figure 2.8: Distribution of Duration of Years Since Well Plugging

2.6 Results

In the following results tables, the sample is restricted to homes in oil and gas counties with no more than 100 wells nearby. The standard errors are clustered at the county x year level, allowing for arbitrary correlation in errors across properties within the same county-year. Columns vary in the fixed effects included in order to account for different sources of unobserved heterogeneity. The first two columns are at the tract level, and the last two columns are at the block level. The temporal fixed effect is either sale year, or I allow for a county specific time trend with sale year x county.

Table A.2 presents results from the aggregate Equation 2.1. The distance cutoff is 2km in Panel A. and 1km in Panel B. These OLS results show that an additional unplugged well is associated with a 0.1 - 0.3% decline in house prices. The effect is small yet statistically significant. For an average treated home with 12.8 unplugged wells nearby at an average home price of $142,614, the loss is around $3650. These results are consistent with the visible disamenity of unplugged wells on the landscape.

Table 2.3: Regression Results: Total Unplugged Wells within 2km

	log(Price)			
	(1)	(2)	(3)	(4)
Total Unplugged	-0.001***	-0.001***	-0.003***	-0.002***
	(0.000)	(0.000)	(0.001)	(0.001)
Characteristics	Yes	Yes	Yes	Yes
Sale Year	Yes	No	Yes	No
Sale Month	Yes	Yes	Yes	Yes
Tract	Yes	Yes	No	No
County x Sale Year	No	Yes	No	Yes
Block	No	No	Yes	Yes
Observations	719,014	719,014	719,014	719,014
R^2	0.62	0.62	0.70	0.70

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ Homes with no more than 100 wells nearby. Standard errors are clustered at the county year level. Active wells refer to wells deemed active by the Department of Environmental Protection (DEP). Abandoned wells include legacy wells and wells that have not been producing for at least 12 months. Plugged wells must have a valid date of plugging prior to the sale year. The distance cutoff is measured in meters.

Table 2.4 presents the results from my main specification, and breaks down the aggregate effect into well status and distance bin. These coefficients represent the percentage change in house price associated with the presence of one additional well of the given status within the distance bin. The comparison is to that of a home with no wells. For active wells, I find relatively little evidence that active wells affect property values within 1.5 km. This could represent a net average effect with some homeowners receiving lease or royalty payments, or with those outside public water service areas bearing the environmental risks and disamenities while those with piped water properties gain, consistent with Muehlenbachs et al. (2015). I find the negative impact of active wells is concentrated in the 1.5-2 km distance bin. For an average treated home with 6.82 number of wells in that distance range, the price decline would be between 0.2 - 0.6% per well. For the average treated home priced at $141,330, the loss would be between $1792 to $5376.

Abandoned wells show consistently negative effects, particularly in Column (3). An additional abandoned well within 500 meters of the property reduces home prices by 0.7% and here the effect of distance is as expected, with persistent yet diminishing effects out to 2 km. The attenuation of these effects in Column (4), which uses county-year fixed effects, suggests that some of the price penalty was driven by county-level shocks correlated with both abandoned well presence and lower housing prices. At the upper bound, a average treated home with 2 inactive wells in the closest bin, the price decline would be around $1976.

Plugged wells are also associated with significant negative effects on property vales, suggesting that remediation does not fully restore market value. This may reflect lingering stigma, incomplete land reclamation, or uncertainty about the effectiveness of plugging. Notably, the effects of plugged wells are larger yet still diminish with distance. For an average treated home with 1.68 plugged wells in the closest distance bin, the price decline could be from $1662 up to $4986.

In practice, homes will be surrounded by a vector of wells within each status distance bin. For example, a treated home with the average number of abandoned wells in each treated bin these estimates imply a price reduction of about 5.3%, corresponding to a loss of around $7621. For an upper bound, if a home had the average number of wells of each status within each distance bin, the loss could be up to 25%, or around $30,00.

I also estimate the same specification using the production-based well status classification rather than the DEP administrative status. Those results are reported in Appendix **??** and are qualitatively similar.

Table 2.4: Regression Results: Binned Linear OLS

	log(Price)			
	(1)	(2)	(3)	(4)
Active Wells				
≤ 500	0.001	-0.001	0.001	0.000
	(0.003)	(0.003)	(0.002)	(0.002)
500–1000	-0.001	-0.002	-0.001	-0.002**
	(0.002)	(0.002)	(0.001)	(0.001)
1000–1500	0.001	0.000	-0.001	-0.001
	(0.001)	(0.001)	(0.001)	(0.001)
1500–2000	-0.003***	-0.003***	-0.006***	-0.006***
	(0.001)	(0.001)	(0.001)	(0.001)
Abandoned Wells				
≤ 500	0.001	0.003	-0.007***	-0.002
	(0.003)	(0.003)	(0.002)	(0.002)
500–1000	0.001	0.003*	-0.003***	0.001
	(0.001)	(0.001)	(0.001)	(0.001)
1000–1500	-0.002	-0.001	-0.003***	0.000
	(0.001)	(0.001)	(0.001)	(0.001)
1500–2000	-0.001*	-0.001	-0.004***	-0.002***
	(0.001)	(0.001)	(0.001)	(0.001)
Plugged Wells				
≤ 500	-0.009**	-0.007**	-0.021***	-0.011***
	(0.004)	(0.004)	(0.004)	(0.004)
500–1000	-0.003	-0.001	-0.017***	-0.008***
	(0.002)	(0.002)	(0.002)	(0.002)
1000–1500	-0.002	0.000	-0.013***	-0.005***
	(0.001)	(0.001)	(0.001)	(0.001)
1500–2000	-0.003**	-0.002	-0.010***	-0.004***
	(0.001)	(0.001)	(0.002)	(0.001)
Characteristics	Yes	Yes	Yes	Yes
Sale Year	Yes	No	Yes	No
Sale Month	Yes	Yes	Yes	Yes
Tract	Yes	Yes	No	No
County x Sale Year	No	Yes	No	Yes
Block	No	No	Yes	Yes
Observations	719,014	719,014	719,014	719,014

Notes:* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ Homes with no more than 100 wells nearby. Standard errors are clustered at the county year level. Each panel corresponds to a different well status. The distance cutoff is measured in meters.

Table 2.5 shows the respective coefficients when reducing the sample to repeat sales and including home fixed effects. These estimates come from within-home variation, and isolate the effect of changes in nearby wells on the same home's price, removing confounding from unchanging home or neighborhood features that are potentially not fully captured by the fixed effects. Restricting the sample to repeat sales and including home fixed effects attenuates the estimated coefficients toward zero. Diminishing marginal effects of additional wells contribute to attenuation: in the cross-sectional specification, the coefficient reflects an average across homes with few and many wells, whereas the within-home specification mostly captures small incremental changes for the same property, which tend to have smaller marginal effects.

Measurement error in both well and property locations may also attenuate estimated effects. For example, historical well coordinates are sometimes imprecise, and parcel centroids do not always reflect the actual position of the home. Such noise in exposure measures makes it harder to detect the true effect of wells, biasing estimates towards zero. Measurement error in well counts could also be a factor that is magnified when using only within-home changes yet still affecting baseline estimates.

In these specifications I treat all wells of a given status as identical, regardless of age. This could be important for plugged wells, where by construction the number of plugged wells can only increase. I explore this concept in Section 2.5.4, where I only allow plugged wells to affect a home for a given number of years. I also explore the assumption that all wells within the first 100 have the same effect.

Table 2.5: Regression Results: Repeat Sales OLS

	log(Price)		
	(1)	(2)	(3)
Active Wells			
$\leq$ 500	-0.002	-0.001	-0.001
	(0.004)	(0.004)	(0.004)
500–1000	-0.004**	-0.005**	-0.002
	(0.002)	(0.002)	(0.002)
1000–1500	-0.006***	-0.005***	0.002
	(0.001)	(0.001)	(0.002)
1500–2000	-0.011***	-0.009***	-0.001
	(0.001)	(0.002)	(0.001)
Abandoned Wells			
$\leq$ 500	-0.002	0.005	0.001
	(0.004)	(0.004)	(0.005)
500–1000	-0.004	-0.001	0.000
	(0.002)	(0.002)	(0.003)
1000–1500	-0.008***	-0.004**	0.001
	(0.002)	(0.002)	(0.002)
1500–2000	-0.012***	-0.007***	0.000
	(0.001)	(0.002)	(0.001)
Plugged Wells			
$\leq$ 500	-0.028***	-0.014*	-0.007
	(0.007)	(0.008)	(0.009)
500–1000	-0.025***	-0.014***	-0.010**
	(0.004)	(0.004)	(0.004)
1000–1500	-0.020***	-0.009***	-0.002
	(0.003)	(0.003)	(0.003)
1500–2000	-0.030***	-0.016***	-0.003
	(0.003)	(0.004)	(0.003)
Property	Yes	Yes	Yes
Sale Year	Yes	No	No
Sale Month	Yes	Yes	Yes
County x Sale Year	No	Yes	No
Tract x Sale Year	No	No	Yes
Observations	438,467	438,467	438,467
R^2	0.83	0.83	0.86

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ Homes with no more than 100 wells nearby. Standard errors are clustered at the county year level. Each panel corresponds to a different well status. The distance cutoff is measured in meters.

Next, I examine how the effect of a plugged well varies with the duration since plugging. First, shows the coefficients on the duration bins when looking at the count of all plugged wells within 2 km. Interestingly, a small yet significant negative effect persists for at least 25 years. This could reflect the trust in more recent plugging operations with a perception of risk with older plugged wells. Very old plugged wells may no longer impact land values if it either takes a very long time to reclaim the land to where it is indistinguishable from baseline land quality, or if there is no longer salience. I present the coefficients on the distance bins of plugged wells from the baseline regression in Figure 2.10 after removing wells that were plugged more than 10 years ago and as expected there is no persistent negative effect. This also means that plugging duration is not correlated with distance to the property.

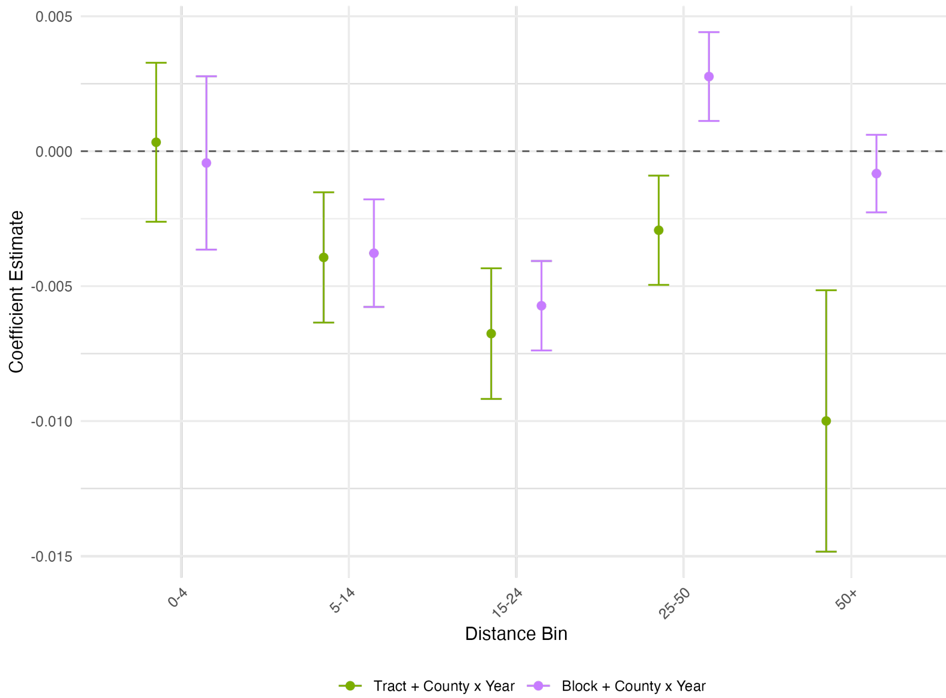

Figure 2.9: OLS Results on Duration of Years Plugged

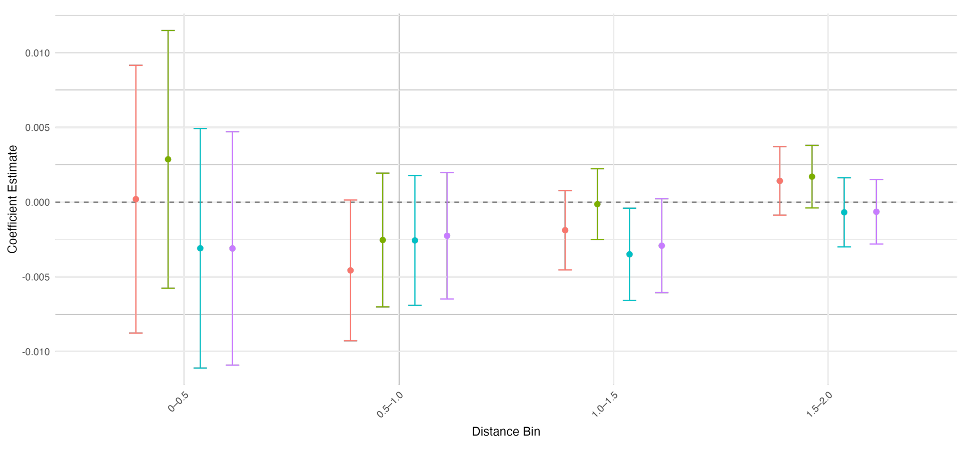

Figure 2.10: Baseline Results After Restricting Plugged Well Exposure

2.7 Subgroup Analysis

To better understand which markets and households drive the estimated capitalization effects, I next explore heterogeneity across subsamples. I re-estimate the model within each subsample using the same geographic and temporal fixed effects and controls. Exploring variation across housing market segments and locations helps clarify which groups drive the baseline results and whether the effects are concentrated in particular contexts. If the effect is concentrated within a particular subsample, pooling all observations can attenuate the estimated coefficient toward zero. All estimates are benchmarked against the baseline specification. Larger confidence intervals in certain distance bins for a given subsample primarily reflects limited sample size and reduced variation in exposure within those cells, which mechanically lowers statistical precision.

Rural vs. Urban Differences in housing density, land use, and local economic structure suggest that oil and gas wells may be valued differently in rural and urban housing markets. In rural areas, land uses are often closely tied to resource extraction, so residents may have stronger economic ties to industry activity, potentially offsetting disamenity effects. At the same time, housing density is lower and wells may be more visually prominent across the landscape. Wells also occupy land that could otherwise be used for alternative investment or development. In contrast, urban areas have a higher housing density and more competitive housing markets, which may capitalize environmental risks more strongly. Rural and urban populations also reflect different demographics, which may further shape risk perceptions and willingness to pay.

I use the Census classification of rural and urban at the block level, defined based on housing units and population. Figure 2.11 shows the 70,741 rural blocks, and 647,932 urban blocks.

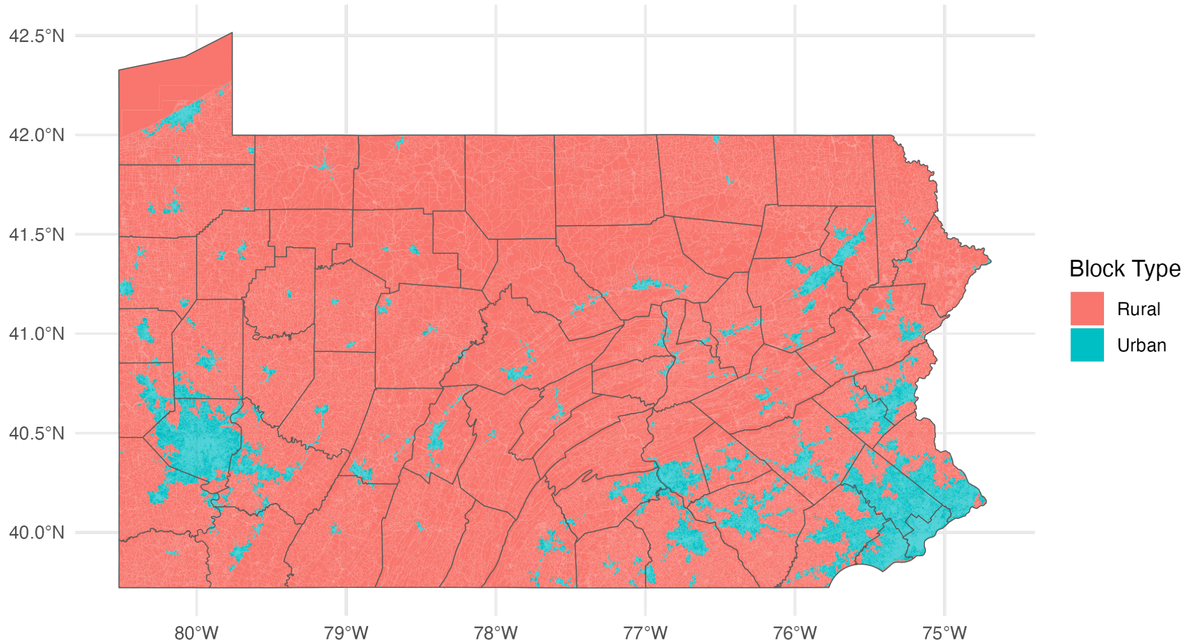

Figure 2.11: Rural and Urban Census Blocks

Figure 2.12 splits the estimates by rural and urban locations. The results indicate little systematic difference between rural and urban areas for active wells. Abandoned wells exhibit more pronounced negative effects in urban areas, particularly at close distances. Similarly, urban areas drive the negative effects for plugged wells, particularly at short distances, which attenuate with distance. This pattern is consistent with land reclamation being most valuable at close distances, yet it also suggests the persistence of perceived risk. Overall, these patterns suggest that the capitalization of legacy well impacts is driven by urban areas, although this partly reflects the greater representation of urban properties in the housing transactions data.

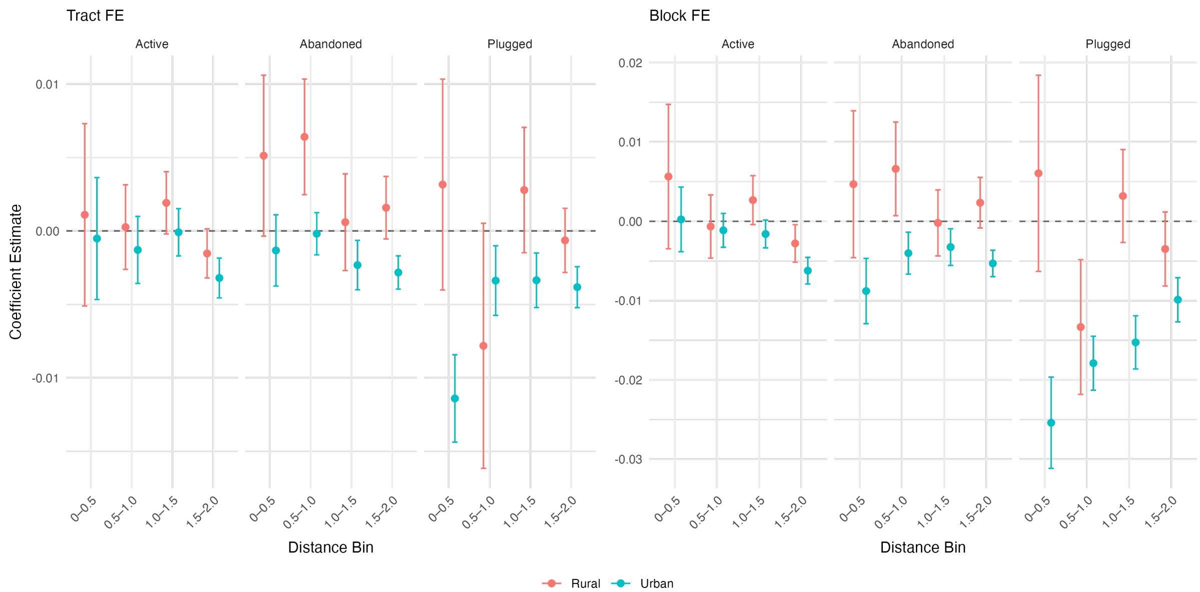

Figure 2.12: OLS Estimates with Rural/Urban Sample Split

Top vs. Bottom Price Quartile Homes Next, I take the top and bottom quartiles of the transaction price distribution. Capitalization effects may differ across segments of the housing market that attract buyers with different wealth levels and preferences. The effect of a well near a higher-value home may be larger if buyers have a greater willingness to pay for risk avoidance and better outside options to substitute into other neighbourhoods. In addition, higher-value properties often embody greater land and amenity value, making environmental disamenities more salient in transaction prices. Differences may also reflect heterogeneity in baseline exposure, if higher-priced homes tend to be located in areas with fewer nearby wells.

The results shown in Figure 2.13 demonstrate consistency across these subsamples. Active and abandoned wells may exhibit slightly more negative effects for higher priced homes, which could suggest greater sensitivity to disamenities; however, the difference is negligible and there is increased noise.

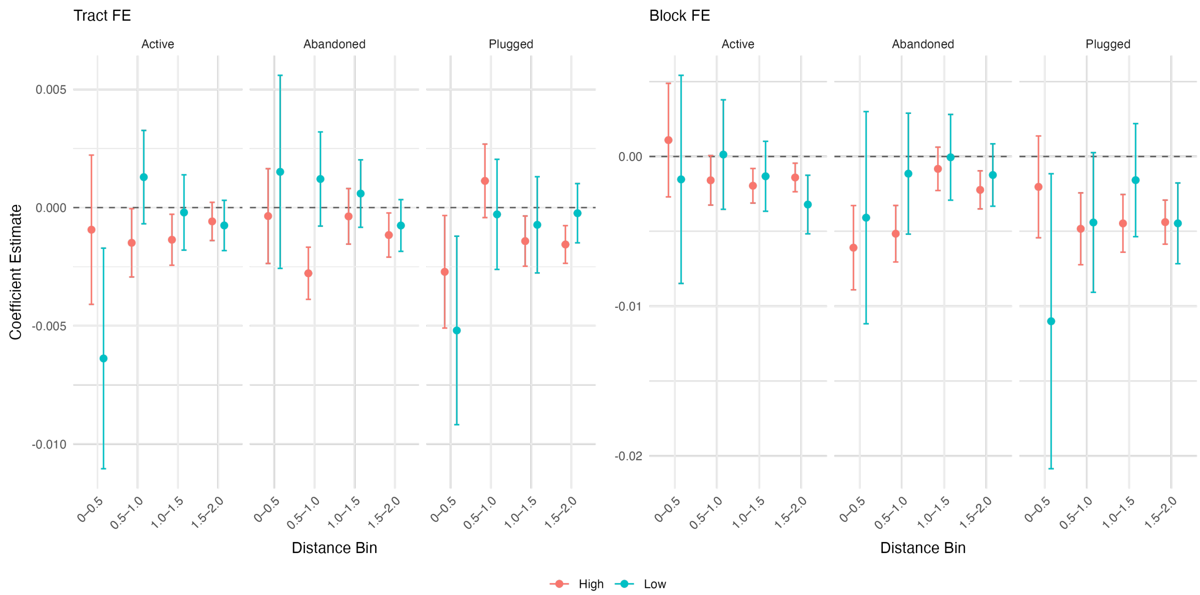

Figure 2.13: OLS Estimates with Top/Bottom Price Quartile Sample Split

Pre/Post 2012 To capture potential policy and industry driven changes, I also split the sample into pre and post-2012 periods. The post-2012 period coincides with changes in regulatory scrutiny, reporting requirements, and plugging activity, which may alter expectations about remediation and long term risk. While these changes may have increased awareness of legacy well risks, tighter regulations may also reduce perceived exposure if households expect that prevention efforts and future remediation will improve conditions.

Figure 2.14 reports the results. Overall, there is little difference across subsamples, suggesting that capitalization of nearby wells is relatively stable over time. Homes sold before and after 2012 are exposed to largely the same legacy wells drilled prior to regulations, so regulatory changes may have limited scope to alter how these risks are priced.

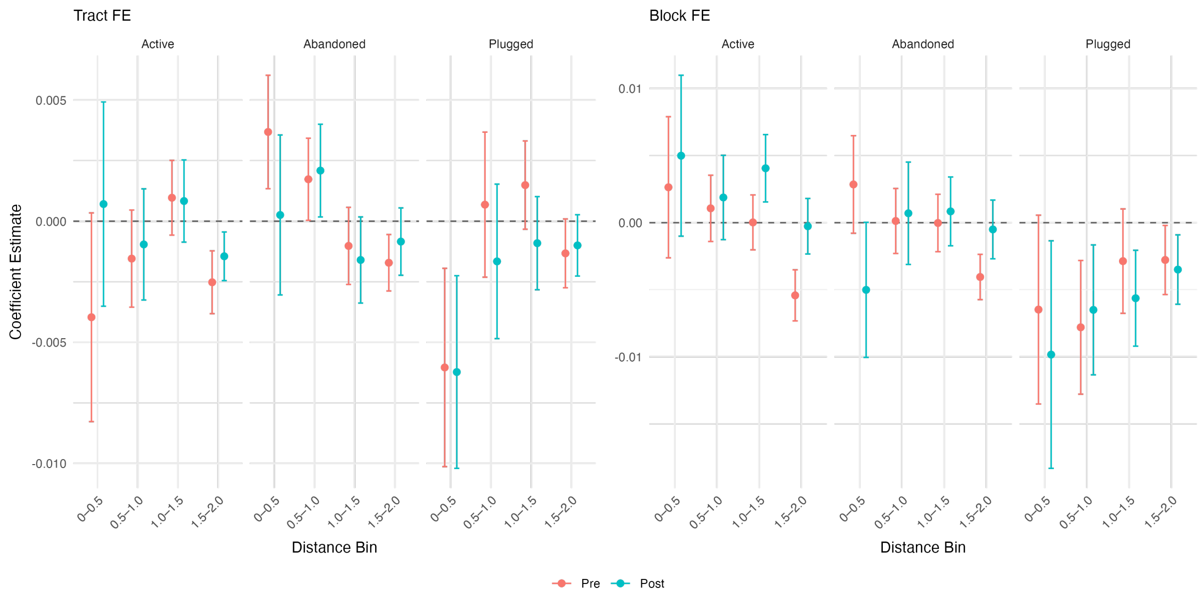

Figure 2.14: OLS Estimates for Pre/Post 2012 Sample Split

These subgroup dimensions capture distinct economic margins along which capitalization may vary, including spatial structure, background exposure, buyer flexibility, information and regulatory salience. Taken together, these subgroup results suggest that the main capitalization patterns are broadly stable across market segments, while highlighting where effects are most pronounced. I next turn to robustness and alternative identification checks.

2.8 Robustness

To assess the sensitivity of my main results, I conduct a series of robustness checks that address considerations related to measurement choices, functional form assumptions, and sample construction.

2.8.1 Alternative Distance Bins

To test whether my price effects are sensitive to arbitrary cutoff choices in the distance bin construction, I estimate my main specification with alternative distance bin definitions as depicted in Figure 2.15. As the perceived disamenity of wells is likely strongest at very short distances, I first allow for more granularity close to a property with finer distance bins. I split the first bin into 100-meter increments while preserving the last three: $(0 - -100], (100 - -200] \ldots (1500 - -2000]$. This specification tests whether the baseline results mask meaningful variation in effects due to coarse spatial aggregation and determines whether the estimated effects decline smoothly with distance or whether there are discrete thresholds at very short ranges. However, this finer partitioning comes at the cost of reduced precision and noisier estimates, since well exposure is spread more thinly across narrower distance ranges. I also consider a "shifted" binning scheme designed to test whether results are mechanically driven by the choice of cutoffs by redefining the bins as $(0 - -250], (250 - -750], (750 - -1250], (1250 - -1750], (1750 - -2000]$.

The results are shown in Figure 2.16 and Figure 2.17 for the finer and shifted distance bins respectively. For the finer bins, despite wide confidence intervals, the effects decline smoothly with distance. There is no evidence of strong localized discontinuities within the 0.5 km threshold. For the shifted bins, the qualitative patterns remain intact, indicating that the results are not sensitive to the precise placement of distance cutoffs.

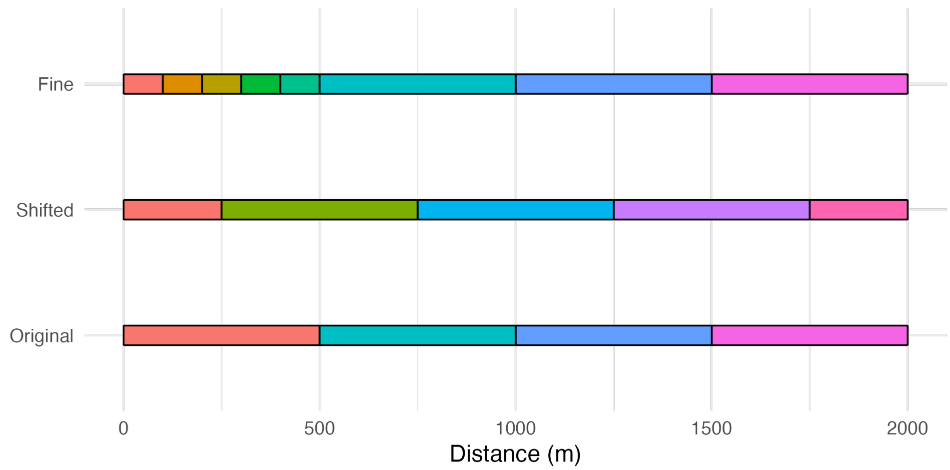

Figure 2.15: Comparison of Alternative Distance Bin Definitions

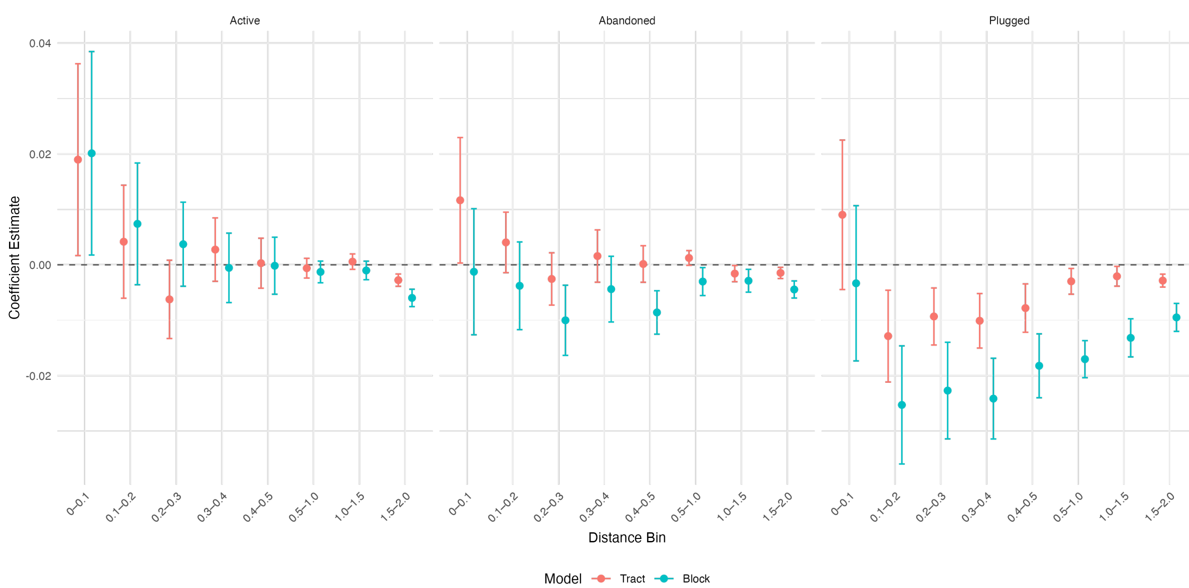

Figure 2.16: OLS Estimates with Finer Distance Bins

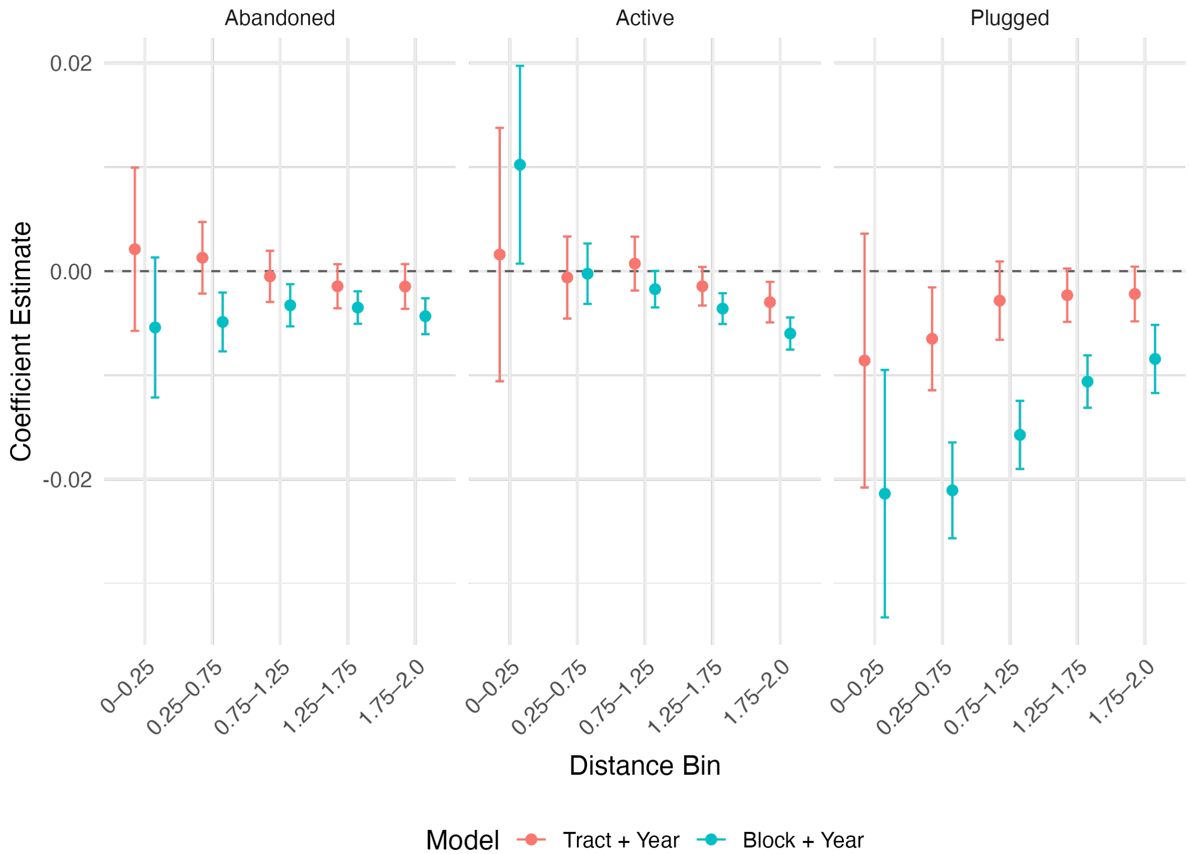

Figure 2.17: OLS Estimates with Shifted Distance Bins

The finer distance bin specification shows a potential positive effect for wells located within 100 meters of a property. One interpretation is that at extremely short distances, proximity captures not only disamenities but also private financial benefits, such as lease payments or royalties, that accrue directly to the homeowner. While I do not have data on lease payments, I can examine the effect of wells located directly on property parcels. Irregular parcel shapes can result in wells located farther away still being situated on the property, while closer wells may lie just beyond the parcel boundary, as demonstrated in Figure 2.18. I spatially join homes and wells to parcel shapefiles for the counties where they are publicly available. However, the resulting overlap is small, with fewer than 900 homes located on parcels containing a well.

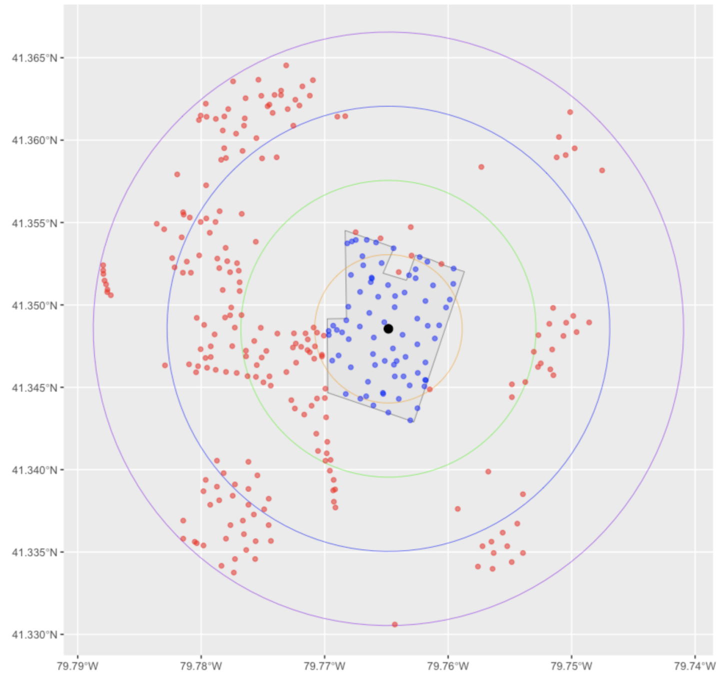

Figure 2.18: Parcel Boundary and Distance Bins

2.8.2 Linear Treatment Effect

My baseline specification assumes that each additional well (up to 100 wells) of a given status within a given distance bin has the same marginal effect on the property value. Since each well presents the same environmental hazard, I assume the effect of each additional well remains constant. However, it is possible the response is non-linear, such as diminishing or compounding effects of additional wells. For example, if the first well represents the largest informational or visual shock to a homeowner, subsequent wells may be perceived as incrementally less consequential, or properties surrounded by an unusually high concentration of wells may be less responsive to well presence. Alternatively, a homeowner may begin to react to the damage after a threshold number of wells is surpassed.

To assess whether the relationship is non-linear, I replace the total well count with a flexible piecewise-linear specification. Specifically, for each well status, I construct a spline over well counts with ten equal intervals. Each segment measures the incremental well exposure within a given range. For simplicity and to preserve sufficient variation, I use the total count of wells within 2 km. Let X_i^S denote the count of wells of status S near property i, and let $\kappa_1 = 10 < \kappa_2 = 20 < ... < \kappa_{10} = 100$ define the interval boundaries. The kth spline segment is defined as:

$$S_k(X_i^S) = \begin{cases} 0, & \text{if } X_i \leq \kappa_{k-1}, \\ X_i - \kappa_{k-1}, & \text{if } \kappa_{k-1} < X_i \leq \kappa_k, \\ 10, & \text{if } X_i > \kappa_k. \end{cases} \tag{2.2}$$

I then estimate the baseline regression with the full set of segments for each status, which allows the marginal effect of an additional well to vary while controlling for the local density.

$$\ln(\text{Price}_{itg}) = \sum_{k=1}^{K} \beta_k^{A}\, S_k(\text{Active}_i) \;+\; \sum_{k=1}^{K} \beta_k^{AB}\, S_k(\text{Abandoned}_i) \;+\; \sum_{k=1}^{K} \beta_k^{P}\, S_k(\text{Plugged}_i) + \\ + \mathbf{X}_i + \gamma_t + \lambda_g + \varepsilon_{itg}. \tag{2.3}$$

Each coefficient estimates the effect of an additional active well within the given count range. If the relationship were linear, the effect of the tenth well would be indistinguishable from that of the thirtieth well, for example, and the estimated coefficients would be similar in magnitude and sign across segments. The results are shown in Figure 2.19. These estimates indicate that marginal effects are broadly stable across most of the well-count distribution, suggesting that a linear approximation captures the dominant pattern in the data. Again, the wide confidence bands at very high well counts are driven by the small number of homes exposed to extreme well densities rather than by systematic non-linear effects. The linear approximation is most relevant at lower well densities, which characterize the majority of observations.

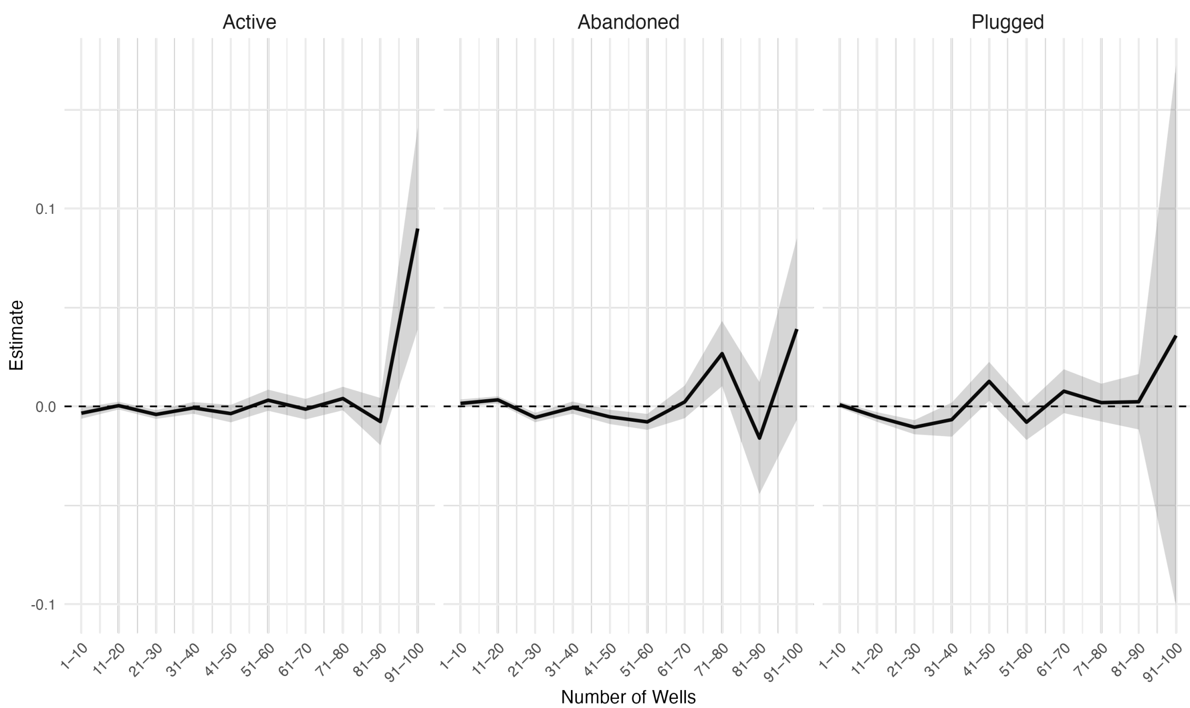

Figure 2.19: Estimates from Spline Specification

2.8.3 Alternative Identification Strategy

A potential threat to identification is endogenous well location and status. Conditional on observables and various time and geographic fixed effects, the assignment of a nearby well to a particular status must not be systematically correlated with omitted factors that also affect property prices. For location, this would mean that where wells are drilled is not systematically a response to local housing market trends. For legacy conventional wells, drilling locations were determined primarily by historical geology and early extraction technology, long before the development of modern residential neighborhoods. However, it must still not be the case that drilling is correlated with gentrification, for example. In the case that drilling is largely determined by geology, this is unlikely. For status, this means that decisions to abandon or plug must not be correlated with local shocks or regulator pressures that would also affect housing prices. Macro conditions such as high prices create incentives for operators to drill new wells, whereas low prices can render some wells unprofitable or lead to operator bankruptcy. This requires that these macro conditions are not indicative of broader economic trends that also shape housing market conditions. This should be accounted for with year fixed effects, and county-by-year fixed effects in the case that trends may differ across counties with varying reliance on the oil and gas sector. It could also be the case that well-specific characteristics which determine status, such as age, are correlated with unobserved characteristics of the local housing market. For example, older neighborhoods with older oil and gas activity could be more likely to have wells approaching the end of their lives. Wealthier homes may also be more likely to be near plugged wells if more responsible operators hold drilling rights in richer areas. Again, these concerns are mitigated by including geographic fixed effects, detailed housing characteristics, and home fixed effects for the repeat sales specifications. Finally, measurement error presents an additional challenge, particularly regarding the geolocation of homes and wells, which is fundamental to the matching process. For homes on large parcels, recorded home coordinates may be somewhat arbitrary, potentially introducing imprecision in well proximity measures. Residual endogeneity motivates the use of an instrumental variables design to further isolate plausibly exogenous variation.

I follow the approach of Shappo (2020), which exploits variation in well activity across counties to instrument for well counts. The instrument is constructed using a shift-share approach. At a given point in time, the probability that any particular well will have a given status is a combination of aggregate conditions such as changes in the regulatory enforcement, or fluctuations in the oil and gas market, and well-specific conditions such as age, type, operator and location. I separate these two components to focus on the shared aggregate conditions. At the same time, the count of wells near each home that could be of each status captures each home's exposure, or the "local share". By combining these global shifts with individual property exposure, I can isolate variation in well status that is plausibly exogenous.

Each of the well counts of each status are considered endogenous regressors and the instrument is a predicted well count. To construct the instrument, I first isolate the well-specific factors from the aggregate factors using a linear probability model. Starting with abandoned, the probability a well j is abandoned is a combination of macro factors ϕ_{gt} and well specific factors γ_{jg}:

$$Pr(Status_{jgt} = Abandoned) = \phi_{gt}^{A} + \gamma_j + \epsilon_{jgt}$$

I estimate this with a linear probability model by first constructing a balanced panel of production where the independent variable is equal to 1 for any year the well is considered abandoned. I use county as the level of geography so $g = c$. The probability a well in a county at a given time is then equal to the macro conditions ϕ_{ct}. Focusing exclusively on the component driven by global factors provides the most general specification to capture the macroeconomic and regulatory environment at the time. This gives a $Pr(abandoned)_{ct}$ at the year and county level, which is equal for all wells in a county at that time. I repeat this process for an indicator of whether a well was active or plugged.

To construct the instrument, I multiply by the number of wells at $t-1$ that could possibly be the status in question in year t. For abandoned, I consider wells that are active, or currently abandoned. For plugged, I consider the set of wells that are active, abandoned or plugged. For simplicity of notation I am abstracting from the distance bins, however in the regression I consider wells closer than 1km, and wells within 1-2km.

The instruments are as follows:

$$\hat{U}_{igt} = Pr(Status_{ct} = Active) * \big(A_{ict-1} + U_{ict-1}\big) = \phi^{A}_{gt} * \big(I_{ict-1} + U_{ict-1}\big)$$

$$\hat{U}_{igt} = Pr(Status_{ct} = Abandoned) * \big(A_{ict-1} + U_{ict-1}\big) = \phi^{A}_{gt} * \big(I_{ict-1} + U_{ict-1}\big)$$

$$\hat{P}_{igt} = Pr(Status_{ct} = Plugged) * \big(A_{ict-1} + U_{ict-1} + P_{ict-1}\big) = \phi^{P}_{gt} * \big(I_{ict-1} + U_{ict-1}\big)$$

The first stages are the regressions of predicted counts on real counts including controls and other instruments:

$$A_{igt} = \beta^{A}\hat{A}_{igt} + \beta^{U}\hat{U}_{igt} + \alpha^{P}\hat{P}_{igt} + \gamma_t + \lambda_g + \mathbf{X}_i + \epsilon_{igt}$$

$$U_{igt} = \beta^{U}\hat{U}_{igt} + \beta^{A}\hat{A}_{igt} + \alpha^{P}\hat{P}_{igt} + \gamma_t + \lambda_g + \mathbf{X}_i + \epsilon_{igt}$$

$$P_{igt} = \beta^{P}\hat{P}_{igt} + \beta^{A}\hat{A}_{igt} + \alpha^{U}\hat{U}_{igt} + \gamma_t + \lambda_g + \mathbf{X}_i + \epsilon_{igt}$$

For an instrumental variable to be valid, it must be relevant and satisfy the exclusion restriction. Relevance requires that the instrument is strongly correlated with the actual number of wells near a home. In other words, aggregate shifts in well abandonment and plugging at the county level must meaningfully predict local well counts. Exogeneity requires that the instrument affects housing prices only through its effect on the actual number of wells, and not through any other channel. In this context, the predicted number of wells is derived entirely from county-level macro shocks which are plausibly exogenous to the unobserved determinants of individual housing prices. By relying on the portion of well variation that is driven by these global factors, the instrument isolates changes in well exposure that are unrelated to local housing market shocks.

The high first-stage F-statistic indicates a strong first stage, presented in Table 2.6. Compared to the OLS results, the IV estimates in Table 2.7 are generally larger in magnitude. This suggests that OLS may underestimate the negative effects of abandoned wells due to attenuation bias. The IV results provide valuable complementary evidence, reinforcing the direction and significance of the effects.

Table 2.6: First Stage: IV Specification

	Act: 1km (1)	Ac: 2km (2)	Ab: 1km (3)	Ab: 2km (4)	Pl: 1km (5)	Pl: 2km (6)
$\widehat{Active}$:1km	0.4265***	0.1875***	0.5845***	-0.1095***	0.0924***	-0.0929***
	(0.0479)	(0.0591)	(0.0490)	(0.0409)	(0.0173)	(0.0253)
$\widehat{Active}$:2km	0.0204**	0.3868***	-0.0101	0.6359***	-0.0168***	0.0190*
	(0.0097)	(0.0512)	(0.0077)	(0.0483)	(0.0043)	(0.0110)
$\widehat{Abandoned}$:1km	0.6233***	0.0998***	0.3878***	-0.1160***	0.0371***	0.1256***
	(0.0326)	(0.0244)	(0.0327)	(0.0225)	(0.0125)	(0.0197)
$\widehat{Abandoned}$:2km	0.0137***	0.6178***	-0.0187***	0.3849***	0.0435***	0.0224*
	(0.0052)	(0.0314)	(0.0048)	(0.0316)	(0.0050)	(0.0117)
$\widehat{Plugged}$:1km	0.2912***	0.0864***	0.4093***	-0.0856***	0.3026***	-0.0071
	(0.0158)	(0.0239)	(0.0200)	(0.0163)	(0.0113)	(0.0129)
$\widehat{Plugged}$:2km	0.0321***	0.3819***	-0.0210***	0.3948***	-0.0123***	0.2272***
	(0.0056)	(0.0216)	(0.0049)	(0.0191)	(0.0032)	(0.0089)
Tract	Yes	Yes	Yes	Yes	Yes	Yes
County x Sale Year	Yes	Yes	Yes	Yes	Yes	Yes
Sale Month	Yes	Yes	Yes	Yes	Yes	Yes
Observations	720,114	720,114	720,114	720,114	720,114	720,114
R^2	0.82172	0.90806	0.86301	0.90686	0.68735	0.78928

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ Homes with no more than 100 wells nearby. Standard errors are clustered at the county year level.

Table 2.7: Regression Results: IV Specification

	log(Price)	
	(1)	(2)
Active: 1km	0.0010	0.0171**
	(0.0009)	(0.0071)
Active: 2km	-0.0020***	-0.0052**
	(0.0005)	(0.0024)
Abandoned: 1km	0.0026***	-0.0162**
	(0.0006)	(0.0082)
Abandoned: 2km	-0.0011**	0.0047**
	(0.0005)	(0.0024)
Plugged: 1km	-0.0027**	0.0085
	(0.0012)	(0.0077)
Plugged: 2km	-0.0008	-0.0113***
	(0.0005)	(0.0037)
Characteristics	Yes	Yes
Tract	Yes	Yes
County x Sale Year	Yes	Yes
Sale Month	Yes	Yes
Observations	720,114	720,114
F-test (2nd stage)		39.006

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$
Homes with no more than 100 wells nearby. Standard errors are clustered at the county year level.

2.8.4 Box-Cox Transformation

A potential concern with hedonic price models is that the assumed functional form of the dependent variable may not adequately capture the distribution of housing prices, which are typically right-skewed. While the log-linear specification is standard in the literature, I also estimate a Box-Cox transformation to test whether an alternative functional form provides a better fit to the data.

The Box-Cox transformation of a positive dependent variable y is defined as

$$y^{(\lambda)} = \begin{cases} \dfrac{y^{\lambda} - 1}{\lambda}, & \text{if } \lambda \neq 0, \\ \ln(y), & \text{if } \lambda = 0. \end{cases}$$

The Box-Cox procedure estimates an optimal transformation parameter λ by maximizing the profile log-likelihood of the model, where $\lambda = 1$ corresponds to the level specification and $\lambda = 0$ corresponds to the log specification. Intuitively, values of λ between 0 and 1 compress the right tail of the price distribution, stabilizing variance and improving residual normality.

I estimate λ to be approximately 0.33, which lies between the level and log specifications. This suggests that the data modestly prefer a power transformation of housing prices that is less aggressive than the logarithm. Appendix Figure A.2 shows the distribution of home prices under a log and Box-Cox transformation and Appendix Table A.1 shows the baseline results. The Box–Cox specification produces coefficient signs, distance gradients, and statistical significance patterns that closely mirror the log price results, indicating that the main conclusions are not driven by the assumed log-linear functional form.

2.9 Interpretation of Results

The implication of these findings lies in the aggregate welfare context, the limitations of the interpretation, and the policy recommendations. The estimates reflect the magnitude by which local housing markets value the presence and status of nearby wells. Compare these values to plugging costs can shed light on the net value to society, or justify allocating funds in the face of high costs. Raimi, Krupnick, Shah, and Thompson (2021) find median decommissioning costs are in the order of $75,000 for plugging and surface reclamation, yet these costs are highly heterogeneous depending on depth, age, surface characteristics and type. On average, there are 114 homes within 2 km of each unplugged well, yet this can easily be in the thousands. Each home only needs to gain around 0.5% of it's value on average for the plugging investment to pay for itself purely through housing capitalization with some much lower. Given the heterogeneity across distance bins and status, estimates suggest capitalization benefits that are likely to exceed this break-even threshold. Even modest recovery in local housing values would therefore be sufficient to justify plugging, even before accounting for additional benefits.

In the long term, these persistent negative capitalization effects from nearby oil and gas wells could imply a reduction in local property tax bases, potentially constraining public revenues or requiring higher tax rates (Weber, Burnett, & Xiarchos, 2016). Furthermore, Harleman et al. (2022) document the lack of investment in the land surrounding abandoned wells which should also be accounted for. There is also just the freeing up of the land for other purposes. Kang et al. (2023) also estimate the opportunity for significant job creation across the country, with up to 20,000 to 46,000 direct job-years to balance the costs. Altogether, these estimates should be interpreted as a lower bound on the full costs of well exposure, particularly when considering broader impacts that evolve over time. There is likely meaningful value in avoiding the period during which a well remains abandoned. Given that these prices also do not capitalize damage due to global emissions, these estimates should be interpreted as partial equilibrium, local welfare effects, not the full social cost of wells.

2.10 Conclusion

This paper documents the long-run costs of oil and gas wells that extend far beyond the drilling phase, with important implications for both private landowners and public policy. Using rich geospatial and housing transaction data from Pennsylvania, I show that wells reduce property values not only while they are active but also persist long after they become inactive, and that plugging does not fully reverse these losses. These results suggest that homeowners face persistent risks from the oil and gas production cycle, and that the private market systematically underestimates the future costs of drilling. The inefficiency arises because operators can externalize long-run liabilities, homeowners fail to anticipate the capitalized risk to their land, and nearby residents bear environmental spillovers.

In this empirical setting, I cannot attribute these prices losses to a particular mechanism. That is to say, these effects could combine percieved risk and realized harm. I do not attempt to identify learning or forward-looking behavior; the estimates reflect reduced-form capitalization of current observable status. There is also the uncertainty aspect or lack of information. Homeowners do not know if or when remediation will occur. In the sense that my results are somewhat low compared to the public discourse show homeowners may under-appreciate or discount the risks, whether due to optimism, lack of information, or constrained choices. As initially outlined, naivety can also not be separated from optimistic homeowners. In other words, we cannot separate a potential lack of perceived risk from a lack of choice or from optimistic expectations that wells will eventually be remediated. From an environmental justice perspective, Proville et al. (2022) shows that marginalized communities are more likely to live near oil and gas activity. Limited household mobility may attenuate capitalization of local disamenities. Null price effects could reflect lack of options. I cannot separate indifference from a lack of information, from people who dislike wells never entering the market, from constrained sorting.

For policy design, the two margins are prevention, or limiting the future accumulation of orphan wells, and remediation, addressing the consequences of the existing stock. Three features of wells are particularly useful for policy design: their open-ended life-cycle, uncertainty over future outcomes, and ambiguity over who ultimately bears responsibility. These features map naturally into two policy margins: prevention and remediation. On the extensive margin, the persistence of price effects underscores the importance of preventative regulation. These policies affect whether and where wells are drilled in the first place and whether operators internalize end-of-life costs at entry. Pennsylvannia policies tend to encourage rapid expansion and drilling and are not precautionary Rabe and Borick (2013). These losses are a result of ex-ante market failures in internalizing these long-term costs. This supports raising bonding requirements to ensure adequate funding is available not only for technical closure, but also for long-term monitoring and site restoration in the case my estimates show that plugging is not perfect. This is also an argument for increasing setback limits, despite the effect on the industry (Ericson, Kaffine, & Maniloff, 2020). On the intensive margin, these estimates can contribute to how to prioritize well plugging. There can be conflicting goals in this process, where in the face of limited budgets, there is a trade-off between the riskiest wells and those closest to populations (Bang, 2023). These estimates quantify the value of plugging the wells closest to populations to compare to the global value of limiting emissions which is left to be done. Finally, these estimates encourage directing impact fees back towards the affected communities. While certain aspects are specific to Pennsylvania policies, mainly the historical landscape, external validity for the persistent long-term effects of abandoned wells is not expected to be unique to Pennsylvania. Empirical studies for other states would shed more light on this.

This paper highlights a central intertemporal inefficiency in the lifecycle of oil and gas wells: the short-run economic benefits of drilling are broadly shared across firms and local labor markets, while the long-run environmental and property value costs are concentrated on nearby landowners and the public sector.

Chapter 3

The Distributional Consequences of Individual Transferable Quotas

3.1 Introduction

Global fish stocks are declining due to over-extraction and an excess of capital, resulting in colossal economic losses (Costello et al., 2016; World Bank, 2017). In recent Canadian history, 50% of the total amount of fish by weight was lost due to overfishing, yet only 18% of critical stocks have plans to support their recovery (Archibald, McIver, & Rangeley, 2020). The percentage of healthy stocks has declined from 55% in 2015 to 27% in 2020 (DW Archibald, McIver, & Rangeley, 2021). The longer this persists, the harder it will be to reverse these damages, raising the risk of potentially irreversible fishery collapse. However, controversial management decisions can also generate tensions between different interest groups trying to secure fishing rights, particularly in resource-dependent local economies. While many regulatory reforms are designed to improve efficiency and conservation outcomes, their distributional consequences are less well understood. In particular, policies that alter access or ownership may reallocate rents across fishers and communities in ways that affect economic inequality. This paper studies how the implementation of one particular policy, individual transferable quotas (ITQs), reshapes the distribution of economic outcomes among heterogeneous resource users. Specifically, we look at the impact of the introduction of this regulation on revenue, effort, and exit among fishers who differ in both skill and outside options.

Individual transferable quotas are seen as a promising solution because they provide a market mechanism for resource allocation (Grafton, 1996). Fishers receive quotas for the regulated species through a designated allocation mechanism, which assigns each fisher a share of the set total allowable catch. These quotas can then be traded at a predetermined market price, which theoretically represents the net present value of all future resource rents accruing from that allocation (Arnason, 2019). This direct link between long-term stock productivity and the current value derived from extracting the resource encourages environmental stewardship, allowing the stock to recover (Van Putten, Boschetti, Fulton, Smith, & Thebaud, 2014). Ownership also removes the “race to fish”, extending the fishing season and allowing for the optimal timing of catch (Birkenbach, Kaczan, & Smith, 2017). This system ensures efficiency through 1) limiting aggregate catches to the socially optimal level and 2) ensuring that fish are caught at the least cost with the quota trade. There is an incentive for the most efficient operators to purchase additional access rights from the more inefficient operators. As less efficient operators exit the fishery, there is a reduction of excess capital and a concentration of catch among the remaining vessels. However, there is an inherent efficiency-equity trade-off. While ITQs can substantially increase aggregate rents, the same market mechanisms that generate these gains may also redistribute rents in ways that reshape inequality across fishers and communities (McCay, Creed, Finlayson, Apostle, & Mikalsen, 1995). It is unclear who gains and who receives these resource rents, and whether this results in a Pareto improvement (Guyader & Thebaud, 2001).

The distributional consequences of ITQs depend on the source of heterogeneity among fishers. For homogeneous agents, the quota trade does not affect inequality because the sellers are perfectly compensated by the quota buyers for the loss of resource rents (Costello & Deacon, 2007). With heterogeneous agents however, the profits are not equally distributed among sellers and buyers of quota. Fishers can differ in both resource-specific productivity and in the availability of outside labor market opportunities, implying that quota may flow either toward the most skillful producers or toward those facing the lowest opportunity costs of fishing. These mechanisms generate conflicting theoretical predictions. Grainger and Costello (2016) find that incomes of the wealthy can decrease when there exists heterogeneity in skill, since ITQs transfer inframarginal rents accrued by the highest skilled to resource rents shared by all. Okonkwo and Quaas (2020) allow for heterogeneity in the outside option and find incomes of the poor decrease after the transition to a property rights regime. In this setting, the poor are the resource users without a profitable "private project", therefore the transition to an ITQ system lowers their resource return. In this paper, we reconcile these competing predictions by allowing fishers to jointly differ in resource-specific productivity and local labour market opportunities, and by testing which dimension governs the reallocation of quota in practice.

Efficiency gains from trade may therefore either evolve from a flow of quota to the most skillful fishers or to fishers with the lowest opportunity costs of fishing (Arnason, 2012). The distinction is relevant because there are increasing concerns that resource privatization can contribute to rising rural inequalities (Brandt & McEvoy, 2006; Da-Rocha & Sempere, 2017), and the loss of quota or fishery access may be especially important in areas with fewer outside options. There are potential negative implications for small-scale harvesters and resource-dependent communities, where the fishing industry provides an important source of employment between harvesting, processing, and aquaculture (Baland & Francois, 2005). The problem is especially severe in remote areas with little access to alternative income opportunities. Furthermore, local ownership of fishing rights can generate spillovers into other sectors of the regional economy, implying that the impacts may extend beyond the fishing sector (Watson, Reimer, Guettabi, & Haynie, 2021). Ultimately, if these regulations disproportionately benefit specific capital-rich users and remove access from rural communities, rational opposition by some users can impact long-term enforcement.

We use the universe of individual-level catch and revenue data for Maritime Québec from 2000 to 2019 and study two ITQ reforms affecting Greenland halibut in 2003 and 2012. To determine the distributional consequences of the ITQ regulation, we first assign each fisher to a quantile along the skill and outside option distribution. Since the regulation affects each fisher differently depending on their catch profile, we create a fisher-level measure of regulation exposure. We use matching methods to find appropriate control fleets to represent counterfactual outcomes among non-regulated fishers. We implement a continuous treatment difference-in-differences approach to study distributional changes in incomes and effort on the intensive margin and exit rates on the extensive margin. Finally, we look at changes in total quota holdings over time.

We find that ITQs increase fishing revenues on the intensive margin, but with substantial heterogeneity across reforms and across fishers. For the 2003 reform, revenue gains are concentrated among lower skilled producers, while high-skill fishers experience little change, consistent with a reduction in within-fishery income dispersion. Revenue gains are larger for fishers with weak outside options, although gains remain positive for fishers in stronger labor markets. For the 2012 reform, we find find no meaningful revenue effects. On the extensive margin, ITQs induce exit primarily among lower-skill fishers. While the probability of exiting does not differ systematically across the outside option distribution, all fishers are impacted by fleet rationalization. Access rights consolidate over time, yet quota holdings do not shift disproportionately towards the high-skilled or away from resource-dependent communities.

Understanding these distributional effects is essential both for evaluating welfare impacts and for assessing the political sustainability of fisheries management regimes. The paper proceeds as follows. In Section 3.2, we describe the Québec fishing industry and provide details on the ITQ regulations. Section 3.2.2 outlines the competing predictions present in the literature. We describe the data in Section 3.3, and Section 3.4 describes the sample construction for the regressions outlined in Section 3.5. Finally, Section 3.6 presents the results, and we finish with a discussion of the potential limitations of this study and further research in 3.7.

3.2 Background

3.2.1 Property Rights, Efficiency, and Rent Distribution in Fisheries

Open-access resource extraction is characterized by excessive capital accumulation and inefficient harvest timing. In such settings, unrestricted entry dissipates economic rents because no agent internalizes the opportunity cost of depletion (Samuelson, 1974; Weitzman et al., 1974). Schlager and Ostrom (1992) argue that without management and exclusion rights, common-pool resources are unlikely to be sustainably governed. Gordon (1954) notably applies this to fisheries, showing that failure to internalize the stock externality leads to over-entry, excessive effort, and inefficient resource depletion, and arguing that regulation or property rights are necessary to restore economic efficiency.[1]

Individual transferable quotas have been widely studied as a policy response to these open-access

[1] In the terminology of Schlager and Ostrom (1992), ITQs confer withdrawal and transfer rights, but not true "ownership" over a specified value of the stock since ownership of fish prior to capture is not well-defined or meaningful in a biological setting. We respect this nuance, and any reference to "property rights" is reflecting these harvesting rights.

failures. Theoretically, individual transferable quotas can address the inefficiencies of open access by assigning tradable property rights over a fixed total allowable catch. ITQs have been shown to generate windfall gains for quota sellers, higher output prices, and a lower probability of a fish stock collapsing (Fox, Grafton, Kirkley, & Squires, 2003; Gunnlaugsson, Saevaldsson, Kristofersson, & Agnarsson, 2020; Isaksen & Richter, 2019). Reimer, Abbott, and Wilen (2014) decompose ITQ rent generation into consolidation effects and incentive effects, showing that in highly overcapitalized fisheries most gains arise from eliminating excess capital as opposed to behavourial changes. At the same time, the efficiency gains from ITQs are neither immediate nor guaranteed. Several studies emphasize that poorly calibrated catch limits and weak enforcement can undermine the expected benefits of quota systems (Arnason, 2012; Chu, 2009). Even when biological targets are met, efficiency gains may be constrained by restrictions on quota transferability or limits to vessel size (Grafton, Squires, & Fox, 2000). Stock recovery for the target species may also come at the cost of negative spillovers for non-ITQ species (Blomquist & Waldo, 2018; Branch, 2009). If a stock is spatially heterogeneous, congestion and redundant search efforts will continue to dissipate rents (Costello & Deacon, 2007; Valcu & Weninger, 2013). Finally, case studies such as the British Columbia halibut fishery highlight how extensive quota leasing, particularly in the presence of asymmetric information, can generate adverse economic outcomes for active harvesters (Pinkerton & Edwards, 2009). Taken together, the empirical evidence emphasizes that capturing the efficiency gains from ITQs requires careful attention to quota design, enforcement, and market rules.

Moving beyond efficiency, there is a large literature on how ITQs redistribute income, rents, and opportunities across users and communities. Da-Rocha and Sempere (2017) show that ITQs can simultaneously increase revenue concentration by reallocating harvest toward productive users, while reducing wealth inequality by transforming fishing rights into tradable assets that generate capital gains for quota owners, at least initially. These effects depend critically on the initial allocation of quotas, including whether rights are grandfathered or auctioned (Doering, Goti, Fricke, & Jantzen, 2016). Removing access can make individuals worse off in the short-run by eliminating a buffer against income shocks where the commons can serve as a form of insurance, yet stock increases in the long-run can compensate through increased employment (Baland & Bjorvatn, 2013; Baland & Francois, 2005). However, the goals of all interest groups must be defined. Conflicts between extractive and non-extractive users, as well as between fishing and conservation interests, may also persist under ITQs, limiting their ability to reconcile competing objectives (Arnason, 2009; Guyader & Thebaud, 2001). Olson (2011) finds that negative impacts from privatization often fall on the less powerful participants which can affect the structure of and viability of communities, specifically for smaller potential entrants. Brandt (2005) find no evidence that any segment of the industry was disproportionately harmed by ITQs when participation is measured at the firm rather than the vessel level. However, subsequent work shows that impacts can actually vary across gear types, home ports, and contractual relationships, implying that some groups gain relative to others even in the absence of widespread adverse effects (Brandt & McEvoy, 2006). The assessments of ITQs depend critically on the normative criteria used to evaluate distributive fairness, including within and across generations, and the policy instruments available to address equity concerns

(Doering et al., 2016).

The distributional benefits often rely on the ability of labour to reallocate to other sectors as there can be a loss of jobs through fleet rationalization (Gislason, 2010). Watson et al. (2021) use variation in fish stocks and prices in Alaska to show that increases in resident fishing earnings raise local employment and income primarily when fishing rights are locally owned, emphasizing the role of resident ownership in determining community-level benefits from resource rents. Hatcher (2014) discusses changes to worker renumeration in an ITQ system. Blomquist and Waldo (2018) address these issues in a Swedish fishery, to show that those that sold their quota largely stay in the fishing sector and there is little labour mobility. Sutherland and Edwards (2022) finds evidence of reduced revenues in rural Alaskan ports, despite less vessel consolidation compared to areas with airport access.

Together, these findings suggest that the distributional consequences of ITQs depend not only on quota design and transfer rules, but also on local labour market opportunities, an issue this paper directly addresses.

3.2.2 Skill versus Outside-Option Heterogeneity: Competing Predictions

The distributional effects of individual transferable quotas depend critically on the source of heterogeneity among resource users. Here we outline the predictions of Grainger and Costello (2016) and Okonkwo and Quaas (2020) who focus on different margins through which the introduction of property rights redistributes profits, yielding contrasting predictions about who gains and who loses.

Grainger and Costello (2016) allow for heterogeneity in resource-specific productivity. Under common-pool or limited-entry management, the most skilled harvesters generate substantial inframarginal rents. These rents arise because high-skill fishers are able to capture a disproportionate share of the harvest during the race to fish. The introduction of ITQs eliminates this race and converts inframarginal rents into resource rents that are capitalized into quota prices and shared across quota owners. As a result, although aggregate rents increase, the highest-skill fishers may experience income losses relative to the pre-ITQ regime unless they receive sufficiently generous grandfathered allocations. In this setting, ITQs can reduce inequality by compressing returns among active harvesters, even as total surplus rises.

In contrast, Okonkwo and Quaas (2020) instead emphasize heterogeneity in outside labour market opportunities. Their dynamic model highlights the role of open-access resources as an income safety net for individuals with poor alternatives outside the resource sector. Under open access, low outside-option individuals earn the common marginal product of labour in harvesting, which may exceed their returns in alternative employment. Privatization restricts access to the resource and reduces labour employed in harvesting, lowering harvesting incomes during both the transition and steady state. Even when resource rents are redistributed equally, these rents may be insufficient to compensate individuals who lose access to harvesting and lack viable alternatives in other industries. In this setting, privatization can increase inequality and reduce welfare for the poorest users, particularly when alternative income opportunities are highly unequal or discount rates are high.

When fishers differ along both dimensions, the distributional effects of ITQs become theoretically ambiguous. In such a setting, the equilibrium allocation of quota reflects the interaction between production efficiency and opportunity costs. If high-skill fishers with strong outside options purchase quota from lower-skill fishers with limited alternatives, ITQs may reinforce inequality. If high-skilled fishers have low outside options, and low-skilled fishers have strong outside options, there may be an efficient selection mechanism into fishing. These competing mechanisms imply that the net distributional impact of ITQs is an empirical question that depends on the joint distribution of skill and outside options or on which dimension of heterogeneity dominates the adjustment mechanism. Aggregate efficiency gains may coexist with heterogeneous income effects across quantiles of skill and labour market access, as well as differential exit responses along each margin. Importantly, ITQs may reduce income dispersion among fishers who remain active in the fishery while increasing overall income inequality once exit and labour market opportunities are taken into account. As we do not observe labour market wages post-exit, our analysis therefore captures labour market reallocation through observed exit behavior rather than realized income outside the fishery, leaving the full welfare consequences of reallocation to future work.

An important distinction between these two frameworks also concerns the initial allocation of rights. In Grainger and Costello (2016), quotas are grandfathered to incumbent fishers based on historical participation, so that resource rents remain closely tied to prior fishing activity. Okonkwo and Quaas (2020) on the other hand analyze a setting in which rights are distributed equally across individuals. These allocation rules are central to their respective results where grandfathering may compensate incumbents for the loss of inframarginal rents, whereas equal allocation fails to compensate those most dependent on the resource. In our setting, the initial allocation of quota is based on historical catch.

3.2.3 Socioeconomics of the Québec Fishing Industry

Maritime Québec comprises three administrative regions: Côte-Nord (North Shore), Gaspésie–Bas-Saint-Laurent (Gaspé–Lower St. Lawrence), and Les Îles-de-la-Madeleine (Magdalen Islands). The area accounts for roughly 5% of Québec's population (DFO, 2015). The industry is highly geographically dispersed, with over 90 remote, resource-dependent communities shown in Figure 3.1, which also reports revenues across census subdivisions. Landings are concentrated on the Gaspé peninsula, with Rivière-au-Renard accounting for the largest value of landings. In 2003, total fishing revenues were distributed across regions as shown in Figure 3.2. The primary species are snow crab, lobster, and shrimp, which together account for 88% of total landed value. Economic conditions are weaker than the provincial average: Maritime Québec has experienced population decline and persistently higher unemployment, suggesting limited outside options for displaced harvesters.

3.2.4 Description of ITQ Regulations

ITQ systems were gradually introduced in Canada starting in the 1990s, transitioning many species away from an open-access competitive regime (McCay et al., 1995). The stated objectives of these regulations range from streamlining fishing operations to creating stability in access to stocks. Regulations are implemented at the administrative-region level and apply to fleets operating in specific areas. Fleets are administratively defined groups of fishers based on primary species and vessel length. Multiple

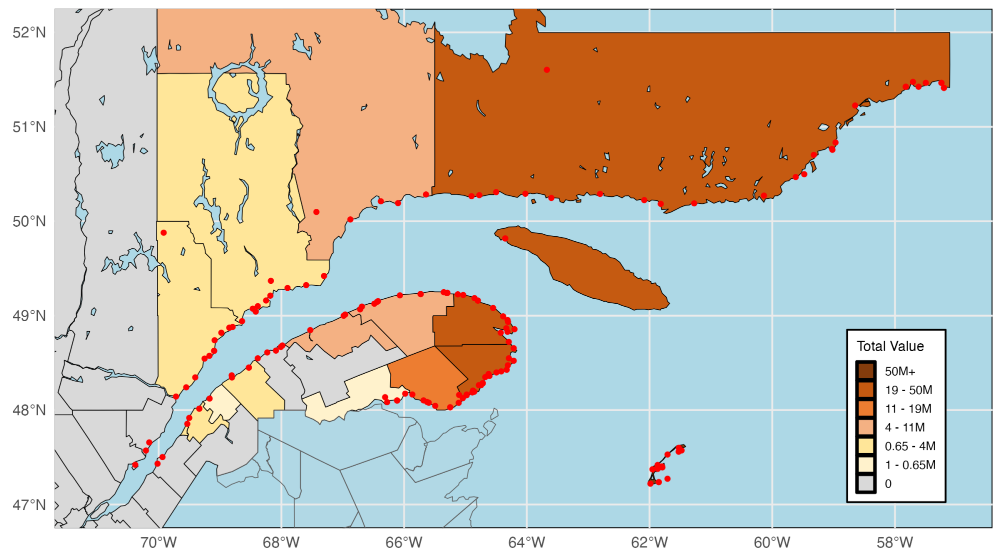

Figure 3.1: Annual Fishing Revenues in Maritime Québec (2015)

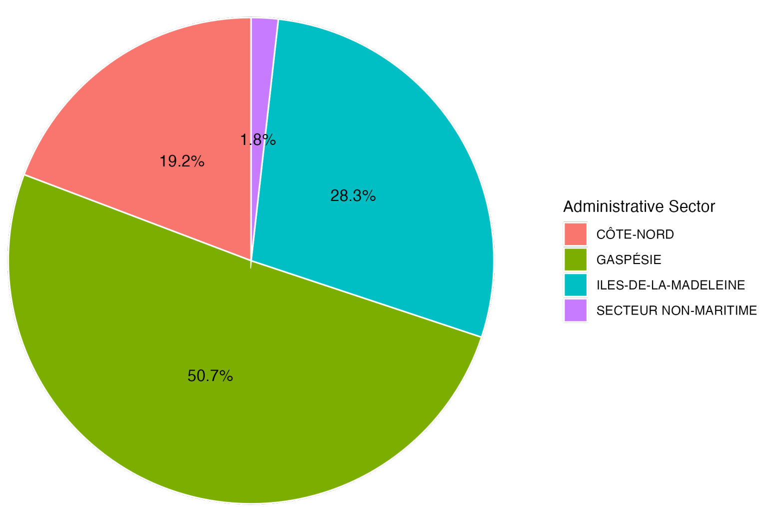

Figure 3.2: Percentage of Total Revenues by Administrative Region in 2003

fleets can fish the same species at the same time, and fleet shares determine the percentage of the total allowable catch (TAC) allocated to each fleet. Under global quota management, such as in a derby system, all fishers compete until the total allowable catch is reached, at which point the fishery is closed. In contrast, under individual transferable quotas, each fisher holds a fixed share of the TAC, which can be permanently or temporarily transferred. Quotas in this setting are initially allocated through a grandfathering system based on historical catch in predetermined base years. Fishers are eligible to receive initial quotas if they hold an active license and meet a minimum landing threshold.

Over our time period of interest, we study two ITQ regulations introduced for Greenland Halibut. First, ITQs were introduced in 2003 for the Gaspé-Upper Middle North Shore, and then in 2012 for the Lower North Shore for fixed-gear vessels of length less than 19.81 meters. The Gaspé-Upper Middle North Shore fleet was subject to individual quotas since the late 1990s, yet these did not become transferable until 2003. The Lower North Shore fleet remained competitive until 2012, although its fleet share is significantly lower. There is a cap on quota holdings of 3.844 percent for the 2012 fleet.

Other species adopted quota systems during the period, but these reforms either affected very few fishers (Atlantic Halibut and Cod in 2012), occurred too late to provide a sufficient post period (Magdalen Islands and Lower North Shore reforms in 2017–2019), or adopted ITQs before our sample period (Snow Crab). Lobster remained competitive and provides a stable comparison fishery. We observe sufficient within-portfolio and across-fleet variation to construct credible comparison groups and isolate the effects of ITQ adoption.

3.3 Data

3.3.1 Fishing Revenues, Effort, and Quota Holdings

For fishing revenues, we use the universe of catch and revenue data for Québec from 2000–2019, provided by the Department of Fisheries and Oceans Canada (DFO). Each observation is the landed weight and value of a specific species caught by an individual on a given trip. We aggregate these data to the individual-year level by summing across trips and species, deflating revenues to 2003 dollars. We observe over 2800 fishers over this time period. Location data include departing and landing communities, and vessel characteristics include length and gear. To measure effort, we calculate total days at sea, total operational days, and the total number of trips taken per year.[2] Table 3.1 reports summary statistics and Figure 3.3 depicts trends over the sample period. Total value of catch is increasing over time, most significantly in Gaspésie, which also has the largest decline in the number of fishers, suggesting a concentration of revenues. The number of fishers is remarkably stable in Les Îles-de-la-Madeleine. The bottom panel shows high and increasing revenue inequality in Gaspésie and Côte-Nord, which are the two administrative regions affected by the ITQ reforms. Figure 3.4 shows that revenues in a given year are highly right-skewed with a long upper tail.

[2] Effort measures proxy for variable operating costs in the absence of data on input expenditures. These include fuel costs that vary with days at sea and distance traveled, crew compensation that is typically paid as a share of revenue or per trip, and maintenance and gear replacement that increase with vessel utilization.

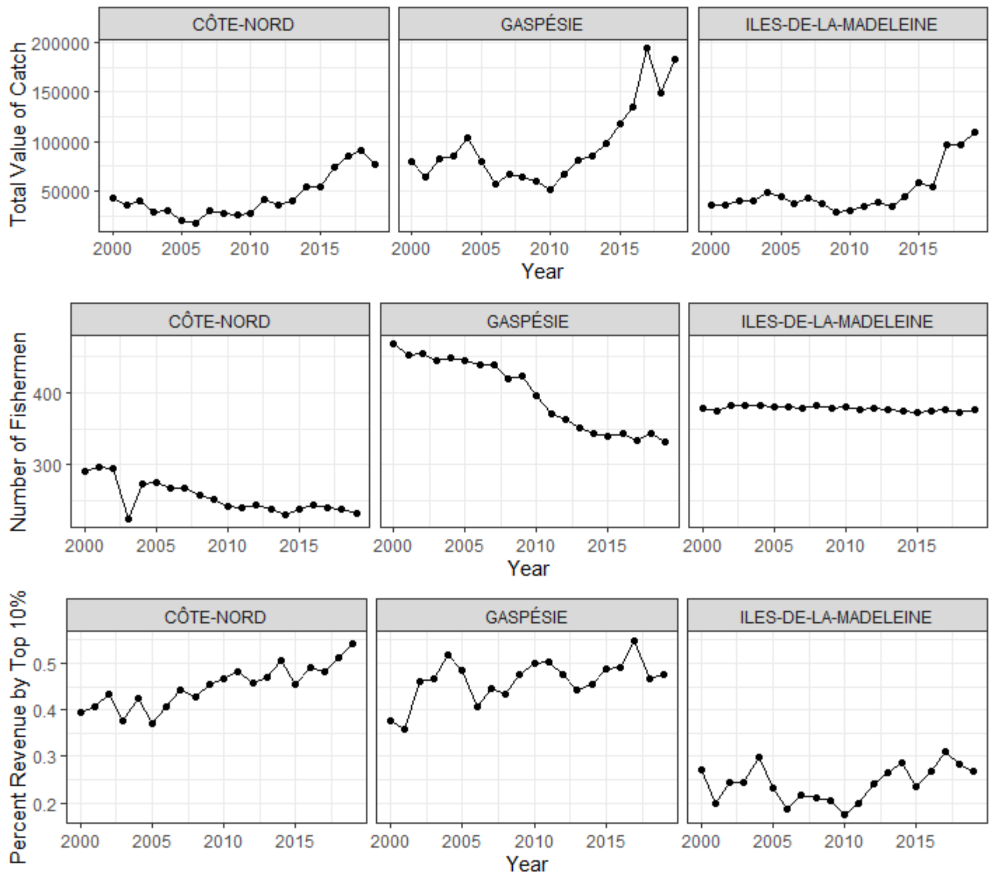

Figure 3.3: Trends in Maritime Quebec

The DFO records annual individual quota holdings for all regulated fleets. We define treatment as receiving an initial quota allocation, and this applies to 85 and 87 fishers respectively in the 2003 and 2012 regulated fleets. We restrict treatment to initial quota recipients to preserve a common and exogenous treatment timing. Initial quota holdings vary from 0.1 to 2 percent of the TAC as shown in Figure 3.5.[3] Figure 3.6 shows the changing distribution of quotas over time, demonstrating the gradual accumulation of larger quota holdings by certain individuals.

Table 3.1: Summary Statistics: Catch and Revenue Data

	Mean	SD	Min	Median	Max
Revenue ($1000s)	142	195	0	84	2716
Vessel Length	36	14	0	38	95
Trip Length	28	27	1	17	362
Operational Days	23	20	1	15	180
Number of Trips	18	16	1	12	188
Number of Species	2.4	1.9	1	2	12

[3] The DFO does not record quota prices, as these are determined through private transactions among fishers.

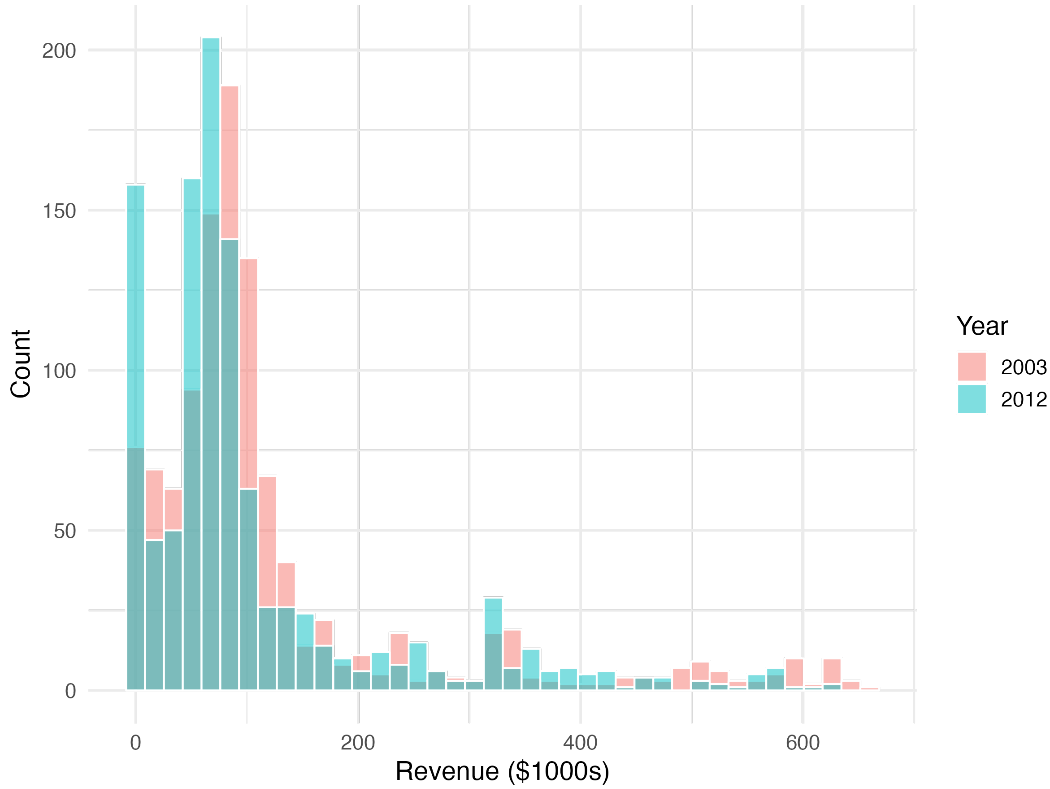

Figure 3.4: Revenue Distribution during 2003 and 2012

3.3.2 Local Labour Market Outcomes

We construct measures of local labor market conditions for Maritime Québec using data from the 2001 Canadian Census. We collect variables for males aged 25 and over most resemble the demographics of the fishing industry. For this analysis, we focus on the Census Division level, which corresponds to Regional County Municipalities in Quebec. This larger geographic area is defined as the "local labour market" of each fisher. These local labor markets are characterized by several indicators of labor market performance. For employment, we use the size of the labour force and the number of individuals employed and unemployed. For income, we use the average, median, and standard deviation of annual earnings. Expected income is calculated as average income multiplied by employment rate. The unemployment rate ranges from less than ten percent to over thirty percent, and average income from around $20,000 to almost $40,000 in 2001 dollars.

3.4 Identification Framework and Sample Construction

First, we construct the key empirical components needed to identify the effects of ITQ regulation on incomes, effort, and exit decisions. The regulations we study were introduced at two different points in time, and they directly affect only a subset of fishers within a subset of fleets. Moreover, even among treated individuals, the intensity of treatment varies depending on the degree to which each fisher was historically engaged in the regulated stock. The empirical design requires a careful calculation of

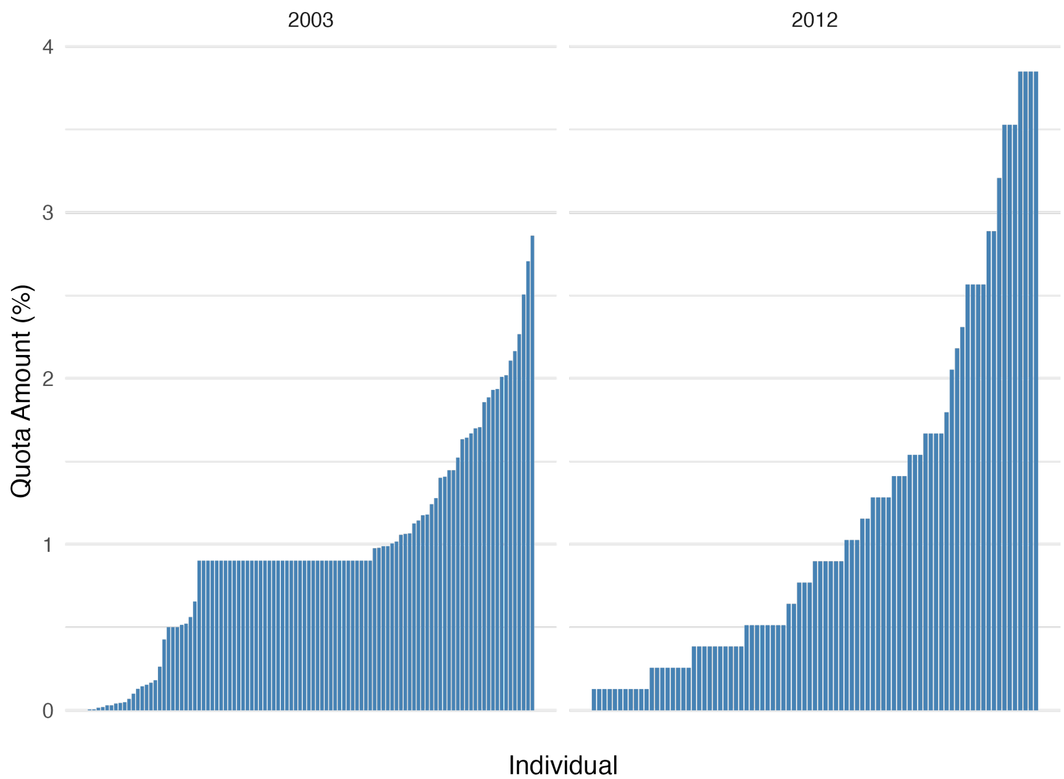

Figure 3.5: Initial Quota Distribution

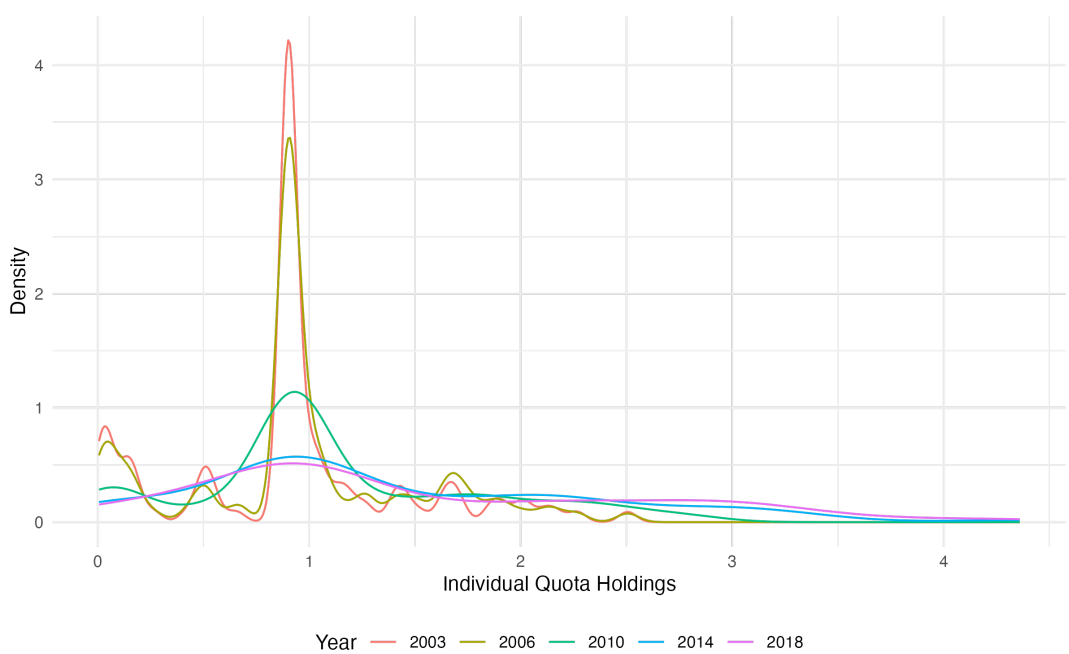

Figure 3.6: Distribution of Quota Holdings Over Time

treatment exposure, assignment of distributional groups, and selecting an appropriate control group that provides a credible counterfactual for treated fishers. These choices allow us to 1) implement a design in which treatment intensity varies continuously, 2) examine heterogeneous impacts across meaningful economic distributions, and 3) compare treated and control fishers who follow similar pre-treatment trends.

3.4.1 Regulation Exposure

Fish stocks are subject to changing regulations over time, and at any given point a fisher may harvest several regulated and unregulated species. This implies that within a fleet, a regulation will affect all fishers to varying degrees depending on their reliance on the regulated stock. Although most individuals typically fish one or two main species, the composition of total revenues is not constant over time and revenues can be highly volatile. We therefore define average treatment exposure in a baseline period. For the 2003 regulation, we use 2000–2002, or 1–3 years prior to the ITQ introduction. For the 2012 regulation, we have a longer period available and use 2006–2011. Exposure is calculated as the contribution of Greenland Halibut to a fisher's total catches during the baseline period. For ease of interpretation, we primarily focus on monetary value, though we also consider the percentage of annual revenues.[4]

Fishers are in the treatment group if they receive an initial share of quota, either in 2003 or 2012. Table 3.2 shows these calculations for the two fleets of interest as a monetary value and percentage of their total catch in the base period. There is substantial within-fleet variation, where some fishers almost exclusively fish Greenland Halibut, yet the mean exposure is far less than 50% for both fleets. The lower value of exposure from the 2012 fleet is explained by the lower fleet-level share.

Table 3.2: Regulation Exposure: Catch-based Measures

Regulation	N	Mean(% Exp)	Max (% Exp)	Mean($ Exp)	Max ($ Exp)
FG 2003	84	32.28	88.41	22,769.31	67,952.06
FG 2012	61	20.24	64.81	10,049.83	35,764.32

3.4.2 Assigning Distribution Quantiles

Skill Distribution Fisher skill is defined as the portion of individual revenue that cannot be explained by observable inputs. By estimating individual fixed effects in a revenue regression controlling for these inputs, we isolate persistent differences across fishers that reflect their underlying ability, experience, or efficiency.

We start with a general production function for individual i at time t:

$$Revenue_{it} = A_i \, K_{it}^{\delta_1} \, L_{it}^{\delta_2} \, S_{it}^{\beta} \, e^{u_{it}}, \tag{3.1}$$

[4]We use revenue-based exposure as our primary measure instead of initial quota share because it captures the economic importance of the regulated stock to the fisher. Although positively correlated through grandfathering, initial quota shares reflect rights allocation rather than economic dependence and exhibit substantially less variation across fishers.

where A_i is total factor productivity capturing efficiency or unobserved skill, K_{it} is capital, L_{it} is labour measured by effort, S_{it} captures species characteristics, and u_{it} is an idiosyncratic shock. Taking natural logs gives the standard log-linear form. Specifically, we estimate the following regression using pre-regulation data:

$$\log(revenue_{it}) = \delta_1 \log(capital_{it}) + \delta_2 \log(effort_{it}) + \beta(species_{it}) + \gamma_t + \alpha_i + \epsilon_{it}, \qquad (3.2)$$

Here α_i is an individual-specific fixed effect capturing persistent differences in skill across fishers. Capital includes vessel length and gear.[5] Effort variables include total number of trips, total operational days, and total days at sea. We also include an indicator for main species fished. We then assign fishers to skill groups based on these fixed effects. This follows Grainger and Costello (2016) in which harvest increases in skill.

The regression is run on all individuals during the base period. Fishers in the treated group tend to be concentrated on the lower end of the distribution of individual fixed effects, as shown in Figure 3.7. The dashed vertical lines indicate the thresholds for four quantiles, however to ensure that both treated and control groups have sufficient numbers of individuals for meaningful comparisons, we instead divide fishers into a low or high-skilled group.

Outside Option Distribution Next, we assign each home port a position in the outside-option distribution, and map individual fishers to this distribution based on their primary fishing location. We first construct a composite index using the census variables summarizing income levels, income dispersion, and unemployment. To ensure comparability across measures expressed in different units, we log and standardize each variable. Positive signs are assigned to measures reflecting stronger labor markets and negative signs to those reflecting weaker conditions. This creates an index expressing local labor market conditions, with higher values indicating stronger outside options for workers. We divide regions into four quartiles and restrict our ranking of geographic areas to locations with active fishing communities. Average incomes are naturally higher in regions closer to major urban centers along the St. Lawrence, but high relocation costs and the specialized nature of fishing limit mobility, so local labor market conditions still meaningfully reflect outside options for fishers. We fix outside-option measures at baseline to avoid confounding treatment effects with temporal local economic shocks.

We define a fisher's "home port" as the most frequented location of departure.[6] These communities are linked to their respective Census Division. Since the majority of locations are small fishing communities, defining the local labour market at a broader level allows for realistic mobility between nearby communities, while still capturing the constraints on longer-distance relocation.

3.4.3 Control Group: Matching Methods

Our empirical strategy requires comparing the outcomes of the treated individuals to those of a control group that is intended to represent their counterfactual evolution in the absence of regulation. Unregu-

[5] When fishers operate multiple vessels, we compute a revenue-weighted average vessel length.

[6] Location of landing was also explored, yet in all cases was identical to the most frequented location of departure.

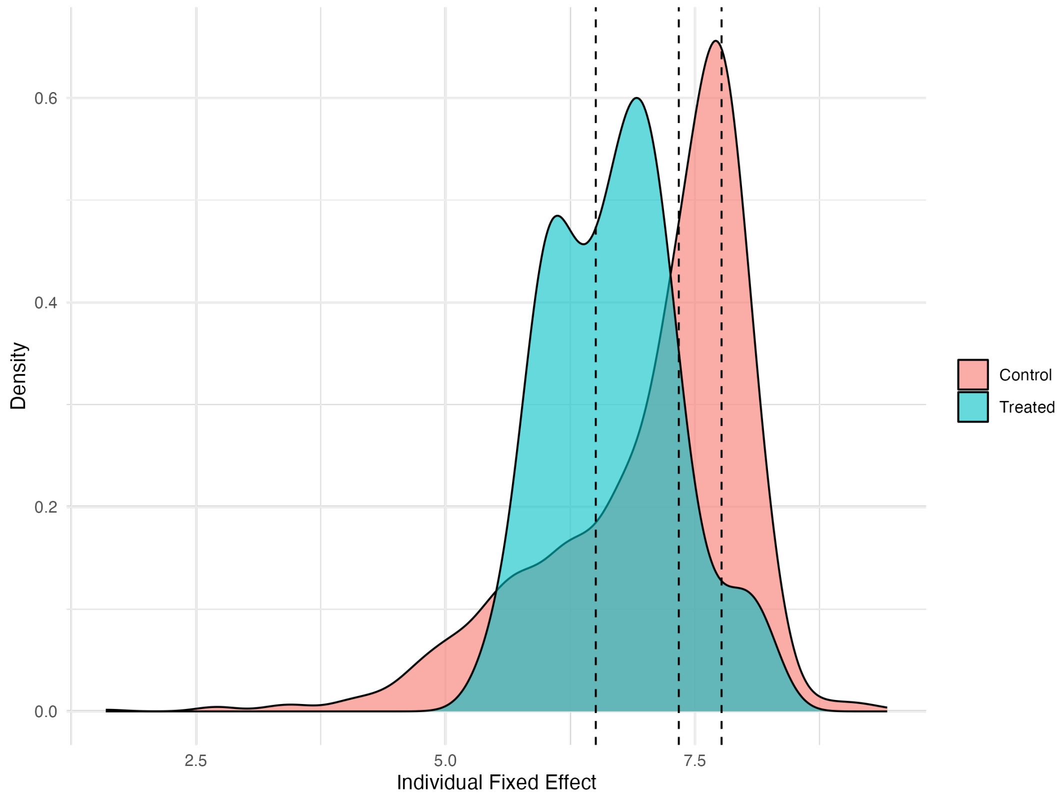

Figure 3.7: Distribution of Individual Fixed Effects for Treatment and Control Groups.

lated fishers far outnumber regulated fishers, so it is unreasonable to expect they are all representative of potential outcomes. We use matching methods to restrict our sample to fishers that represent this valid counterfactual. We will match on pre-treatment and time invariant characteristics, which highlight areas of the covariate distribution where there is sufficient overlap between the treatment and control groups. Furthermore, as treated individuals are present in the base years by construction, matching ensures we are not including fishers who enter post-regulation. We run the following regression, which essentially estimates the propensity score.

$$Treat_i = (\overline{revenues})_i + (\overline{effort})_i + (\overline{vessellength})_i + (\#species) + \epsilon_i \tag{3.3}$$

We run two slightly different variants of nearest-neighbour propensity score matching. In both cases, we estimate each fisher's probability of treatment and select control fishers who were not exposed to an ITQ reform but had similar ex-ante likelihood of being treated. In the first specification, we match one control to each treated fisher using only average base-period revenues and vessel length. In the second specification, we include the full set of pre-treatment covariates. We flexibly match an average four controls to each treated fisher where each treated fisher can have a minimum of one and maximum of six matches. This variable-ratio matching allows treated fishers with many close matches to be matched to more controls, while limiting matches for treated fishers with fewer comparable controls, improving overall balance.

Table 3.3 shows the results of this matching process. The richer matching specification achieves balance with no statistically significant differences between treated and matched control fishers along the key dimensions. Vessel length is a particularly important matching variable, since eligibility for the regulation is explicitly defined by vessel length thresholds. The number of species is less meaningful, as it can reflect bycatch rather than targeted fishing behavior. However, it remains informative about the breadth of a fisher's catch portfolio and the extent of diversification across species. We therefore treat the number of species as a control rather than as a primary matching variable. Overall, the matching procedure substantially improves balance on the covariates that govern selection into the regulated fleets.

Table 3.3: Covariate Balance Across Three Matched Control Groups

Covariate	Treated Mean	Method 1	Method 2	Method 3
Panel A: 2003 Regulation				
Vessel Size	41.01	38.12 (0.00296)	41.68 (0.675)	37.93 (0.00444)
Mean Log Revenue	11.28	10.75 (7.45e-07)	11.26 (0.869)	11.14 (0.262)
Mean Effort	51.91	24.51 (3.35e-18)	29.07 (1.25e-09)	36.62 (1.41e-07)
Mean Number of Species	5.14	2.02 (6.27e-27)	2.44 (5.26e-18)	2.54 (1.89e-21)
Panel B: 2012 Regulation				
Vessel Size	35.66	32.21 (0.00133)	35.56 (0.957)	30.88 (0.0003)
Mean Log Revenue	10.58	9.94 (8.59e-06)	10.58 (0.99)	10.10 (0.00122)
Mean Effort	27.79	22.36 (0.00216)	22.24 (0.0456)	19.27 (7.19e-06)
Mean Number of Species	3.00	1.95 (1.07e-10)	2.05 (2.18e-05)	1.84 (1.45e-11)

Notes: P-values are in parentheses.

3.5 Regression Specifications

3.5.1 Two-Way Fixed Effects Model

Here we present the baseline specification, which is a two-way fixed effects model exploiting within-individual variation over time and allowing the treatment effect to vary with the intensity of treatment exposure as defined above. It can also be interpreted as a difference-in-differences model with heterogeneous treatment intensity.

We first look at the average effect of the regulation as follows for individual i under regulation j at time t:

$$\log(y_{ijt}) = \beta\big(ITQ_{jt} \times Exp_i\big) + \gamma_t + \alpha_i + \epsilon_{ijt} \tag{3.4}$$

On the intensive margin, the outcome variables are revenues and effort. On the extensive margin, we look at the probability of being active in the fishery, by using a balanced panel where the dependent variable equals one if a fisher is actively fishing in a given year and zero otherwise. We regress these outcomes on the interaction between exposure and an indicator for the ITQ regulation, which changes from 0 to 1 in the post period. Here β is the coefficient of interest, which represents the marginal effect of an additional unit of exposure, where one unit corresponds to an increase of $1000 of baseline exposure. We include individual and year fixed effects throughout.

For the distributional impacts, we run the same regression while allowing β to differ across the income and outside option distributions:[7]

$$y_{ijt} = \sum_{k=1}^{4} \beta_k \mathbb{1}\{q_i = k\}\big(ITQ_{jt} * Exp_i\big) + \gamma_t + \alpha_i + \epsilon_{ijt} \tag{3.5}$$

The two main assumptions for identification in this model are: (1) exogenous treatment assignment conditional on observables and (2) unaffected and stable control units. Although ITQ regulations are not randomly assigned, the use of multiple base years to calculate historical catch ensures that fishers cannot strategically alter their behaviour to manipulate initial quota allocations. Even if the policy is anticipated, the timing of implementation is plausibly exogenous to short-run individual outcomes, given the lengthy administrative process required to establish these regimes.

While control units may be affected through spillover effects or changes in stock abundance, fishers are still subject to fleet shares, where each fleet cannot collectively catch over a pre-defined amount of the total allowable catch. Spillover effects are further mitigated by the institutional separation across fleets and management areas, which limits direct competition over the same local stocks. For unregulated species, this may be violated if treated fishers sell their quotas and begin to fish other species from the portfolios of the control group with a greater intensity. In practice, though, this appears uncommon: sellers of quotas typically exit the fishery altogether, potentially because existing gear investments make it suboptimal to immediately shift effort toward alternative species.

The model is estimated separately for the 2003 and 2012 ITQ regulations to avoid the complications associated with staggered adoption. In particular, the 2003 cohort cannot serve as a control post-treatment yet before the 2012 treatment. Estimating them independently also allows us to evaluate whether the impacts are similar or heterogeneous across the two settings, as these ITQ regimes were implemented in different fleets and under different management conditions. In this case since

[7] It is important to note the difference between this specification and a quantile regression. A quantile regression estimates the effect of ITQ exposure at different points of the conditional distribution of revenues, examining whether the treatment effect differs for low, median, or high outcome individuals. Our approach demonstrates average effects within pre-defined groups based on pre-treatment characteristics. Quantile regression can reveal whether ITQs disproportionately benefit the highest-earning fishers, for example, but it does not identify which specific individuals receive these gains, which is precisely what we are interested in.

each regulation is introduced in a single year for a single cohort, the resulting design fits a conventional difference-in-difference framework in which two-way fixed effects provide valid estimates and avoid the complications associated with staggered adoption. To examine dynamic adjustment to the regulations however, we estimate separate and pooled event-study specifications.

3.5.2 Event Study

First, the separate event studies demonstrate the evolution of the treatment effect over time. We look at 3 periods before the regulation and 6 periods after, as these are the years where we have individuals from both fleets. The specification is as follows, with $\bar{t}_j$ representing the year of regulation for the two fleets:

$$y_{ijt} = \sum_{d=-3, d\neq -1}^{6} \beta_d \mathbb{1}\big(t - \bar{t}_j = d\big) \times Exp_i + \gamma_t + \alpha_i + \epsilon_{ijt} \tag{3.6}$$

Next, we estimate a pooled event-study specification in which both regulations are analyzed jointly. In this setting, treatment timing is staggered, and we implement the Callaway and Sant'Anna (2021) estimator to obtain dynamic average treatment effects. For the 2003 regulation, the valid controls are the 2012 treatment cohort and the never treated, and for the 2012 regulation the valid controls are only the never treated. This allows us to directly compare our separate-regulation estimates with those produced by a modern difference-in-difference estimator designed for staggered adoption, providing a transparent robustness check. The main limitation is that the Callaway and Sant'Anna framework requires a binary treatment indicator and cannot incorporate continuous treatment intensity. As a result, these pooled event-study estimates assume equal treatment intensity. Nonetheless, these pooled results provide complementary evidence on the timing and magnitude of the policy's effects.

3.5.3 Validity of the Continuous Treatment Difference-in-Differences Design

Identification in this setting requires that revenue trends do not systematically differ across fishers with higher versus lower baseline exposure to the regulated species. In a continuous treatment difference-in-difference model, exposure must not proxy for underlying differences in revenue trends in the years prior to regulations. To assess whether exposure shares predict outcome trends prior to the introduction of the ITQ, we regress pre-regulation changes in revenues and effort on the exposure measure using only pre-treatment years:

$$\Delta \log(y_{ijt}) = \theta Exp_i + \epsilon_i$$

Here $\Delta \log(y_{ijt}) = \log(y_{ij,t=2002}) - \log(y_{ij,t=2000})$. For both the 2003 and 2012 ITQ introduction, there is no suggestive evidence of pre-trends where θ is statistically insignificant at zero in all specifications. Incomes are highly volatile yet this is true for raw and winsorized revenues. Note that although our treatment intensity is constructed from baseline revenue shares, this does not pose an identification concern. By construction, exposure is mechanically correlated with pre-period *levels* of revenue, but the model framework requires only that fishers with higher baseline exposure do not exhibit systematically different pre-treatment *trends* in revenues relative to lower-exposure fishers. This is also a complementary test before running the event-study, which can also be used to visually examine pre-trends.

Finally, we regress pre-treatment observables on the exposure measure as a balance check. Since the matching procedure already restricts the sample to a control group with comparable baseline characteristics, we find no correlation between exposure and these covariates as expected. Exposure is not systematically correlated with vessel size, suggesting no pre-existing systematic differences in capital.

Together, the low concentration of exposure shares and the relative absence of differential pre-trends provide empirical support for the validity of our identification strategy.

3.6 Results

3.6.1 Intensive Margin: Revenues and Effort

Here we include our OLS results for the regulation implemented in 2003. The first column for each outcome includes all fishers[8] and the second column uses the result of the preferred matching method.[9]

Table 3.4 demonstrates the average effect of ITQs on revenues among active fishers. We find that an additional $1 of exposure increases annual revenues by $1.28 - $1.58. This represents the expected revenue increases post-ITQ. We find no effect of ITQ exposure on total fishing days, suggesting that revenue gains are not driven by more days at sea, however this does not rule out reductions in effective effort along other unobserved margins such as distance traveled.

Table 3.4: Intensive Margin Results: 2003 Overall

	Total Revenues		Total Fishing Days	
	(1)	(2)	(3)	(4)
ITQ Exp	1.278***	1.580***	0.000	0.000
	(0.250)	(0.279)	(0.000)	(0.000)
Individual FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Observations	21,664	6,360	21,664	6,360
R^2	0.77	0.71	0.75	0.63

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ The first column for each outcome reflects the full sample, the second column restricts the sample to the matched fishers.

[8]Fishers who purchase quota later or are in the other treatment group are still excluded.

[9]These results are robust to adjusting the parameters of the matching method so we present the most flexible here

Table 3.5 examines heterogeneity in revenue outcomes across the skill distribution. Revenue gains are concentrated among low- and medium-skill fishers, who experience increases of approximately \$1.43–\$2.08 and \$1.24–\$1.56 per dollar of exposure, respectively. The effect for high-skill fishers is statistically insignificant yet negative, indicating no comparable revenue gains.[10] These results are consistent with a flattening of the skill gradient in earnings among those remaining in the fishery. Again, we find no effect on observable effort.

Table 3.5: Intensive Margin Results: 2003 Skill Distribution

	Total Revenues		Total Fishing Days	
	(1)	(2)	(3)	(4)
ITQ Exp × Skill: Low	1.432***	2.080***	0.000	0.000
	(0.230)	(0.294)	(0.000)	(0.000)
ITQ Exp × Skill: Medium	1.244***	1.558***	0.000	0.000
	(0.309)	(0.326)	(0.000)	(0.000)
ITQ Exp × Skill: High	-6.188	-3.625	-0.002	-0.002
	(6.427)	(7.287)	(0.004)	(0.004)
Individual FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Observations	16,142	6,333	16,142	6,333
R^2	0.73	0.72	0.74	0.64

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ The first column for each outcome reflects the full sample, the second column restricts the sample to the matched fishers.

Finally, Table 3.6, presents the results along the outside option distribution. Fishers operating in regions with weaker labor markets experience larger revenue gains from ITQ exposure of \$2.27–\$2.731 compared to those in stronger labor markets who gain from \$0.938–\$1.274, however both are significant and positive. These results indicate that revenue gains accrue to fishers in more remote resource-dependent labor markets, consistent with a compensating effect for those with weaker outside options.

Taken together, these findings show that on the intensive margin, the 2003 ITQ regulation increased fishing revenues, with gains distributed unevenly across skill levels and local labor market conditions. Since the revenue gains on the intensive margin are concentrated on the lower end of the skill distribution, these patterns are consistent with a reduction in within-fishery income dispersion.

[10] It is possible this driven by the limited number of high-skilled treated fishers, which reduces the identifying variation in this subgroup.

Table 3.6: Intensive Margin Results: 2003 Outside Option Distribution

	Total Revenues		Total Fishing Days	
	(1)	(2)	(3)	(4)
ITQ Exp × Labour: Low	2.271***	2.731***	0.000	0.000
	(0.486)	(0.558)	(0.000)	(0.000)
ITQ Exp × Labour: High	0.938***	1.274***	0.000	0.000
	(0.301)	(0.319)	(0.000)	(0.000)
Individual FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Observations	19,035	6,028	19,035	6,028
R^2	0.73	0.70	0.74	0.64

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ The first column for each outcome reflects the full sample, the second column restricts the sample to the matched fishers.

For the 2012 regulation, we estimate the same specifications and find no significant effects, shown in Table 3.7, Table 3.8, and Table 3.9. We find no economically meaningful average revenue effects, nor do we observe differential effects across the skill distribution. There is evidence that fishers operating in areas with stronger labor market conditions experienced revenue declines following the 2012 ITQ, while revenue effects for fishers in weaker labor markets are positive yet insignificant. Again, these are intensive margin revenues and may capture voluntary reallocation away from fishing in response to opportunity costs. The absence of strong effects may reflect lower average exposure to the policy, a smaller fleet share of the regulated species, or differences in underlying structural dynamics.

Table 3.7: Intensive Margin Results: 2012 Overall

	Total Revenues		Total Fishing Days	
	(1)	(2)	(3)	(4)
ITQ Exp	0.070	0.724	0.000*	0.000
	(0.439)	(0.487)	(0.000)	(0.000)
Individual FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Observations	21,804	5,099	21,804	5,099
R^2	0.77	0.72	0.74	0.60

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ The first column for each outcome reflects the full sample, the second column restricts the sample to the matched fishers.

Table 3.8: Intensive Margin Results: 2012 Skill Distribution

	Total Revenues		Total Fishing Days	
	(1)	(2)	(3)	(4)
ITQ Exp × Skill: Low	-7.229	-4.706	-0.001*	-0.001*
	(4.049)	(4.004)	(0.001)	(0.001)
ITQ Exp × Skill: Medium	0.203	0.836	0.001*	0.000
	(0.524)	(0.567)	(0.000)	(0.000)
ITQ Exp × Skill: High	-0.141	0.403	0.000	0.000
	(0.391)	(0.512)	(0.000)	(0.000)
Individual FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Observations	19,227	5,099	19,227	5,099
R^2	0.74	0.72	0.72	0.60

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ The first column for each outcome reflects the full sample, the second column restricts the sample to the matched fishers.

Table 3.9: Intensive Margin Results: 2012 Labour Distribution

	Total Revenues		Total Fishing Days	
	(1)	(2)	(3)	(4)
ITQ Exp $\times$ Labour: Low	0.128	0.827	0.000	0.000
	(0.457)	(0.507)	(0.000)	(0.000)
ITQ Exp $\times$ Labour: High	-1.490***	-0.725*	0.001	0.001
	(0.278)	(0.403)	(0.001)	(0.001)
Individual FEs	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes
Observations	18,828	5,056	18,828	5,056
R^2	0.74	0.72	0.72	0.60

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ The first column for each outcome reflects the full sample, the second column restricts the sample to the matched fishers.

Figure 3.8 presents dynamic average treatment effects estimated using the (Callaway & Sant'Anna, 2021) framework, which accounts for the staggered introduction of the 2003 and 2012 ITQ regulations and uses appropriate not-yet-treated units as controls. The coefficients trace out the evolution of revenues relative to the year of treatment, or event time = 0. The post-treatment trajectory shows a persistent and increasing positive effect on revenues, consistent with a gradual adjustment process following ITQ implementation. The wide confidence intervals in later years reflect fewer overlapping observations for the two cohorts. Importantly, because this framework treats the policy as a binary intervention, it abstracts from heterogeneity in treatment intensity. These event-study magnitudes correspond to the effect of a full intensity binary treatment and mechanically appear larger than the continuous-treatment estimates reported earlier. Given the results from the separate baseline regressions, these dynamic revenue gains are likely driven primarily by the 2003 regulation rather than the 2012 reform.

3.6.2 Extensive Margin

Here we present the extensive-margin results, which examine whether ITQs induce exit from the fishery. Figure 3.9 plots the cumulative share of remaining fishers in the treatment and control groups, normalizing the number of individuals in 2003 to one. While there is a baseline rate of exit in both groups, exit is systematically higher among fishers exposed to the ITQ regulation.

The regression results are presented in Table 3.10. Overall, ITQs induce exit from the fishery as expected. ITQs are shown to disproportionately induce exit among lower-skill fishers. This pattern is consistent with efficiency-enhancing consolidation through the reallocation of activity from marginal fishers toward more productive participants. Exit rates are relatively similar across labor markets, sug-

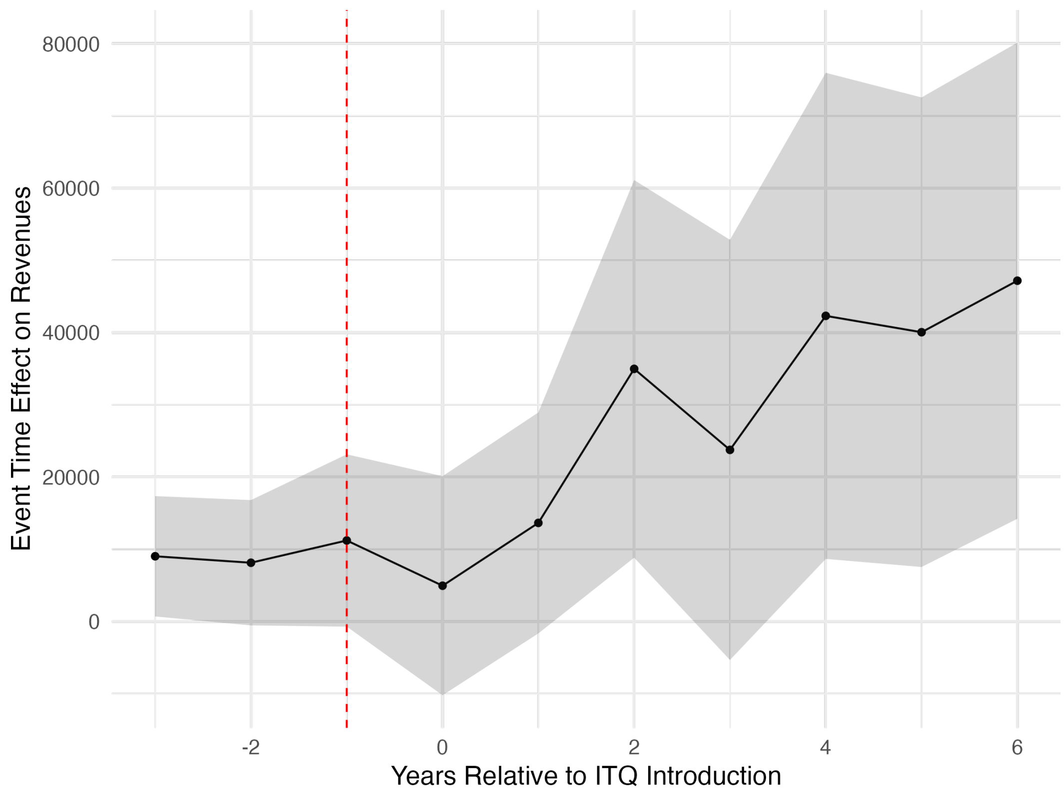

Figure 3.8: Event Study Results for the Pooled Regression

gesting that consolidation is driven more by productivity differences than by differential reallocation opportunities. Thus, while exit is not disproportionately concentrated in rural areas, fleet rationalization represents a pervasive adjustment throughout the sector.

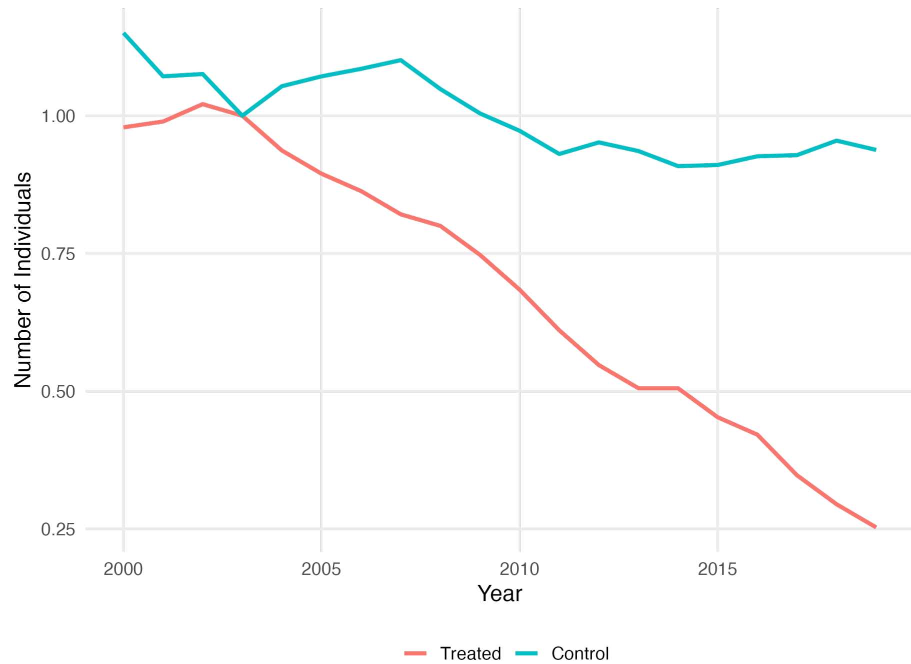

Figure 3.9: Number of Active Fishers in Treatment and Control (2003 = 1.00)

Table 3.10: Extensive Margin Regression Results

	Active = 1					
	(1)	(2)	(3)	(4)	(5)	(6)
ITQ	-0.335***	-0.059				
	(0.034)	(0.038)				
ITQ × Skill: Low			-0.091**	-0.071*		
			(0.036)	(0.039)		
ITQ × Skill: High			0.000	0.020		
			(0.110)	(0.111)		
ITQ × Labour: Low					-0.356***	-0.102
					(0.067)	(0.071)
ITQ × Labour: High					-0.358***	-0.177*
					(0.090)	(0.090)
Individual FEs	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes
Observations	54,100	9,700	27,560	9,700	31,940	4,920
R^2	0.49	0.59	0.60	0.59	0.50	0.58

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ The first column for each specification reflects the full sample, the second column restricts the sample to the matched fishers.

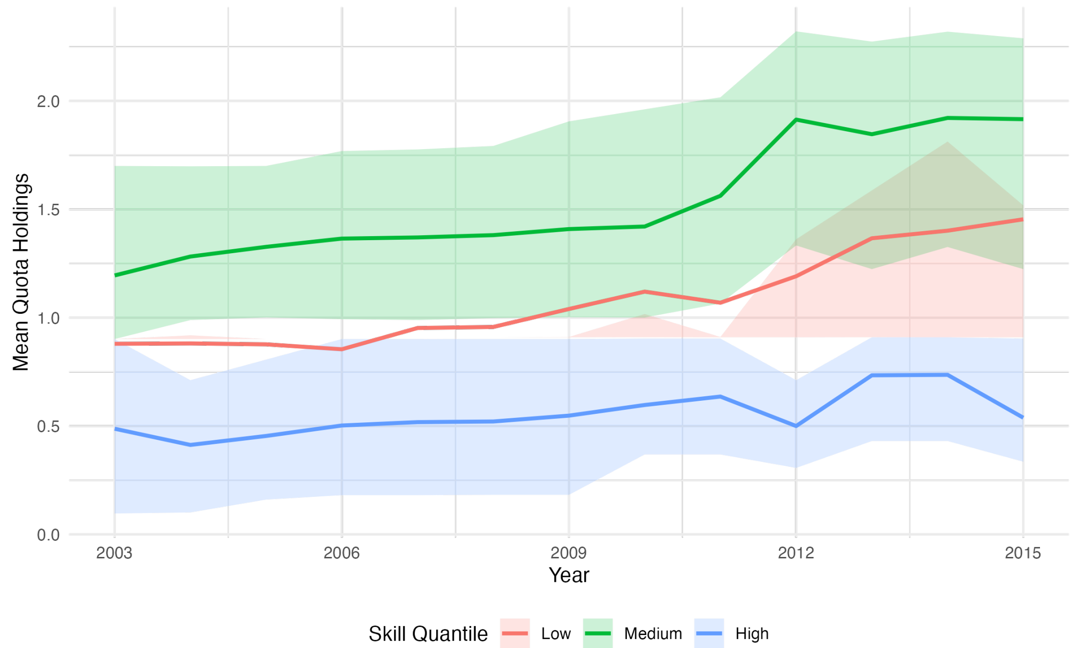

Figure 3.10: Quota Holdings by Skill Quantile

3.6.3 Quota Holdings

Next, we examine quota holdings across each distribution over time. In the introduction year, the highest quota percentage held was around 2.5 percent. Over the years this increased to almost 4.5 percent, with many individuals holding over 3 percent.

Figure 3.10 plots mean quota holdings over time by skill group, with shaded bands indicating the interquartile range. On average, high-skill fishers do not accumulate additional quota following the introduction of ITQs, but rather the low and medium skill fishers increase their mean quota holdings over time. This reallocation provides a mechanism for the observed compression of the skill gradient in revenues, however widening interquartile range shows high variation among groups.

Figure 3.11 shows analogous patterns by outside-option group. Fishers operating in weaker labor markets experience a larger increase in mean quota holdings over time than those in stronger labor markets, indicating that quota reallocation favors participants with weak outside options. While this suggests that fishing communities did not lose an important source of income, however there is considerable dispersion.

3.7 Discussion

Overall, ITQs reduce between-skill inequality in average revenues by concentrating gains among lower skilled fishers. Revenue gains are also disproportionately larger for fishers with weak outside options, indicating revenues are not leaving remote communities for active fishers. On the extensive margin, ITQs induce selective exit among low-skill fishers suggesting efficiency gains. Although exit does not differ systematically across the outside option distribution, fleet rationalization affects fishers across

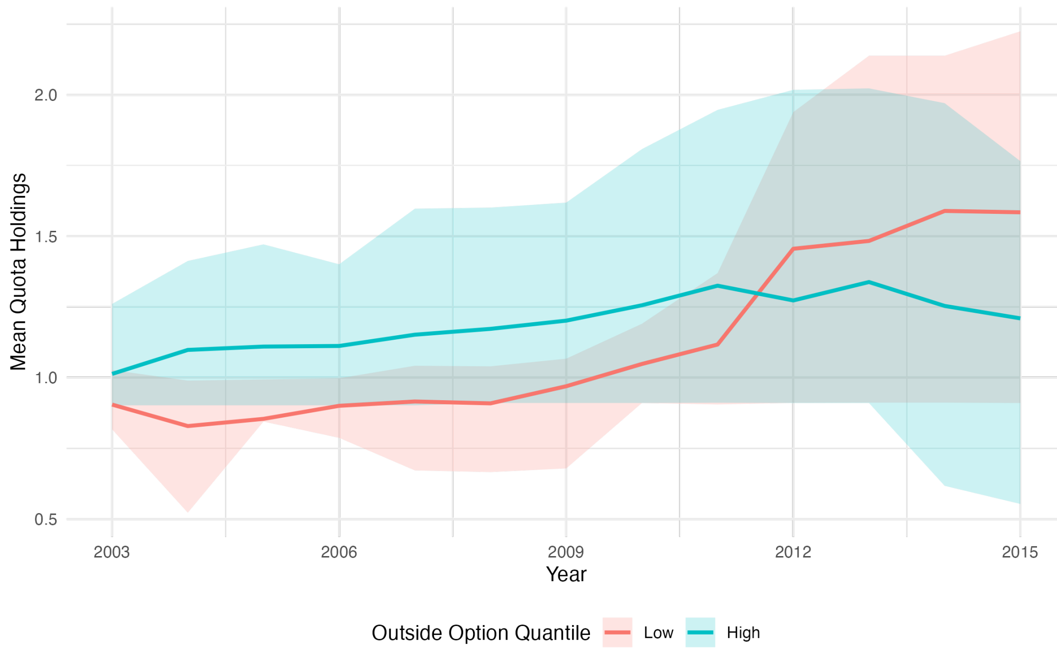

Figure 3.11: Quota Holdings by Labour Market Quantile

all labor markets, so the overall equity implications depend on the post-exit outcomes of displaced participants.

Our results align more closely with the predictions of Grainger and Costello (2016) than with those of Okonkwo and Quaas (2020). Adjustment to ITQ adoption operates primarily through skill-based selection: lower skill fishers are more likely to exit, while remaining lower skilled fishers experience the largest revenue gains. High-skill fishers exhibit stable participation and revenues, consistent with a compression of returns as inframarginal rents are converted into shared resource rents. In contrast, we find little evidence that outside-option heterogeneity governs exit responses to the extent predicted by Okonkwo and Quaas (2020). Exit is not concentrated in weak labor markets, but rather revenue gains are larger for fishers in weaker labor markets. These patterns indicate that heterogeneity in fishing skill, rather than outside options, is the dominant margin shaping post-ITQ adjustment in our setting. The combination of grandfathered quota allocation, caps on individual holdings, and limited geographic mobility emphasizes the skill-based compression mechanism.

While our results shed light on adjustment to the transition toward property rights in maritime Québec, their broader policy implications remain to be explored. In particular, outcomes may be shaped by institutional features of Québec's ITQ program, including caps on individual quota holdings that limit the extent of consolidation. A natural extension of our analysis would be a comparison to British Columbia fisheries where there exists no upper bounds on quota ownership, to study how unrestricted reallocation affects efficiency and inequality. These findings should be valid within the institutional context of Québec's fishing industry and therefore informative for future ITQ implementations in the region. At the same time, aggregate effects are likely to be larger in fisheries where the regulated species represent a greater share of fishers' revenue portfolios. In our setting, groundfish account for a relatively small fraction of total catch, which may attenuate both intensive and extensive margin responses. In addition, limited information on the extent of quota leasing and quota prices constrains our ability to fully characterize market dynamics and rent redistribution.

We leave for future work the incorporation of richer labor market measures at finer spatial scales and further investigation of the mechanisms underlying exit and reallocation. It is important to note that similar exit rates across labor markets do not imply similar welfare consequences of exit. Even if exit is not disproportionately concentrated among fishers with weak outside options, those who exit from low outside-option regions may experience substantially larger income losses than those who exit from stronger labor markets. Without observing post-exit earnings, we cannot assess whether ITQ-induced exit generates unequal welfare losses across labor markets. Our results therefore speak to the distribution of exit behavior, but not necessarily to the distribution of post-exit outcomes. We also abstract from general labour market equilibrium effects, including whether increased exit from fishing affects local wages through changes in labor supply, or whether sufficiently large income losses induce migration away from remote communities. Finally, for future entrants to the industry, quota purchases may represent an important barrier to entry.

Future work is also needed to understand the mechanisms driving the results, namely changes in fisher behaviour. Fishers may adjust along several intensive margins, including the distance traveled, fishing location, timing of trips, targeting of specific species, and the composition of their catch portfolios. ITQs may reduce effective effort by lowering congestion, shortening trips, or reallocating effort toward higher-margin activities. Our results therefore capture the net effect of these adjustments on revenues and participation, but cannot disentangle the relative importance of changes in spatial behavior, species targeting, or capital utilization. Quantifying these mechanisms is important for the long-run success of these policies.

3.8 Conclusion

In this paper, we provide an empirical analysis of the distributional impacts of individual transferable quotas in Maritime Québec. We leverage the introduction of the regulation for Greenland Halibut in 2003 for Gaspésie and then in 2012 for Côte-Nord to study how the transition to property rights affects revenues, effort, and activity. Using a comprehensive dataset covering the universe of catch and revenue outcomes, we offer new evidence on how ITQs affect fishers with heterogeneous productivity and heterogeneous outside labor market opportunities.

Our results show that the 2003 ITQ led to significant revenue gains, with the largest increases accruing to fishers in the lowest skill quantiles and those with the lowest outside options. At the same time, while lower-skill fishers were also more likely to exit in the post-ITQ period, exit was not concentrated among fishers facing weak outside options, suggesting industry-wide fleet rationalization. While there is evidence within-fishery inequality decreased, the overall equity implications depend on the post-exit outcomes of displaced participants. Importantly, quota holdings did not systematically shift away from remote communities, indicating that resource rents largely remained local in the medium run. The weaker results for the 2012 reform highlights that the distributional and efficiency consequences of ITQs depend critically on institutional design, fleet structure, and the economic importance of the regulated species within fishers' revenue portfolios, rather than on the adoption of property rights alone.

As more Canadian fisheries move towards ITQ management schemes, quantifying their impacts is crucial in informing policy to limit economic losses, ensure prospering fishing communities, and allow for long-term sustainability of the fishing industry in Canada. Carefully designed ITQ policies can raise aggregate revenues and support lower-skilled fishers without disproportionately harming vulnerable fishing communities.

Chapter 4

Mapping Risk: Predicting the Geography of the Potential Orphan Well Burden

4.1 Introduction

Orphan oil and gas wells represent a growing environmental and fiscal liability across the United States. Without proper plugging, wells abandoned by their operators pose serious risks to air, water, and soil quality, and place significant cleanup costs on state governments (Kang et al., 2023). In addition to methane leaks and water contamination, the presence of these non-producing wells reduce local infrastructure investment and disrupt ecosystems (Harleman et al., 2022; Mitchell & Casman, 2011; Nallur et al., 2020). Historical production has left a large and costly landscape of unmanaged legacy wells that policymakers are actively trying to address (Boutot et al., 2022). Pennsylvania alone is home to more than 300,000 documented unplugged wells with tens of thousands currently classified as orphaned or abandoned, and many more idle with uncertain futures (Jahan et al., 2025). The scale of the problem has prompted unprecedented public investment, and under the 2021 Bipartisan Infrastructure Law, Pennsylvania received more than $25 million for well-plugging programs (Pennsylvania Department of Environmental Protection, 2024). Yet despite these increased clean-up efforts, policymakers still lack credible forecasts of how many additional wells are likely to become orphaned in the coming decades and where those liabilities will arise. Preventing the accumulation of additional orphan wells is critical to mitigating environmental risk and future public expense.

This paper addresses that gap by developing a predictive framework to identify the subset of Pennsylvania's existing wells most at risk of becoming orphaned. Using detailed well-level production and ownership data, I first construct a naive, production-based classification of the current well stock. I aggregate these categories to the operator level to highlight the concentration of high-risk wells among small, low-production firms. This provides a transparent depiction of the state's aging well portfolio. Second, I estimate the probability that a given well becomes orphaned within a fixed time horizon by using a logistic model that incorporates well age, cumulative production, recent production decline, operator size, drilling vintage, and market conditions. The results show that orphaning risk is driven primarily by structural features of well life cycles and operator scale. Finally, I map the spatial distribution of these high-risk wells to provide a statewide view of emerging environmental liabilities. These at-risk wells are disproportionately located in older conventional oil fields in northwestern and southwestern Pennsylvania, rather than in the newer shale gas regions of the state. This spatial pattern suggests that revenues from the Act 13 impact fee are not directed toward the regions facing the greatest future orphan-well liabilities since the fee is assessed only on active unconventional wells for the first fifteen years.

By comparing forecasts of future orphan-well risk with information on local revenue allocations, this paper highlights how environmental liabilities are distributed across Pennsylvania. The results highlight how existing policy frameworks may leave portions of the state exposed to long-run cleanup obligations that are not matched by current revenue streams. More broadly, the framework developed here offers a scalable approach for other producing states to anticipate and manage long-term liabilities from the industry.

The remainder of the paper is organized as follows. Section 4.2 provides background on well life cycles and the regulatory and fiscal context of orphan wells in Pennsylvania. Section 4.3 outlines the operator's decision to abandon a well. Section 4.4 presents the data sources, Section 4.5 introduces the empirical framework for predicting orphanhood risk and Section 4.6 provides the results. Section 4.7 discusses the fiscal alignment and compares the results to the current orphan well inventory, and Section 4.8 concludes.

4.2 Background

4.2.1 Well Status Classification in Pennsylvania

A well is considered active when it produces positive quantities of oil or gas, or zero quantity with expected future production. If a well ceases production for twelve consecutive months but remains under an operator's control, it is classified as inactive, which can be authorized for up to five years. After this initial period, PADEP can grant extensions on a year-to-year basis, and failure to obtain an extension requires the well to be returned to active use or permanently plugged as inactivity is intended to be a temporary state[1]. Wells that cease production and do not receive inactive status are considered abandoned. Abandonment may occur for a variety of reasons including economic, mechanical, or changes in the owner's financial viability. While an abandoned well may still have a legally responsible owner and receive limited oversight, the end of active operations increases the risk of deterioration, regulatory noncompliance, and increases the likelihood that the well ultimately becomes an orphan. When an operator officially ceases production, the well must be properly decommissioned. Decommissioning refers to the permanent retirement of an oil or gas well, which entails plugging the wellbore to prevent fluid migration and restoring the surface site. Decommissioning costs vary systematically with well depth, age, type, surface characteristics, and location (Raimi et al., 2021). The large upfront cost of decommissioning incentivizes well operators to postpone plugging, either temporarily or indefinitely.

[1] 58 Pa. Cons. Stat. § 3203

Discrete regulatory categories of well status offer limited information, as wells lie along a continuous spectrum of production, age, ownership, and operational activity. At one end are active wells that remain under private-firm responsibility and at the other are orphan wells, which represent realized public liabilities. Between these extremes lies a substantial stock of inactive or abandoned wells that are still legally owned but vary widely in maintenance and environmental risk (Dilmore, Sams III, Glosser, Carter, & Bain, 2015). Abandoned wells present the same burden as orphan wells, differing only in the state's ability to assign legal responsibility. Accordingly, this paper focuses on the set of unplugged wells, active or inactive, as the universe of potential future liabilities. By classifying these wells according to their probability of transitioning into abandonment or orphanhood, the analysis identifies where Pennsylvania's next generation of environmental liabilities is likely to emerge.

4.2.2 Production Profiles and the Composition of Pennsylvania's Well Stock

The production lifecycle of an oil or gas well follows a characteristic decline pattern that shapes both its economic viability and its eventual environmental legacy. Conventional and unconventional wells differ sharply in the magnitude and timing of that decline, with important implications for the trajectory of orphan well liabilities in Pennsylvania.

Conventional wells dominated Pennsylvania's drilling landscape for over a century, typically exhibiting low initial production rates followed by a long gradual exponential decline. The large number of low output historical wells comprises a large portion of the current official orphan well inventory. Unconventional wells, by contrast, exhibit a dramatically different production profile where horizontal drilling techniques generate high initial output followed by a steep decline (Carter, Harper, Schmid, & Kostelnik, 2011). As a result, unconventional wells have much shorter productive lifespans, and nearly all new wells since 2010 fall into this category. This shift creates ambiguous implications for future orphaning, where faster decline may hasten decommissioning and limit long-run orphan growth if obligations are met, but the large stock of high-cost, rapidly declining wells also raises the risk of a future wave of orphaning as revenues fall quickly. Plugging decisions arise much sooner for unconventional wells due to the compressed window between peak revenue and end-of-life.

From a policy perspective, Pennsylvania's historical legacy ensures that the current orphan problem is concentrated in older conventional fields, while the emerging risk lies with newer unconventional gas wells (Dilmore et al., 2015). The temporal and spatial overlap of these two periods with legacy oil in the northwest and modern gas in the northeast and southwest, results in the challenge of managing a backlog of legacy liabilities while anticipating a shorter-cycle wave of future liabilities (Carter et al., 2011). Understanding the contrasting production dynamics of these well types is therefore essential for forecasting both the timing and geography of Pennsylvania's long-run environmental burden.

4.2.3 Entry, Production, and Exit

Operators choose where and when to drill, and then extract the resource, taking prices as given. In the canonical Hotelling framework, oil and gas extraction decisions are governed by an intertemporal tradeoff between extracting a finite resource today and leaving it in the ground for future extraction (Hotelling, 1931). This framework implicitly assumes that producers can freely adjust extraction paths over time in response to price incentives. However, Anderson, Kellogg, and Salant (2018) emphasize that economic decisions in oil and gas production are governed primarily by the discrete decision to drill or not, rather than continuous adjustment of quantity each period that the well is active. Production from existing oil wells is largely constrained by reservoir pressure and mechanical decline, so there is little room for optimization along the intensive margin. Unlike in standard Hotelling models, producers cannot profitably defer extraction in anticipation of higher prices. As a result, they show that price dynamics primarily affect drilling decisions with little influence on production timing or well retirement, with exceptions for extremely low prices. This can also apply to expected prices and price uncertainty, where firms delay irreversible drilling until expected returns exceed a threshold that rises with uncertainty (Kellogg, 2014).

However, the extensive margin also includes the end-of-life decision once wells approach economic exhaustion. At this stage, firms must choose whether to continue operating, temporarily idle the well, or permanently plug. Muehlenbachs (2015) shows that temporary inactivity often reflects a strategic delay of costly and irreversible cleanup rather than a meaningful option to resume production. Postponing preserves both the opportunity cost of funds and the option value associated with waiting for price increases or technological changes. Delay may be privately optimal for the operator but socially costly through environmental risks, foregone land uses, and heightened probability that liabilities ultimately fall on the state. Furthermore, price volatility, learning, and favorable shocks do little to revive inactive wells, implying that regulatory design governs the timing of decommissioning.

A well may become uneconomical before production reaches zero, once fixed operating costs and regulatory obligations are taken into account, yet operators can postpone plugging by producing small quantities to preserve active status or in anticipation of improved future conditions. Weber et al. (2021) derive a minimum production threshold below which wells are unlikely to be viable even under favorable prices, showing that regulatory frameworks classifying wells as active based on any positive production can allow uneconomic wells to remain open indefinitely.

Together, these papers imply that oil and gas markets are best understood as threshold-based decisions over drilling and exit, while production paths are determined by physical constraints rather than forward-looking price optimization. Consequently, abandonment and plugging decisions are governed primarily by regulatory incentives rather than market forces, making the management of well decommissioning fundamentally a policy problem rather than a market-driven outcome.

4.2.4 Fiscal and Regulatory Environment

As well decommissioning is governed by regulation, the fiscal institutions that collect drilling-related revenues and finance environmental remediation play a central role in shaping long-run outcomes.

Act 13 of 2012 reshaped the fiscal treatment of unconventional gas development in Pennsylvania. Rather than adopting a typical severance tax on the volume of extracted gas, the legislature created an annual "impact" levied on each unconventional well that is drilled or producing in a given year. The revenues are distributed at the county level, with allocations based on the number of unconventional wells adjusted for population and other factors. In recent years, annual collections have been around \$150–\$250 million. The fee schedule depends on well age and the annual average natural gas price with declining obligations over time. The impact fee is assessed annually for only the first fifteen years after a well is spud and does not depend on output. While the fee generates substantial short-run revenue, (Black et al., 2018) show it primarily affected leasing rather than drilling behavior, raising concerns about whether it adequately addresses long-run liabilities from aging wells. Rabe and Borick (2013) argue that Act 13 prioritized rapid resource extraction and short-term economic gains over environmental protection and liability. Rather than adopting precautionary policy tools, the impact fee is low and present biased, raising questions about the durability of Pennsylvania's approach.

A separate but related instrument is well bonding, which is intended to ensure that operators internalize at least a portion of future plugging and reclamation costs. Operators must post bonds, with amounts varying by well type and operator scale. For conventional wells, firms may post a relatively modest per-well bond or a blanket bond covering all wells up to a capped amount, while for unconventional wells the required bond depends on total wellbore length and the number of wells operated, with statutory caps that keep total bonding obligations well below estimated plugging costs. Recent analyses have concluded that these bond amounts fall far short of the expected costs of plugging and site restoration, particularly for older conventional wells, implying substantial residual risk for taxpayers if operators default or dissolve (Boomhower, 2019; Mitchell & Casman, 2011).

This paper addresses the spatial and temporal aspects of this fiscal policy. First, by forecasting which active and inactive wells are most likely to become orphaned, I provides a forward-looking measure of expected environmental liabilities at the county level. Second, I assesses the degree of spatial misalignment between where the state collects drilling-related revenues and where long-run cleanup costs will fall.

4.3 Operator Decision to Decommission

This section outlines operator behavior as a discrete choice over well status, in line with Anderson et al. (2018). I abstract from continuous production choice, treating production paths as given conditional on operation. I introduce a simple framework following Weber et al. (2021) that characterizes the operator's decision to remain active, idle, or permanently decommission a well. This perspective informs my empirical analysis in identifying and predicting future uneconomic wells.

For a given well i at time t, a well faces output price p_t. If the well is operating, production q_{it} generates revenue $p_t q_{it}$. There are fixed costs F_{it} and variable operating costs which are a function of quantity $C_{it}(q_{it})$. Decommissioning requires a one-time plugging fee of P_i. Operators discount future revenues by β and V_{it} is the continuation value of keeping the well unplugged. Each period the operator chooses to keep the well active and producing q_{it} (determined outside of the model), idle with $q_{it} = 0$, or plug the well.

Active Given an operator chooses to operate, revenues are

$$\pi_{it}^A = p_t * q_{it} - C_{it}(q_{it}) - F_{it}$$

.

A breakeven production value Q^* can be defined such that producing $q_{it} < Q^*$ leads to negative revenues. Given that any non-zero production is sufficient to maintain active status, operators may choose to operate at a loss if they anticipate more favourable conditions to preserve the continuation value of the well:

$$p_t * q_{it} - C_{it}(q_{it}) - F_{it} + \beta V_{it} \geq -P_i$$

Idle As the well nears the end of its life, variable costs can increase if it is harder to extract the diminishing resource. Hence the revenues may not cover the variable costs, yet the operator may choose to incur the fixed costs to avoid plugging:

$$-F_{it} + \beta V_{it} \geq -P_i$$

Leaving a well unplugged also leaves the option open to produce in the future. Wells that remain idle indefinitely are functionally abandoned, but regulatory frameworks can permit this status without triggering decommissioning.

Plugged Plugging generates no revenue and eliminates all future revenue opportunities. The operator pays P_i, and $V_{i,t+1} = 0$.

Essentially this can be summarized with the following value function:

$$V_{it} = \max \left\{ \pi_{it}^{A} + \beta V_{i,t+1}, \pi_{it}^{Idle} + \beta V_{i,t+1}, -P_i \right\}.$$

Taken together, the model clarifies the incentives underlying operators' decisions. First, higher plugging costs unambiguously reduce the likelihood of decommissioning. Second, higher current prices raise the value of keeping a well nominally active for the current and future revenues, making operators more likely to continue producing small quantities rather than initiate plugging. Once again, prices do not alter the extraction path of an operating well but can affect the operator's decision to keep a well active at the end of its life. Finally, greater optimism about future conditions from anticipated higher prices, falling operating costs, or regulatory leniency, increases the continuation value of the well and further slows decommissioning.

Assuming a deterministic production path, quantity in this setting essentially reflects the well's stage in its life cycle rather than a decision variable. High-producing wells will almost always remain active, and production becomes increasingly relevant for older wells as revenues decline relative to costs. Heterogeneity in well characteristics, geology, and operating conditions, implies that economic exhaustion does not occur at a common age across wells. The economic life of a well varies substantially across operators and locations because the cost function is highly well-specific and non-linear in age. An output price that is economical for one well may be highly uneconomic for another.

My empirical approach effectively identifies wells for which the continuation value implied by observed production and costs is sufficiently low that abandonment or orphaning becomes likely under existing regulatory incentives.

4.4 Data

4.4.1 Oil and Gas Production Reports

The oil and gas production data comes from Pennsylvania's Department of Environmental Protection (PADEP). I build on the compiled dataset constructed in Chapter 2, which aggregates and merges annual, semi-annual, and monthly production reports over 1980–2024. I create a balanced panel under the assumption that any post-spud year with missing production data reflects idleness, unless a plugging date is reported. I restrict the sample to unplugged oil, gas, or combined oil and gas wells. Natural gas is reported in thousand cubic feet (MCF) and oil in barrels. I convert the natural gas quantity to barrels of oil equivalent (BOE) to directly compare production volumes across wells and operators. Unless otherwise stated, all quantities are in units of BOE. Each well in a given year can be described using well-specific characteristics, production variables summarizing current and historical activity, and operator characteristics constructed by aggregating and tracking the production of all wells managed by that operator. These variables are all constructed from the annual production recorded for each well and are described in detail in Table 4.1.

The majority of wells are non-producing at some point over the period of interest. Production decline is defined as the percentage change in production relative to a previous period. This measure becomes undefined when the lagged value is zero; however this could represent a transition from zero to zero, or zero to any amount of positive production. While these two cases are very different, representing either persistent inactivity or reactivation, the distinction is not relevant since periods of lagged zero production do not provide meaningful information on a well's depletion trajectory. Only when a well is producing does a percentage decline meaningfully signal diminishing reservoir pressure or resource exhaustion. I recode these values to zero so that the observations are retained in the regression.

Overall, production variables capture the key dimensions of a well's lifecycle and provide the foundation for the predictive analysis that follows.

Operators are of particular importance as the behavior and operational capacity of firms ultimately determine whether wells are maintained, transferred, plugged, or abandoned. Tracking operator characteristics over time such as number of wells they own, how many of each status, and production intensity, provides insight into their economic viability. Operators managing large, actively producing portfolios are more likely to have the resources to comply with plugging obligations, whereas small or declining operators face greater bankruptcy risk and may leave wells orphaned. There are over 3700 unique operators over time, yet in any given year a larger share do not own producing wells as shown in Figure 4.1. This is a defining feature of Pennsylvania's oil and gas industry.

Unconventional wells differ sharply from older conventional wells, with high initial output and rapid decline rather than low, stable production. This pattern reflects Pennsylvania's transition since around 2008 from small-operator vertical drilling to horizontally drilled shale wells that now dominate new development. As a result, today's landscape reflects an aging stock of legacy conventional wells and a newer, highly productive generation of horizontal shale wells as demonstrated in Figure 4.2. Given the compressed production cycle of unconventional wells, a key question is whether this accelerates

Table 4.1: Variables for Classification and Prediction

Variable	Description
Well Characteristics	
Unconventional	Unconventional indicator
Age Variables	
Age	Well age in years
Number of Productive Years	Number of years since first production
Production Variables	
Annual Production	Annual oil or gas production in BOE
Daily Production	Average oil or gas production per day in BOE
Production Days	Total oil or gas production days in the year
Production Decline	% change in production from previous year
Cumulative Production	Sum of production up to date
Total Production (5 yr.)	Total production over previous 5 years
Non-Zero Production (5 yr.)	Number of non-zero production years over previous 5 years
Operator Characteristics	
Number of Wells	Annual number of wells operated
Total Quantity	Total annual quantity (BOE) over all wells operated
Total Quantity (5 yr.)	Total production by operator over previous 5 years
Percentile	Annual rank of an operator's production among producing operators
Number of High-Producing Wells	Total annual active well count
Number of Low-Producing Wells	Total annual marginal well count
Number of Idle/Zero Production Wells	Total annual idle well count
Number of Orphan Wells	Total annual orphan well count

abandonment rates, or whether recent plugging efforts have successfully slowed the accumulation of new orphan wells, leaving older legacy wells as the main source of ongoing liability.

I also assemble annual crude oil and natural gas price series from the U.S. Energy Information Administration (EIA). For natural gas, I use the Henry Hub spot price, reported in dollars per million British thermal units (MMBtu), which is the standard benchmark for U.S. gas markets. For crude oil, I use the West Texas Intermediate (WTI) spot price at Cushing, Oklahoma, measured in dollars per barrel. For wells that produce both oil and natural gas in varying proportions, I construct a weighted price that reflects the relative importance of each resource for that well based on the production share. While the oil price series dates back to 1986, the gas price is only available beginning in 1997, so specifications that include price are restricted to observations from 1997 onward.

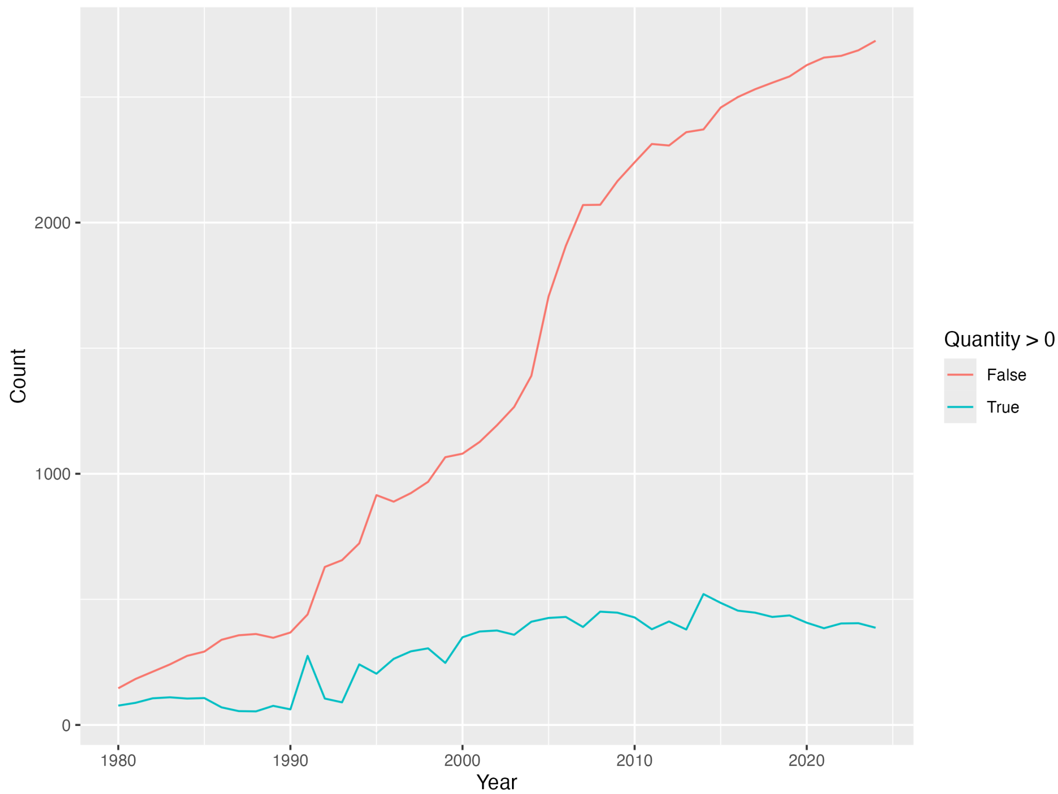

Figure 4.1: Producing and Non-Producing Operator Count Over Time

4.4.2 Act 13 Impact Fee

I compile annual impact fee collection and disbursement data from the Pennsylvania Public Utilities Commission (PUC). These data provide total fee payments by operator and detailed county and municipality level allocations by year since the onset of the regulation in 2012. The allocations are linked to the number of unconventional wells located within the respective boundaries, with a portion allocated statewide. I aggregate these data to the county-year level to measure total revenue received.

4.5 Empirical Strategy

4.5.1 Naive Classification

For a naive classification of the current stock of unplugged wells, I develop a rough screening rule for wells following Boomhower, Shybut, and DeCillis (2018). I assign each well to one of the following categories: likely already orphaned, high risk of becoming orphaned, marginal, idle, or active. First, I take all wells with zero production over the last five years as potentially orphaned. If the well's operator has no other producing wells during that period, the well is assumed to be orphaned. If the operator has positive production but the average production rate across all wells is less than 5 BOE/day and primarily operates marginal or idle wells, the well is classified as high risk of becoming an orphan. If the operator has a higher average production rate, the well is classified as low risk. The remaining wells

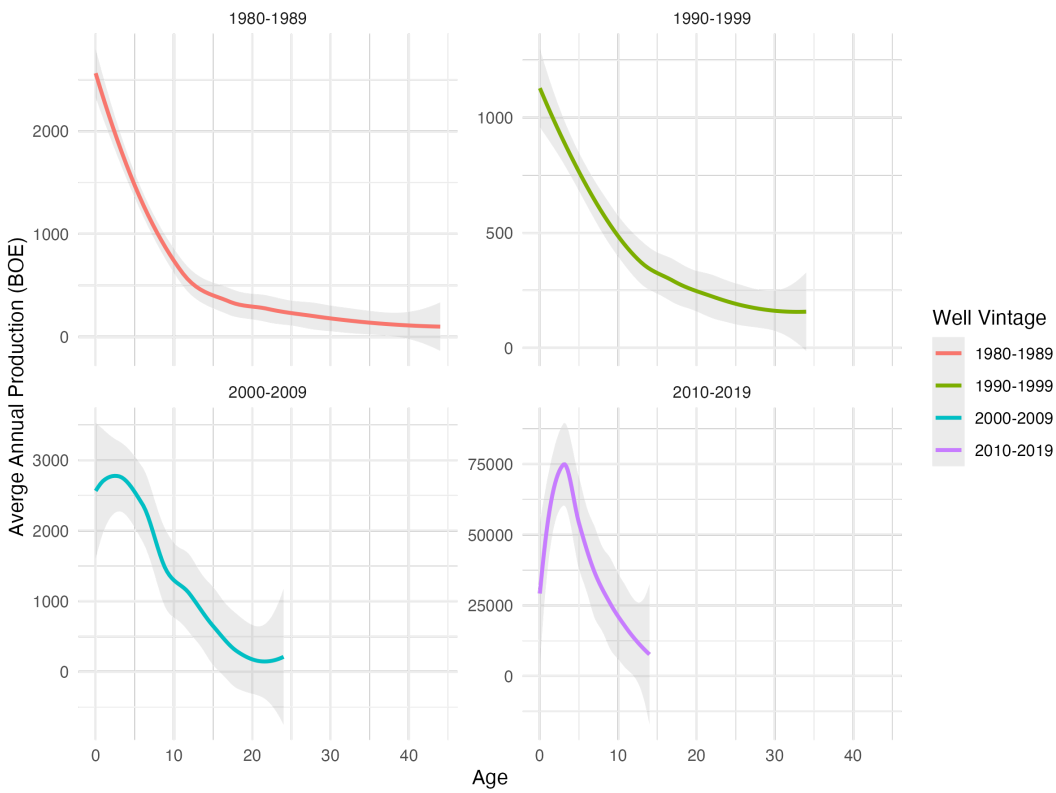

Figure 4.2: Average Annual Production by Well Vintage

are classified using current production. Marginal wells produce between 1 and 5 BOE per day, and idle wells are less than 1 BOE per day, with many completely inactive producing zero. High-producing wells with quantities of more than 5 BOE per day are considered active.[2]

I translate the daily production threshold of 5 BOE per day into an equivalent annual threshold of 1,825 BOE per year and use this annual measure in the main analysis.[3] Annual production ultimately represents viability: even if a well produces at high rates for only part of the year, low total annual output still signals likely exhaustion.

These production-intensity thresholds are calibrated primarily to conventional oil wells and are correspondingly less restrictive when applied to unconventional gas wells with higher production profiles. However, for the purposes of this naive classification these thresholds are applied uniformly across the full well life cycle, so I adopt relatively lenient cutoffs to avoid overstating inactivity or economic exhaustion at any single production level.

[2]Weber et al. (2021) estimate an economic breakeven threshold for conventional wells in Pennsylvania by comparing operating costs with projected gas prices. They find that under highly optimistic assumptions, wells become uneconomical when producing below 0.5 Mcf per day, or 0.01 BOE. This best interpreted as a lower-bound rather than a practical screening cutoff. I retain the 5 BOE/day threshold to capture economically marginal wells more broadly; using the stricter cutoff would only reclassify idle wells as marginal. The choice of threshold also does not impact the "likely orphan" category, and therefore does not affect the logit predictions.

[3]Although daily output provides a finer measure of production intensity, many wells lack reliable production-day records. I show that using daily rather than annual thresholds produces little change in the resulting classifications or aggregate counts in Appendix Table B.1.

4.5.2 Logit Model

The naive classification labels wells based on recent production patterns is inherently descriptive. It offers little insight into which active or inactive wells are likely to become future liabilities. To forecast Pennsylvania's future orphan-well burden, I estimate a predictive model that uses full production histories to relate current well characteristics to the probability of abandonment within the next five years.

I use historical patterns in production, decline trajectories, and operator attributes of likely orphan wells to predict future patterns. First, I create a binary indicator Y_{it} equal to one if the well becomes a likely orphan or high-risk well in the next five years of the observation year, measured using the above criteria.[4] This model is conditional on being observed for the past 5 years for certain explanatory variables and for the future 5 years for the dependent variable so the first and last four observations per well are omitted. This restricts the sample for this regression to 1984–2019.[5]

I then estimate a logistic regression that maps well-level and operator-level covariates to the probability of near-term abandonment. The covariates outlined above include production attributes that are known to correlate with well retirement decisions: well age, recent annual production decline, cumulative lifetime production, daily output, five-year production history, and operator characteristics such as size and well status portfolio. Year or county fixed effects are not included as that would absorb the cross-sectional variation essential for forecasting. I incorporate oil and gas prices in the logit to better capture how future market conditions may influence operators' decisions.

Specifically, I estimate the following:

$$\begin{aligned}\Pr(Y_{it+5}) = \beta_0 + \beta_1 Age_{it} + \beta_2 Decline_{it} + \beta_3 \log(Cum.Prod._{it}) + \beta_4 \log(5yr.Prod_{it}) \\ + \beta_5 DailyProd._{it} + \beta_6 SmallOp._{it} + \beta_7 Price_t + \epsilon_{it}.\end{aligned}$$

I apply the estimated coefficients to all active and inactive (idle and marginal) wells in 2024, generating well-level predicted probabilities of orphan status over the next five years. These predicted risks form the basis for the subsequent mapping and aggregation analysis. The probabilities range from near zero for recently drilled unconventional wells operated by large firms to substantially higher values for older, low-producing conventional wells managed by small operators.

This approach fully reflects uncertainty, incorporates the joint influence of age, decline patterns, operator characteristics, and production history, and avoids arbitrary threshold choices present in the naive classification. It also makes the estimates additive and comparable across regions, enabling a direct link between expected future liabilities and current policy instruments.

An important caveat is that here I focus on wells with five consecutive years of zero production as "likely orphaned," reflecting their functional abandonment in an operational sense. Some of these

[4] While the DEP maintains an inventory of verified legacy orphan and abandoned wells, these records lack production histories and therefore cannot be used to predict future orphaning risk. They are not included in the estimation sample, but because they represent confirmed state liabilities, their counts are added separately to the total orphan-well burden alongside the model-based forecasts when explicitly stated.

[5] If a well is spud in a later year, the first four years of production still need to be removed.

wells may eventually meet the formal regulatory definition of an orphan well if their operators become insolvent or relinquish responsibility, while others may be held by active operators but effectively left idle without a realistic prospect of reactivation. In either case, prolonged non-production represents a real addition to the stock of long-run liabilities: these wells impose the similar risks and remediation costs on either the operator or the state. These wells are expected to transition into a status that ultimately requires public or private remediation resources.

4.5.3 County Level Aggregation

The predicted probabilities generated by the logistic model provide a well-level measure of near-term abandonment risk. To translate these individual predictions into a statewide assessment of Pennsylvania's future orphan-well burden, I aggregate the well-level probabilities to the county level and map the resulting spatial distribution.

$$ExpectedOrphans_c = \sum_{i \in c} p_{it+5}$$

4.6 Results

4.6.1 Naive Classification

I present the results of this naive classification for 2024 in Table 4.2 which is based on production alone. Nearly all active wells are horizontal (98 percent), relatively young with an average age of 9.65 years, and produce substantial volumes, averaging over 100,000 BOE or 360 BOE per day. This emphasizes that 5 BOE per day functions as a conservative lower bound on meaningful production - in practice, wells classified as active in the sample are not clustered near this cutoff but instead produce at substantially higher rates.[6] By construction marginal wells average fewer than 2 BOE per day, while idle wells produce essentially zero. These wells are considerably older, with mean ages of over 25 years. Wells that are likely orphaned are the oldest in the inventory with an average of 32 years and have the longest operational histories.

I further break down the production-based potential orphan category to include characteristics of the operator as previously outlined in Table 4.3. The majority of wells without production over the past 5 years are also managed by operators with no recent production, suggesting these wells may have no viable operator. This aligns with the fact that they are among the oldest conventional wells in the state. A second group consists of likely orphans whose operators exhibit weak operational capacity and may be unable to absorb future plugging liabilities. For the purposes of prediction, these high-risk wells are treated as effectively orphaned. The final group are wells that may meet the production-based definition of likely orphan, yet are held by operators with substantial, ongoing production portfolios. These younger unconventional wells are considered less likely to impose future costs on the state. For

[6] Raising the active threshold to 10 BOE/day would reclassify only 267 wells as marginal, and increasing the threshold further to 20 BOE/day would reclassify only 585 wells.

Table 4.2: Naive Classification for 2024: Production-Based

Status	N	% Horiz.	Annual	Daily	Total 5-Year	Age	Prod. Yrs.
Potential Orphan	15,065	2.38	0.00	0.00	0.00	33.52	27.81
Idle	67,200	1.03	100.28	0.34	245.28	28.35	24.63
Marginal	8,870	3.55	592.67	1.88	764.63	24.81	23.21
Active	11,778	98.62	108034.83	367.17	88090.67	9.58	9.33

Notes: Columns (2) and (3) use raw well counts. Columns (4)–(6) refer to average production in BOE among wells of the given status. Column (7) refers to average years since spud year, while column (8) is average years since first production.

Table 4.3: Operator-Based Refinement for Likely Orphans

Status	N	% Horiz.	Age	Prod. Yrs.	Op. % Active
Likely Orphan	9,570	0.05	34.69	28.87	0.00
High-risk Orphan	5,141	1.93	32.50	26.60	0.03
Low-risk Orphan	354	72.03	16.81	16.46	0.81

Notes: Column (2) uses raw well counts. Columns (3)–(5) refer to average production in BOE among wells of the given status. Column (6) refers to average percentage of active wells of the operator.

the purposes of prediction, this small percentage of low-risk wells are not considered orphaned.

Although the naive classification is primarily descriptive, it offers simple forward-looking insights. Idle and marginal wells are substantially older and produce far less than active wells, and together represent more than 75,000 wells statewide. If even 5–10 percent of these wells follow the same trajectory as current orphan wells, Pennsylvania could face 3,000–7,500 additional abandoned wells in the coming decade. Notably, idle and marginal wells are only a few years younger than today's orphan wells. A large number of wells already produce less than 1 BOE/day, a threshold at which wells historically exhibit high abandonment rates. These naive projections underscore the scale of potential future liabilities and motivate the more formal predictive modeling approach that will use this information.

I present operator summary statistics in Table 4.4 which use the naive classification and reveal an interesting industry structure in Pennsylvania. The group of operators with above-median production is small at 369 firms, but on average they manage more than 230 wells each and account for virtually all statewide production. Nearly 90 percent have at least one idle well, over half manage marginal wells, yet only a quarter hold high-producing wells. Importantly, 51 percent of them also have at least one well that has not produced over the past five years, suggesting a likely orphan well if the producer could not be held liable. The below-median production operators average fewer than five wells and overwhelmingly consist of single-well entities where 264 firms own exactly one well. Their production volumes are negligible, and nearly all of them operate exclusively idle wells. The zero-production operators represent small legacy operators that collectively manage wells with no measurable output. These firms own an average of 5-6 wells and 90 percent of these operators have a likely orphan well.

Table 4.4: Operator Summary Stats for 2024

Stat	Above Median	Below Median	Zero
Number of Operators	363.00	363.00	2246.00
Mean Number of Wells	243.13	4.54	5.16
Number of Wells = 1	23.00	263.00	1593.00
Total (1000 BOE)	3517.94	0.03	0.00
% Have Abandoned Well	51.52	6.89	92.30
% Have Idle Well	90.08	100.00	9.17
% Have Marginal Well	53.17	0.00	0.00
% Have Active Well	24.24	0.00	0.00
% Only Idle	22.87	93.11	7.70

Notes: Above (below) median refer to operators with annual quantity in the top (bottom) half, and zero production refer to operators with zero annual quantity.

This underscores the strong association between very small operators, aging conventional wells, and abandoned infrastructure.

Taken together, these patterns show that Pennsylvania's future orphan well burden is highly concentrated among the thousands of very small, very low-production operators who collectively manage the bulk of aging, non-producing wells, while nearly all economically meaningful production is concentrated among a few hundred large operators.

The results above classify wells in 2024, but the same classification can be applied to all years in the sample. Figure 4.3 shows transitions across statuses over time and highlights the strong persistence of well conditions, with most movement reflecting gradual production decline. However, the transition matrix also shows that historical status alone has limited predictive power for identifying future orphan-well risk, reinforcing the need for a modeling framework that incorporates richer well-level and operator-level characteristics.

4.6.2 Logit Model Prediction

Table 4.5 reports the logistic regression estimates linking well and operator characteristics to the probability of becoming "likely orphaned" within five years. Since future oil and gas prices are unknown at the prediction horizon, the baseline specification excludes prices. Column (2) includes prices, which I use to generate predictions under low and high price scenarios based on the historical minimum and maximum observed prices.

The signs of the coefficients are consistent with expected patterns in well life cycles. Wells with greater age, fewer recent production observations, and steeper production declines exhibit sharply higher abandonment risk. Higher cumulative and recent production are strongly protective, reflecting that productive wells are rarely left idle long enough to orphan. Operator characteristics also matter: wells owned by smaller operators face systematically higher near-term abandonment risk. Unconventional wells have markedly lower predicted orphan risk, consistent with their younger age and higher annual

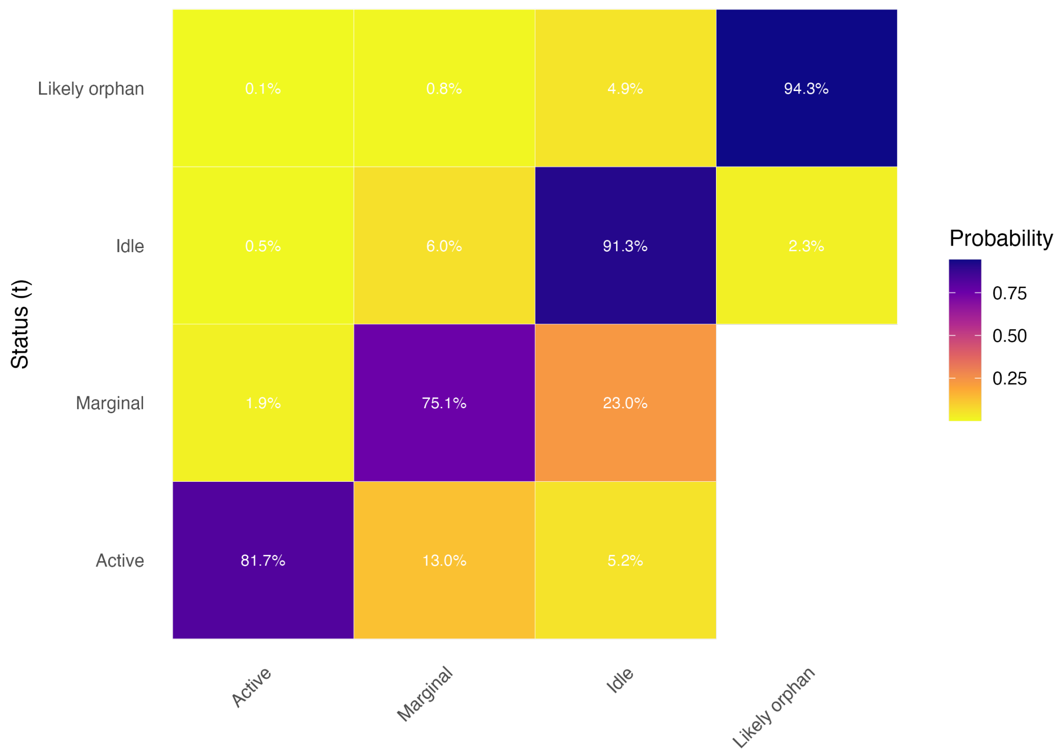

Figure 4.3: Transition Matrix

output.

The predictive model is then used to forecast orphaning risk for the current inventory of wells in 2024. Using the characteristics of each active or inactive well and the coefficients from the logit equation, the model yields a well-specific probability of becoming orphaned within the next five years. I apply the inverse logit function to map the linear predictor from log-odds into a probability between zero and one. I present the distribution of predicted probabilities in Figure 4.4. The distribution of predicted abandonment probabilities is right-skewed: most wells have low risk while a subset have substantially higher predicted probabilities.

In Table 4.6 I summarize the logit results by my previous classification. Current high-producing wells have the lowest average probability of abandonment, highlighting how they are managed by the largest operators with little risk of bankruptcy. Idle wells have the highest average probability as expected due to the persistence of low output and smaller operators. Overall, within the next five years the expected number of additional wells added to the current legacy burden is over 3,500. This is the sum of all well-level probabilities. This is in addition to the "likely orphan" category of wells, which was over 14,700. The limited sensitivity of the predictions to price scenarios is expected, as previously outlined. Price variation has little effect on abandonment risk except among currently active wells.

Table 4.5: Results from Logistic Regression

	Orphan in 5 Years = 1	
	(1)	(2)
Well age	0.003***	0.005***
	(0.000)	(0.000)
Production years	-0.010***	0.001
	(0.001)	(0.001)
One-year production decline (%)	-0.003***	-0.004***
	(0.000)	(0.000)
Log cumulative production	-0.039***	-0.083***
	(0.004)	(0.005)
Log 5-year production	-0.371***	-0.353***
	(0.004)	(0.004)
Mean Daily production	-0.008***	-0.030***
	(0.002)	(0.005)
Operator active-well count	-0.001***	-0.001***
	(0.000)	(0.000)
Operator production percentile	-2.034***	-2.015***
	(0.012)	(0.013)
Prices		-0.002***
		(0.000)
Observations	1,278,835	1,204,348

Notes: Standard errors in parentheses. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table 4.6: Prediction based on Logit Results

Status 2024	Mean	Max	Std. Dev.	Future Expected Orphans	Low Prices	High Prices
Idle	0.052	0.767	0.083	3345.964	3502.931	3232.967
Marginal	0.019	0.321	0.040	183.433	189.812	176.824
Active	0.001	0.090	0.003	7.574	3.802	3.644

Notes: Predicted probabilities between 0 and 1 are calculated from the logit model. Column (5) reports results from the specification excluding prices. The high (low) price scenario takes the maximum (minimum) historical oil and gas price, respectively.

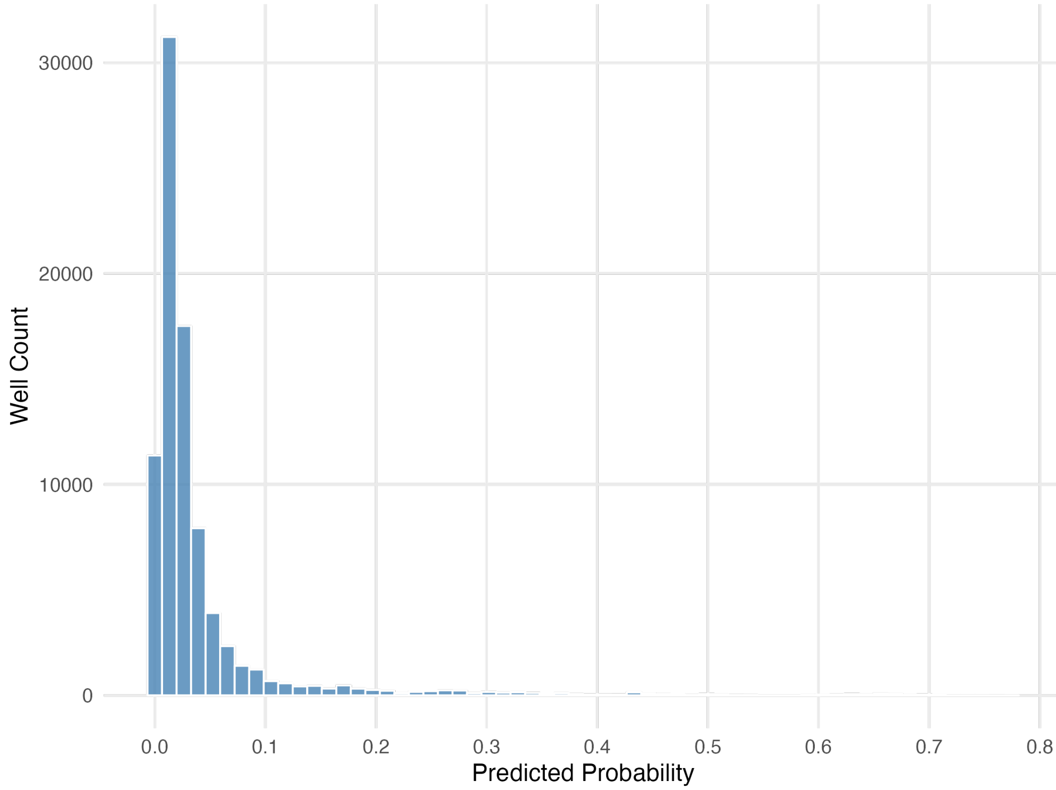

Figure 4.4: Predicted Probabilities for all Wells

4.6.3 Spatial Aggregation

I aggregate these probabilities to the county level. Table 4.7 ranks the county-level expected orphan-well burden from highest to lowest. As expected, the projected orphan-well burden is highest in counties with extensive drilling activity, simply reflecting the larger underlying stock of wells. However, the spatial concentration of the most likely future orphans are clustered in specific regions with historical conventional drilling such as McKean, Venango, Warren, and Forest rather than those with a high number of active wells, such as Indiana county.

4.7 Discussion

4.7.1 Fiscal Alignment under Act 13

Although the impact fee was originally designed to address short-run externalities associated with unconventional drilling, comparing these revenues to the long-term orphan-well liabilities remains informative and highlights the misalignment between where short-term revenues are generated and where long-term remediation burdens ultimately may arise. Counties such as Washington, Susquehanna, Bradford, and Greene receive the largest annual impact fee allocations in the northeast and southwest. Figure 4.5 visualizes this contrast by overlaying predicted orphan-well risk on county-level impact fee revenues for 2024. The wells classified as "likely" orphans are assigned probability one, while all other

Table 4.7: Expected Orphan Well Burden by County

GEOID	County Name	Likely Orphans	Number of Unplugged Wells	Expected Orphan Wells
42123	Warren	1535	10030	732.66
42083	McKean	919	8730	518.41
42121	Venango	2943	3067	359.00
42053	Forest	811	3889	241.90
42065	Jefferson	816	4515	241.83
42063	Indiana	809	10254	231.73
42005	Armstrong	640	7517	185.81
42031	Clarion	682	2776	155.66
42049	Erie	1308	1374	129.33
42129	Westmoreland	401	5096	114.06
42033	Clearfield	243	3844	88.39
42019	Butler	384	1119	74.72
42085	Mercer	205	2965	70.96
42039	Crawford	271	2780	70.85
42105	Potter	265	616	62.20
42051	Fayette	254	2982	60.43
42125	Washington	376	3078	43.24
42047	Elk	1067	1378	38.64
42059	Greene	517	2545	38.35
42003	Allegheny	237	987	28.01
42021	Cambria	33	463	14.75
42027	Centre	32	652	12.69
42035	Clinton	23	449	7.89
42007	Beaver	51	204	6.98
42073	Lawrence	16	211	2.35
42111	Somerset	30	50	1.68
42117	Tioga	105	905	0.91
42081	Lycoming	36	1039	0.90
42015	Bradford	26	1641	0.54
42023	Cameron	14	122	0.49
42115	Susquehanna	4	2029	0.47
42131	Wyoming	1	340	0.21
42113	Sullivan	4	185	0.03
42013	Blair	0	6	0.02
42061	Huntingdon	0	3	0.01

Notes: Unplugged refer to idle and marginal wells.

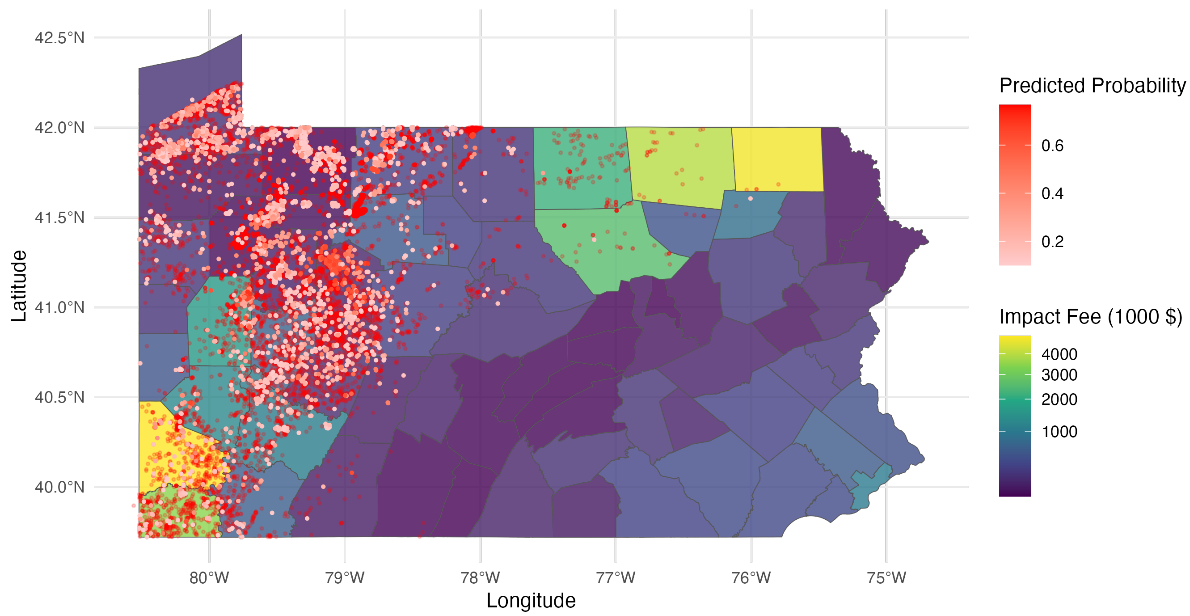

Figure 4.5: Future Orphan Well Burden across Pennsylvania and Impact Fees

unplugged wells are weighted by their predicted probabilities of abandonment. The map shows that high expected orphan burdens are concentrated primarily in northwestern Pennsylvania, where legacy conventional wells are oldest and least productive. These counties receive little impact fee revenue. Greene County in the southwest shows the strongest alignment between impact fee revenues and expected orphan-well burden.

While the current allocation structure is well suited to addressing immediate shale development impacts, it is less aligned with the geography of long-run remediation risk. As a result, Pennsylvania's current funding framework does little to prepare for the long-term costs of well abandonment. A policy mechanism explicitly aligned with the spatial and temporal distribution of these liabilities is needed to ensure adequate and sustainable support for the predicted orphan well burden.

4.7.2 Comparison to DEP Inventory

The official inventory contains data on all current and legacy wells known to the DEP. A comparison of the official reported status and the production-based naive classification reveals large inconsistencies between wells recorded as "Active" and their observed production patterns. A significant number of these wells exhibit activity far more consistent with idle or even orphan wells. Among wells recorded as Active, fewer than 12,000 also appear as Active under the naive classification. In contrast, more than 66,000 DEP Active wells are classified as Idle, and an additional 10,700 are classified as Likely Orphan wells producing zero output over the past five years. Taken together, over four-fifths of wells labeled Active by DEP show little or no recent production, suggesting that the operational meaning of "Active" in regulatory records diverges sharply from the economic reality. This mismatch most likely reflects structural features of Pennsylvania's regulatory regime, where operators face limited pressure to either

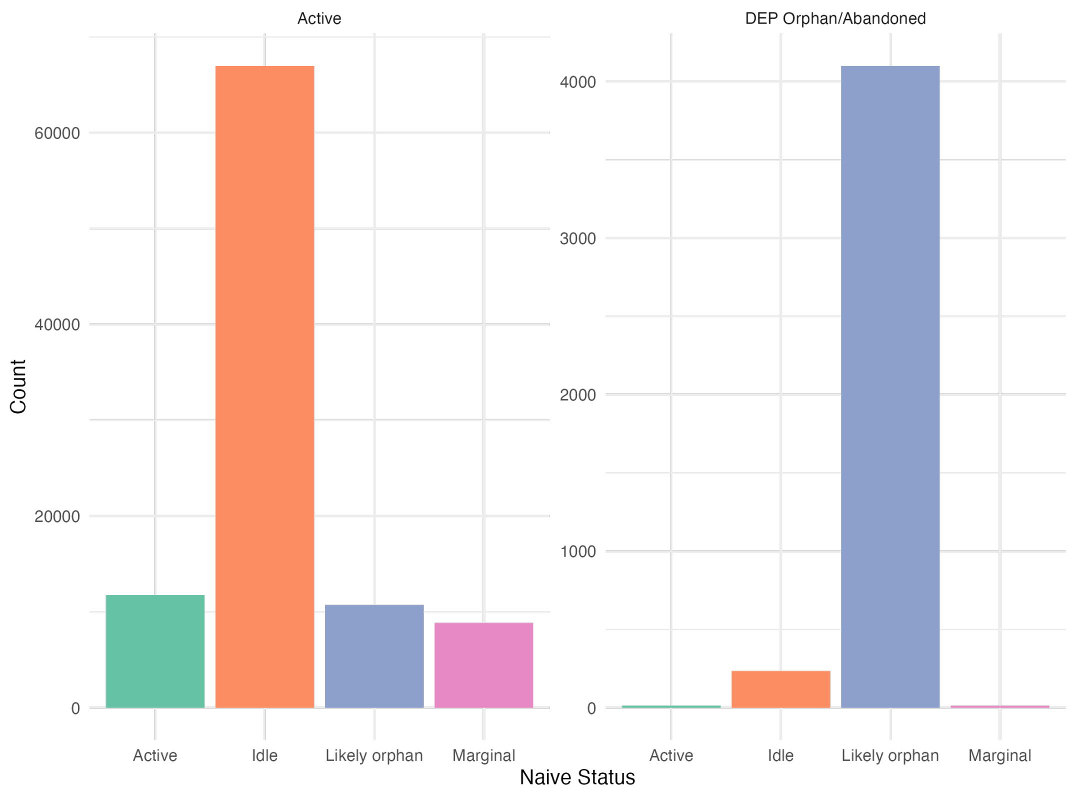

Figure 4.6: Naive Status Distribution of DEP Inventory Categories

produce from or properly decommission non-producing wells, and many wells remain in a nominally "active" status long after ceasing production.

If DEP records define tens of thousands of wells as "Active" despite negligible output, the formal inventory substantially overstates the operational viability of the state's well stock and understates both the scale and urgency of long-term liabilities. Figure 4.7 shows implied landscape of abandoned wells compared to the DEP categories. The DEP abandoned and orphan lists remain small relative to the much larger pool of wells that meet behavioral criteria for abandonment but are not administratively recognized as such.

Using the predictive probabilities for wells in the DEP "Active" category, the expected number of orphan wells that could be added to the burden within five years ranges from around 9,080 to 14,253. The lower bound is obtained by summing predicted probabilities across all wells, while the upper bound assigns probability one to all wells classified as "likely orphans." Together, these comparisons indicate that Pennsylvania's regulatory inventory substantially understates the scale of its non-producing well problem.

4.8 Conclusion

This paper develops a forward-looking assessment of Pennsylvania's orphan well liabilities by linking detailed production histories, operator characteristics, and regulatory classifications to the probability

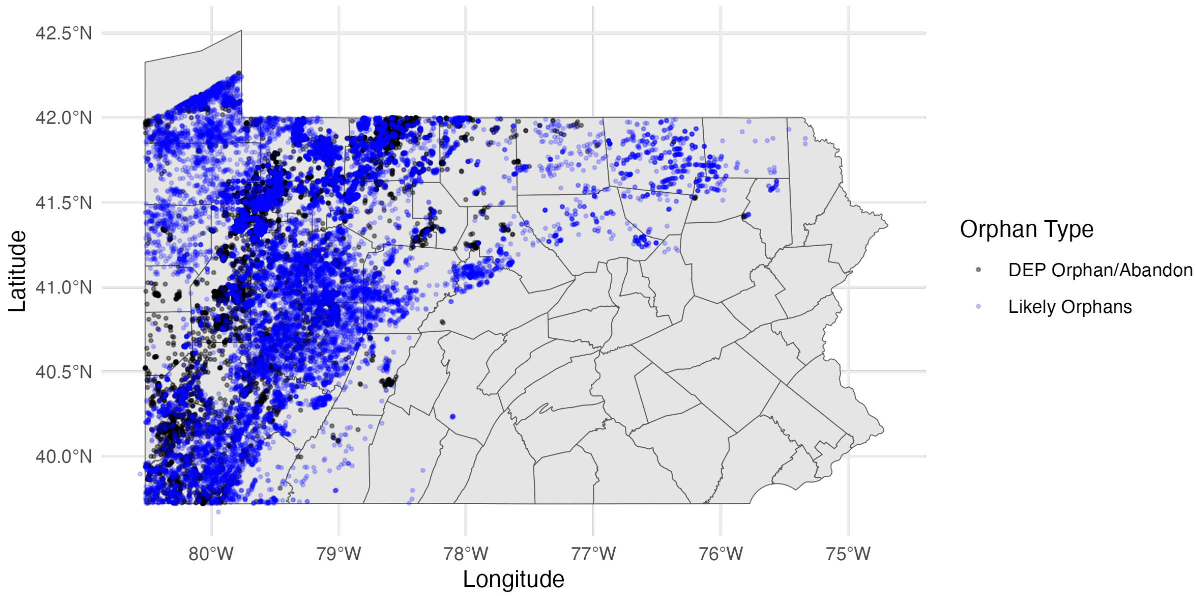

Figure 4.7: DEP Orphan Wells vs. Naive Classification

that currently active or inactive wells will become orphaned in the near future. The core patterns uncovered by the model align closely with intuitive features of well life cycles: long-idle, low-production, and aging conventional wells exhibit the highest abandonment risk, while recent unconventional wells operated by large firms rarely approach orphanhood in the short run. In this sense, sophisticated prediction confirms that Pennsylvania's abandoned well problem is driven not by failures among active, high-producing wells, but by the vast, persistent stock of legacy wells that have effectively ceased to be economically viable but are not recognized by the DEP. By quantifying the expected number of orphan wells embedded in the current well inventory, this paper highlights the extent to which Pennsylvania's long-term environmental liabilities stem from wells that are already orphaned but not administratively recognized as such.

The results reveal a structural misalignment between the geography of future liabilities and the distribution of revenues under Act 13. Impact-fee collections are concentrated in counties with ongoing unconventional development, while the burden of future orphaning lies predominantly in older conventional regions that receive little fee revenue. This divergence between short-run revenue flows and long-run cleanup costs implies that Pennsylvania's existing fiscal framework internalizes immediate drilling externalities but leaves intertemporal environmental liabilities largely unfunded. Addressing this gap will require complementary policy mechanisms.

Ultimately, these estimates represent the expected future orphan-well burden based on predicted transitions and are in addition to the existing stock of currently orphaned wells. Furthermore, these forecasts represent risk-based classifications rather than certain outcomes, as operator behavior, financial conditions, and regulatory enforcement can still lead some high-risk wells to be properly plugged before orphan status is realized.

Taken together, the findings demonstrate that credible forecasts of orphan-well risk are feasible using naive classifications and predictive models, but the deeper value lies in demonstrating the scale and geography of a longstanding legacy problem. Predictive frameworks of this kind can help policymakers anticipate emerging liabilities, target resources effectively, and design fiscal institutions that better align revenue collection with long-term environmental responsibility.

Chapter 5

Conclusion

In this dissertation, I explored three topics in resource economics dealing with mitigating economic and environmental costs present in these industries, either quantifying and predicting long-term risk or measuring distributional impacts. Together, the chapters highlight a central tradeoff in resource-based industries: while extraction generates substantial short-run private benefits, the environmental and economic costs often unfold slowly, fall on different groups of individuals, and are poorly understood or internalized by existing institutions.

In Chapter 2, I show that oil and gas wells impose persistent long-term impacts on the housing market across the life cycle of a well. I find a significant loss of property value arising when wells become inactive or abandoned and only partial recovery after plugging for recently plugged wells. Chapter 3 examines the distribution of gains from the introduction of major fisheries management reform, demonstrating that although ITQ systems raise overall rents, these benefits accrue unevenly and depend crucially on heterogeneous skills and local labor-market opportunities. Chapter 4 develops a predictive framework to spatially map out future environmental liabilities and identify wells that are either already abandoned, or at high risk of becoming orphans across Pennsylvania counties.

Several avenues for future research emerge from this work. One important avenue is to identify the mechanisms through which resource extraction and regulation affect dynamic economic and environmental outcomes, specifically the long-run distributional effects of conservation policies in resource-dependent local labour markets. Future research will also examine post-remediation land use and economic activity following well plugging, including redevelopment and ecosystem restoration.

Broadly the chapters of this dissertation can be summarized as quantifying the persistent stigma of environmental risk when facing delayed liability, the reallocation of rents across heterogeneous agents following a market-based regulation, and finally predicting environmental default risk in capital-intensive industries with declining productivity and limited responsibility. It is the author's hope that the insights of these chapters inform the management of natural resources in ways that promote efficient and sustainable use and improve the welfare of the populations that depend on and benefit from the resources in question. While economic decisions shape patterns of resource use, environmental conditions in turn shape economic activity, making sustainable management central to long-run welfare.

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Appendix A

Appendix to Chapter 2

A.1 Individual County Contributions

To examine whether these results are being driven disproportionately by any single county, I re-estimate the baseline specification repeatedly, sequentially excluding each of the top fifteen counties by size in turn. This produces the distribution of coefficient estimates shown in Figure A.1 that can be compared to the full-sample estimate, which is shown with the dashed line.

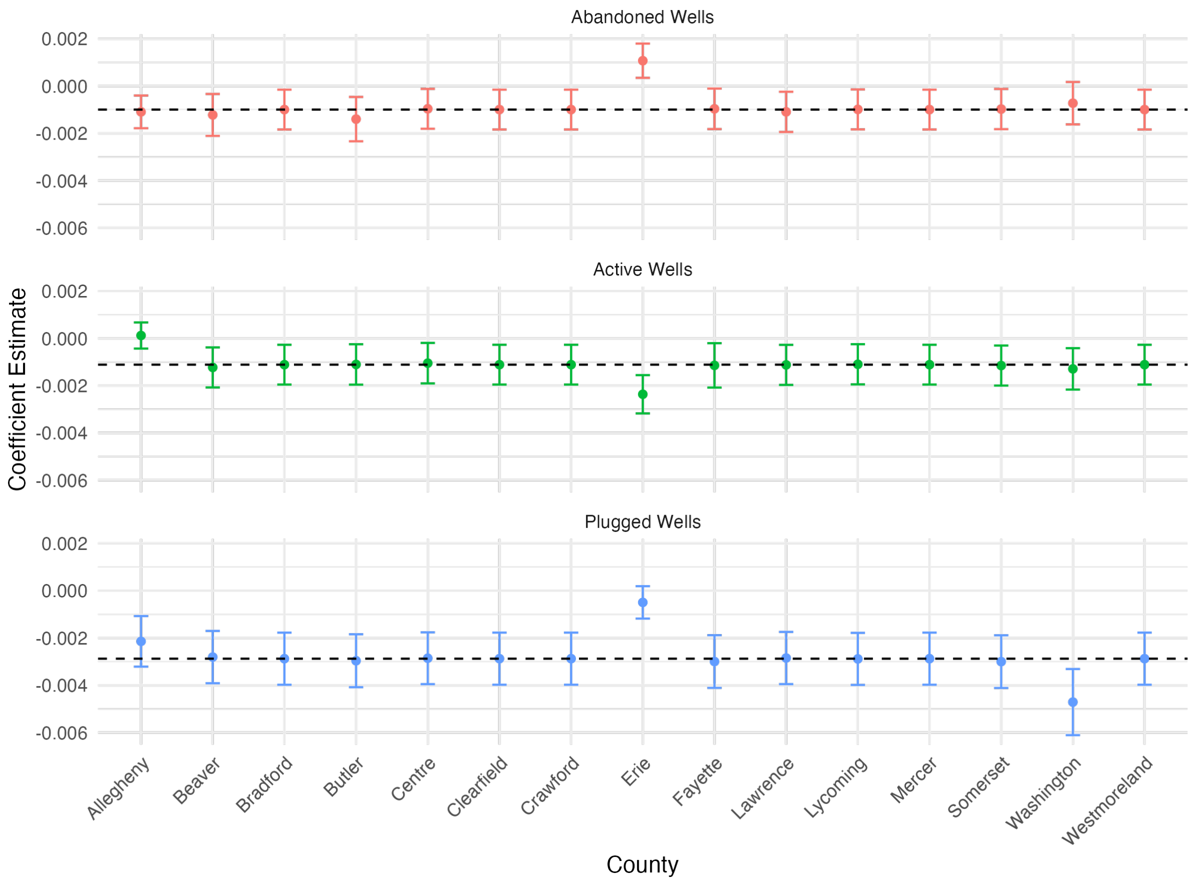

Figure A.1: Individual County Contributions

The results remain largely stable across iterations with the exception of Erie County. This is unsurprising given its substantial contribution to both well counts and transaction count, and reflects its informational importance rather than anomalous behavior. Erie drives the negative impact of abandoned and plugged wells, yet active wells have more negative impact when omitting.

A.2 Box-Cox Transformation

Note that when price is transformed using the Box–Cox specification, the coefficients are not directly interpretable as percentage effects. However, the signs and significance levels can still be compared to the log-linear results to assess robustness.

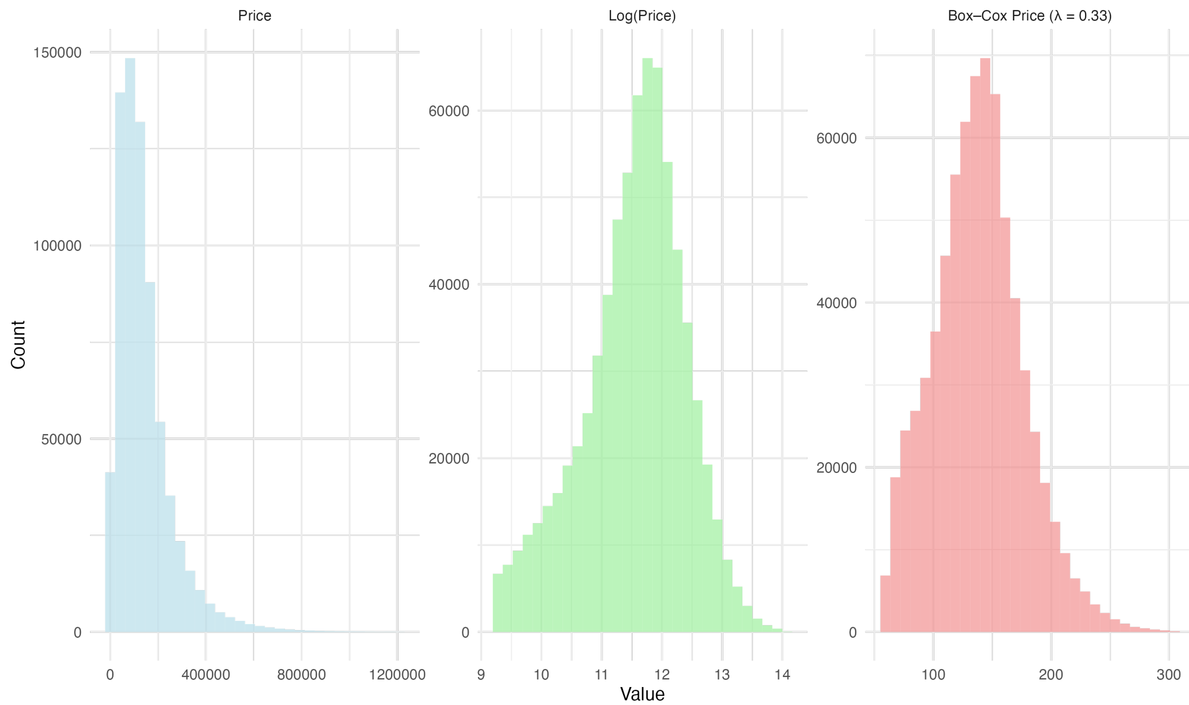

Figure A.2: Distribution of Home Prices Under Log and Box-Cox Transformations

Table A.1: Baseline Regressions with Box-Cox Transformed Prices

	price_boxcox			
	(1)	(2)	(3)	(4)
≤ 500	0.053	-0.036	0.024	-0.041
	(0.067)	(0.061)	(0.085)	(0.086)
500–1000	-0.043	-0.098***	-0.080*	-0.128***
	(0.038)	(0.037)	(0.043)	(0.044)
1000–1500	0.016	-0.023	-0.051	-0.070*
	(0.030)	(0.028)	(0.038)	(0.038)
1500–2000	-0.124***	-0.143***	-0.254***	-0.250***
	(0.025)	(0.024)	(0.036)	(0.040)
AB≤ 500	0.029	0.110**	-0.385***	-0.133*
	(0.047)	(0.046)	(0.080)	(0.069)
AB500–1000	0.020	0.082***	-0.192***	-0.021
	(0.029)	(0.028)	(0.057)	(0.046)
AB1000–1500	-0.071**	-0.037	-0.131***	-0.007
	(0.034)	(0.032)	(0.048)	(0.043)
AB1500–2000	-0.078***	-0.043**	-0.213***	-0.113***
	(0.022)	(0.020)	(0.036)	(0.034)
PL≤ 500	-0.350***	-0.282***	-0.902***	-0.439***
	(0.060)	(0.058)	(0.122)	(0.115)
PL500–1000	-0.121**	-0.037	-0.747***	-0.338***
	(0.051)	(0.047)	(0.077)	(0.058)
PL1000–1500	-0.114***	-0.032	-0.584***	-0.233***
	(0.040)	(0.030)	(0.078)	(0.059)
PL1500–2000	-0.137***	-0.077***	-0.438***	-0.200***
	(0.026)	(0.021)	(0.058)	(0.046)
Characteristics	Yes	Yes	Yes	Yes
sale_yr FEs	Yes	No	Yes	No
sale_mo FEs	Yes	Yes	Yes	Yes
FIPStract FEs	Yes	Yes	No	No
FIPS-sale_yr FEs	No	Yes	No	Yes
FIPSblock FEs	No	No	Yes	Yes
Observations	719,014	719,014	719,014	719,014
R^2	0.67	0.67	0.74	0.75

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ Homes with no more than 100 wells nearby. Standard errors are clustered at the county year level. Active wells refer to wells deemed active by the Department of Environmental Protection (DEP). Abandoned wells include legacy wells and wells that have not been producing for at least 12 months. Plugged wells must have a valid date of plugging prior to the sale year. The distance cutoff is measured in meters.

A.3 Baseline Regression Using Imputed Production-Based Status

Table A.2: Baseline Regressions with Production-Based Well Status

	log(Price)			
	(1)	(2)	(3)	(4)
Total Unplugged	-0.001***	-0.001***	-0.003***	-0.002***
	(0.000)	(0.000)	(0.001)	(0.001)
Characteristics	Yes	Yes	Yes	Yes
Sale Year	Yes	No	Yes	No
Sale Month	Yes	Yes	Yes	Yes
Tract	Yes	Yes	No	No
County x Sale Year	No	Yes	No	Yes
Block	No	No	Yes	Yes
Observations	719,014	719,014	719,014	719,014
R^2	0.62	0.62	0.70	0.70

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ Homes with no more than 100 wells nearby. Standard errors are clustered at the county year level. Active wells refer to wells deemed active by the Department of Environmental Protection (DEP). Abandoned wells include legacy wells and wells that have not been producing for at least 12 months. Plugged wells must have a valid date of plugging prior to the sale year. The distance cutoff is measured in meters.

A.4 Maybe Actual Parcel Thing

Appendix B

Appendix to Chapter 4

B.0.1 Daily Threshold

Marginal decrease, turn to idle. Essentially this means that daily decently translates to annual.

B.0.2 Alternative Activity Threshold

Table B.1: Naive Classification for 2024: Production-Based

Status	N	% Horiz.	Annual	Daily	Total 5-Year	Age	Prod. Yrs.
Likely orphan	15,065	2.38	0.00	0.00	0.00	33.52	27.81
Idle	65,858	0.89	99.11	0.29	226.09	28.31	24.61
Marginal	10,033	3.20	532.74	1.64	640.97	25.67	23.63
Active	11,957	97.94	106427.87	362.01	86863.01	9.67	9.40

Notes: Columns (2) and (3) use raw well counts. Columns (4)–(6) refer to average production in BOE among wells of the given status. Column (7) refers to average years since spud year, while column (8) is average years since first production.